Clinical Practice Guideline by Infectious Diseases Society of America (IDSA): 2025 Guideline on Management and Treatment of Complicated Urinary Tract Infections: Introduction and Methods

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Introduction

Reasons to Update the UTI Guidelines

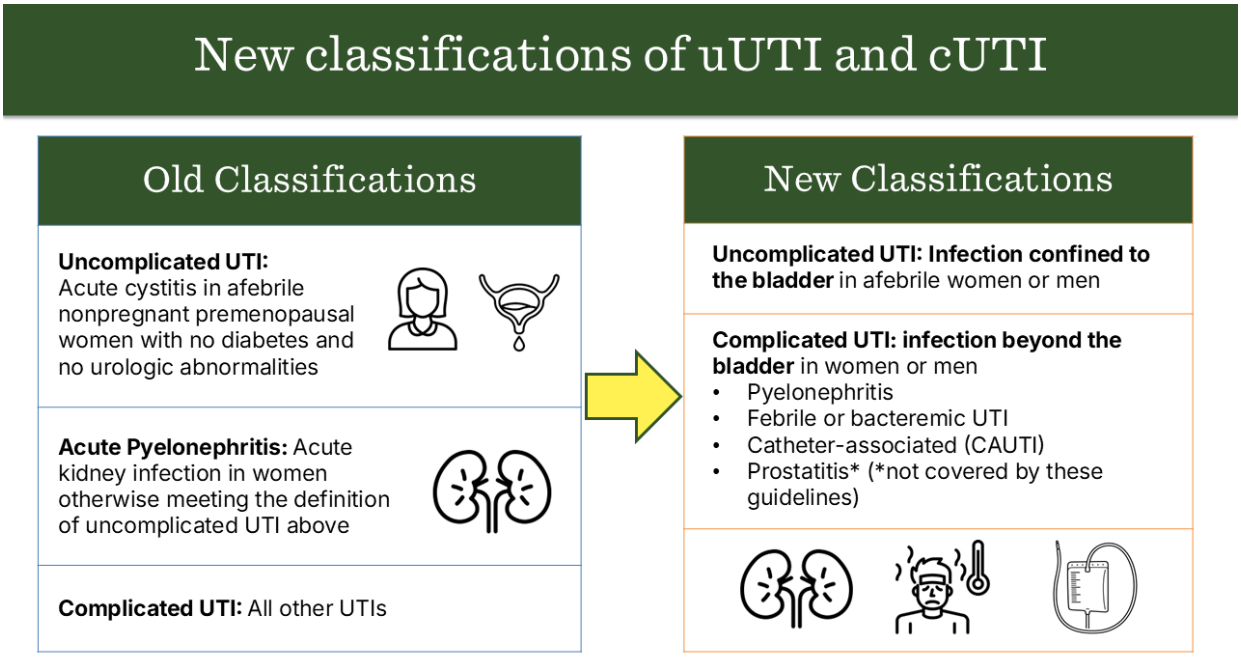
Magnitude of the burden of urinary tract infection: Urinary tract infection (UTI) is one of the most frequent reasons for clinic, emergency room, and hospital visits, with an estimated annual global burden of 404.61 million cases and 236,790 deaths in 2021.[1] Consequently, UTI is one of the most common infections managed by primary care clinicians, urgent and emergency care clinicians, hospitalists, general internists, gynecologists, and geriatricians. In 2021, an estimated 7.7 million visits to United States (US) emergency departments were for genitourinary issues, the majority presumably for UTI.[2] A summary of the National Inpatient Sample (NIS) database in 2018 reported that among 35.5 million hospitalizations in the US, 2.8 million had any UTI ICD-10 diagnosis code, and 626,520 met their definition of complicated UTI.[3] Of the complicated UTI diagnoses, only 126,520 (20%) had coding documentation of a urinary catheter and thus met their definition of catheter-associated UTI (CAUTI).

UTI is also the leading cause of gram-negative bacteremia, accounting for 48% of gram-negative bacteremia reported by 24 US hospitals in 2021.[4-6] Many but not all healthcare-associated UTIs are related to urinary catheters. A point prevalence survey of patients in US acute care hospitals in 2015 found that UTI accounted for 9.1% of all healthcare-associated infections, with an estimated 62,700 healthcare-associated UTIs annually.[5] Of these healthcare-associated UTIs, 62% were related to an indwelling transurethral urinary catheter, thus meeting the surveillance definition for CAUTI. Unfortunately, estimates of the burden of complicated urinary tract infections (cUTIs) are uncertain, for several reasons. Documentation of urinary catheter use is often missing, and thus database studies tend to underestimate CAUTI. On the other hand, CAUTI may be over diagnosed in asymptomatic patients with bladder colonization. National surveillance for healthcare associated infections in acute care settings requires reporting of CAUTI but not of UTIs that develop in the absence of a urinary catheter, so the true incidence of UTI cases occurring in hospitalized patients is hard to quantitate.[6]

Need to expand the scope of prior UTI guidelines: The prior version of the IDSA UTI guidelines focused on uncomplicated cystitis and pyelonephritis in women, omitting complicated UTI (cUTI) and UTI in men.[7] Since the publication of those guidelines, many randomized, controlled trials assessing new antimicrobials for cUTI in both women and men have been published. Although UTI is more common in women, m. Women have a lifetime risk of 53% of experiencing UTI. While UTI is uncommon in men prior to age 50, their lifetime risk is a nontrivial 14%.[8] Risk of experiencing a UTI increases with age in both sexes.[9] Given the aging US population, UTI in men is a salient issue, as is UTI in women. Fortunately, a reasonable evidence base now exists to support guidelines for treatment of cUTI in men and women. As a caveat, throughout these guidelines, “women” and “men” refer to persons’ biological sex (assigned at birth), as data from transgender persons are lacking in UTI clinical trials, and genitourinary anatomy is likely relevant for UTI risk and treatment response.

The bacteria that are isolated from UTI are becoming increasingly resistant to antibiotics: Gram-negative urinary organisms collected from outpatients across all regions of the United States now have antimicrobial resistance rates above the thresholds recommended for using antibiotics as empiric treatment of UTI in the 2010 guidelines.[7] Specifically, these guidelines recommended avoiding empiric use of trimethoprim-sulfamethoxazole for acute cystitis if the local resistance rates of uropathogens are known to exceed 20% and avoiding empiric use of fluoroquinolones alone for pyelonephritis if the resistance of community uropathogens to fluoroquinolones is known to exceed 10%. Enterobacterales (formerly Enterobacteriaceae) is the order of gram-negative bacteria that includes many of the organisms typically causative of UTI (i.e., *Escherichia coli*, *Klebsiella* species, *Proteus* species) but does not include *Pseudomonas* species. Among Enterobacterales urinary isolates from US adult patients in ambulatory settings from 2018-2020, resistance was 21.6% to fluoroquinolones, 22.4% to trimethoprim-sulfamethoxazole, and 21.6% to nitrofurantoin. Corresponding inpatient urinary isolates had resistance rates of 27.5%, 25.4%, and 27% to fluoroquinolones, trimethoprim-sulfamethoxazole, and nitrofurantoin, respectively.[10] In the face of these concerningly high rates of resistance, the evidence needed to guide empiric choice of antibiotics for treating UTIs needs to be reevaluated.

Figure 1.0. Comparing prior and updated classifications of uncomplicated and complicated UTI



Need to update the classifications of uncomplicated and complicated UTI: We perceived a need to update the classifications of uncomplicated and complicated UTI to better align with clinical practice, become more congruent with the available data on male UTI, and better guide management decisions. The clinical distinctions between subcategories of complicated UTI, such as pyelonephritis and febrile UTI, are not standardized or clearly understood from a pathogenesis point of view.[11,12] We therefore focused our revised classifications of uncomplicated and complicated UTI on the presence or absence of localized or systemic symptoms, particularly fever, that would suggest the infection had progressed beyond the bladder. We also focused the revised classifications on factors that would be readily apparent to the treating clinician at the point of care (e.g., vital signs and catheterization) rather than factors that might not be apparent without a urologic evaluation (e.g., anatomic abnormalities or urinary retention). These updated classifications align with the current clinical terminology of simple cystitis versus complicated UTI. Our updated classifications also align the European Urologic Association's (EUA) recently published classifications of UTI as localized or systemic.[13] We likewise agree with the EUA that risk factors for treatment failure (and subsequent complications) are relevant to both uncomplicated (localized) or complicated (systemic), and the clinician must consider each patient's risk factors in addition to determining whether the infection is presenting as a localized or systemic infection. Furthermore, all patients diagnosed with UTI need to have signs or symptoms related to the organisms in the bladder (or spreading beyond the bladder); otherwise the patient actually has asymptomatic bacteriuria (ASB) and does not need antibiotic treatment.

Classifications

Classifications of complicated and uncomplicated UTI: Our understanding of what constitutes uncomplicated and complicated urinary tract infection has evolved since the 2010 publication of the IDSA guidelines (**Figure 1.0**). Clinical practice and considerations for management of the infectious syndrome have largely driven these changes. The classifications that we have chosen for the purposes of these guidelines focus on whether or not the infection is likely to be confined to the bladder, on initial evaluation. If so, the infection is defined as an uncomplicated UTI. Systemic signs of illness, such as fever or bacteremia, suggest that the infection has extended beyond the bladder and is thus defined as a complicated UTI. Evidence of renal parenchymal involvement (e.g. costovertebral angle tenderness), or a prostatic abscess, would also demonstrate that the infection has extended beyond the bladder. Subsequent discovery of obstruction in the upper urinary tract, such as from a stone, would render the infection complicated. Therefore, our categorization focuses on the extent and clinical severity of the illness, which in turn will drive management considerations.

Patients who have underlying urologic abnormalities (such as benign prostatic hypertrophy or a cystocele), diabetes, or immunocompromise are not automatically classified as having a complicated UTI. If such patients appear to have infection confined to the bladder and are not systemically ill, their infection would be considered an uncomplicated UTI (uUTI). By this classification, both men and women can have uncomplicated UTI, but choice of antibiotics and duration of treatment for uncomplicated UTI may differ for men and women. Duration of treatment for uUTI will be addressed in the subsequent uUTI guidelines. While we acknowledge that patients with underlying urologic abnormalities, diabetes, or immunocompromise may be at higher risk for severe infection and may merit closer observation, these beliefs should be balanced against the lack of robust data indicating said populations benefit from longer antibiotic duration for UTI. On the other hand, a patient with new onset hydronephrosis related to obstruction along the urinary tract would be expected to have a cUTI with systemic symptoms.

In general, acute prostatitis, should be considered and excluded before classifying UTI as uncomplicated in a male patient, as prostatitis can affect choice of therapy and duration.[14] Our clinical questions and accompanying search strategies did not encompass prostatitis, epididymitis, or orchitis, and the management of these conditions is outside the scope of these guidelines.

Our updated classifications should help avoid the difficulty in clinical practice of determining which patients have an underlying urologic abnormality (such as chronic urolithiasis), as such information is often unavailable at the point of care when treating UTI.

Box 1: Complicated UTI classifications for guidelines purposes (intended to guide treatment not diagnosis)

- Clinical presentation:
  - Complicated UTI is accompanied by symptoms which suggest an infection extending beyond the bladder, including:
    - Fever
    - Other signs or symptoms of systemic illness (including chills, rigors, or hemodynamic instability)
    - Flank pain
    - Costovertebral angle tenderness
  - Pyelonephritis is encompassed in complicated UTI.
  - UTI with systemic symptoms associated with transurethral, suprapubic, or intermittent catheterization is encompassed in complicated UTI.
- Populations:
  - Patients with complicated UTIs may have an indwelling urinary catheter, neurogenic bladder, urinary obstruction, or urinary retention as an underlying condition.
  - These guidelines are not intended to apply to bacterial prostatitis, epididymitis, or orchitis.

Conversely, patients with long-term indwelling urinary catheters, stents, or percutaneous nephrostomy tubes and symptoms related to the presence of bacteria in the urinary tract are considered to have a complicated UTI, as the presence of a medical device/foreign body is readily apparent and impacts multiple aspects of the treatment approach. For our purposes, urinary catheters include transurethral (Foley) catheters, suprapubic catheters, and daily intermittent use of urinary catheters because all can be associated with bladder colonization, biofilm formation, bladder/urethral trauma, and chronic sediment or stones in the urinary tract. However, we acknowledge that a subset of patients with CAUTI without systemic symptoms may be treated as simple cystitis, particularly those in which the catheter has only been in place for a short while, thus minimizing the bladder injury. In a patient with a long-term urinary catheter who does not have signs or symptoms of infection beyond the bladder, the clinician should carefully consider whether the patient has UTI or asymptomatic bacteriuria (ASB). ASB is covered in the IDSA guidelines on this topic.[15] Urinary catheters can cause urethral and bladder discomfort, which needs to be distinguished from the bladder pain of cystitis.

"note that the classifications of uncomplicated and complicated UTI chosen for these guidelines are intended to guide management of UTI, not diagnosis. Details of our classifications appear in **Figure 1.0** and **Boxes 1 and 2**.

Our classifications of uncomplicated and complicated UTI differ from the Food and Drug Administration (FDA) guidance definitions. The FDA guidance defines uncomplicated UTI as occurring in a female patient with lower urinary tract symptoms and pyuria, and microbiologic confirmation via culture is recommended. Many of the clinical trials that form the evidence base for these guideline recommendations were designed per the FDA definitions. Therefore, trials designed to study a new drug for uncomplicated cystitis excluded men, and trials focused on complicated UTI may have enrolled men with simple, acute cystitis. Another important implication of our updated classifications is that studies focused on women with "uncomplicated pyelonephritis," including cases of mild pyelonephritis that could be managed in the outpatient setting, became part of our evidence base for management of complicated UTI. Pyelonephritis in a premenopausal woman without other urinary tract abnormalities is clearly a different scenario than pyelonephritis in an older woman with a staghorn calculus; nuances of the management of pyelonephritis will be addressed under the text of the clinical questions. Implications of these definitional differences are discussed under each guideline recommendation.

Box 2: Uncomplicated UTI classifications for guidelines purposes (intended to guide treatment, not diagnosis)

- Clinical presentation:
  - A clinical syndrome characterized by local bladder signs and symptoms such as dysuria, urgency, frequency, and suprapubic pain.
  - Uncomplicated UTI is presumed to be confined to the bladder and is defined by absence of signs or symptoms which suggest an infection extending beyond the bladder:
    - No fever, unless explained by a non-UTI cause
    - No other signs or symptoms of systemic illness (including chills, rigors, or unstable vital signs), unless explained by a non-UTI cause
    - No flank pain
    - No costovertebral angle tenderness
- Populations:
  - Uncomplicated UTI can occur in females or males, patients with underlying urologic abnormalities, patients with immunocompromise, and persons with diabetes. Recurrent UTI can be uncomplicated.
  - Patients with urinary catheters (including transurethral, suprapubic, and intermittent catheterization), stents, and percutaneous nephrostomy tubes generally do not have uncomplicated UTI.
  - These guidelines are not intended to apply to bacterial prostatitis, epididymitis, or orchitis.

A few other points of clarification may help in understanding these updated classifications and resulting guideline recommendations. Hospitalized patients can develop an uncomplicated UTI, so the term "healthcare-associated UTI," typically referring to a UTI acquired in a healthcare setting, does not define whether the UTI is uncomplicated or complicated.[16] Additionally, recurrent episodes of UTI can be uncomplicated or complicated. Our guidelines address empiric treatment of individual episodes of UTI without addressing preventive strategies for recurrent UTI. Uncomplicated UTI will be addressed in a subsequent set of guidelines.

Please note that these classifications are intended to help clinicians choose the best course of action, but there are cases in which the clinical presentation might not completely fit into one of these two categories. Clinical judgment is required for application of these categories and guidelines to patient care.

Definitions of “male” and “female”: UTI research is largely limited to cisgender male-bodied persons and female-bodied persons and largely fails to distinguish between gender identity and biological sex. Reference in the literature to “men” or “males” and “women” or “females”, often used interchangeably, suggests gender identities that participants may or may not have for themselves and omits people with nonbinary identities. The terms “male” and “female” are sex-based terms that refer to binary biological characteristics and exclude those whose bodies do not fit the binary. Acknowledging these limitations in language and past research, we will use language employed in original research studies. Throughout the text, we will use terms employed by the original studies to describe their participants when describing the research literature.

Scope: Population

The population encompassed in these guideline recommendations is adults with complicated urinary tract infections (cUTI). Patients with renal transplantation or who are pregnant, lactating, or neutropenic are not specifically addressed in this guideline, because these populations were excluded from the clinical trials that form the evidence base for these guidelines. Clinicians must use their judgment to determine whether and when our recommendations can be generalized to individual patients in these populations. That said, we do not have any reason to believe that lactating women should be treated for a different duration of antibiotics, or that the IV to oral switch data would not apply to them. For antibiotic choice in lactating women, excretion into the milk is a factor to consider, Advice on antibiotics choices during lactation can be found through the American College of Obstetrics and Gynecology guidelines[17] and the online Drugs and Lactation Database (LactMed®).[18]

Clinical trials contributing to the evidence base typically limited enrollment to ages 18 and up, leading to exclusion of adolescents and children. Other populations rarely included in randomized, controlled trials of cUTI management include catheterized patients, patients with pyogenic infection (abscesses) of the urinary tract, men with acute prostatitis, patients with renal failure, patients who are immunocompromised for reasons other than neutropenia, patients with urinary obstruction, patients with nephrostomy tubes or urinary stents, and patients with urinary stones. Data on cUTI outcomes in these populations are, therefore, very limited. Of note, few studies attempted to determine if men with febrile UTI had prostatic involvement, nor is there a single widely agreed-upon approach for ruling out prostatitis in men with febrile UTI. Details of which cUTI treatment duration trials included men with prostatitis versus excluded men with prostatitis are discussed in clinical question 3 (duration of treatment for cUTI) and the supplementary materials for clinical question 3. Our guidelines also recognize that several populations were not formally studied due to limited identification of these study populations in trials, including patients with external (condom) catheters and transgender or gender diverse persons; these groups may have been represented within study populations yet not

documented as such. Finally, many patients who are treated for cUTI do not actually have a UTI and instead have another etiology for their presenting symptoms, including fever or sepsis.

Scope: Audience

Appropriate treatment of cUTI requires a correct diagnosis; however, diagnosis of cUTI, including overdiagnosis of cUTI in people with asymptomatic bacteriuria, will be covered in a future set of guidelines. We focused this initial guideline on cUTI treatment because this infection is common and the IDSA has not previously published clinical practice guidelines for cUTI. Additionally, many new drugs have recently received FDA approval for cUTI, and their registration trials provide an appropriate evidence base for cUTI guidelines. New treatment recommendations for uncomplicated UTI are in progress and will be subsequently released. For management of asymptomatic bacteriuria, please refer to the IDSA guidelines on management of asymptomatic bacteriuria.[15]

The intended target audience for these guidelines is the healthcare clinicians who most often treat cUTI in the emergency department, urgent care clinics, and hospital wards, including hospitalists, primary care clinicians, and emergency room clinicians. We also considered the perspectives of infectious disease specialists and pharmacists. The four practice questions this guideline addresses are: 1) how to best approach selection of empiric antibiotics for cUTI, 2) whether and when antibiotics for cUTI can be switched from the parenteral to oral route, 3) the appropriate duration of antibiotic treatment for cUTI, and 4) the role of imaging for evaluation of a patient presenting with suspected cUTI. Clinical questions 1, 2, and 3 will be presented here; clinical question 4, focused on the role of imaging, will follow later.

Methods

Clinical Practice Guidelines

Clinical Practice Guidelines are statements that include recommendations intended to optimize patient care by assisting practitioners and patients in making shared decisions about appropriate healthcare for specific clinical circumstances. These are informed by a systematic review of evidence and an assessment of the benefits and harms of alternative care options. The “IDSA Handbook on Clinical Practice Guideline Development” provides more detailed information on the processes followed throughout the development of this guideline.[19] The guideline was developed using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) framework to prioritize clinical questions, identify patient-centered outcomes, and conduct an evidence synthesis.[20] The guideline panel used the Evidence-to-Decision framework to develop recommendations and provide implementation considerations for clinical practice.[21]

Guideline Panel Composition

The IDSA leadership selected the two co-chairs of the guideline panel. The co-chairs selected a total of 17 panelists from among volunteer applicants, including physicians and a pharmacist. Expertise represented among our panelists included infectious diseases, medical microbiology, hospital medicine, primary care, emergency medicine, gynecology-obstetrics, urology, and pharmacology. Panelists were diverse in gender, geographic distribution, and years of clinical experience. An IDSA guideline methodologist oversaw all methodological aspects of the guideline development. IDSA staff oversaw all administrative and logistical issues related to the guideline panel.

Disclosure and Management of Potential Conflict of Interest

All expert panel members complied with the IDSA policy on conflict of interest (COI), which requires disclosure of any financial, intellectual, or other interest that might be construed as constituting an actual, potential, or apparent conflict. Evaluation of such relationships as potential conflicts of interest was determined by a review process that included assessment by the Standards and Practice Guideline Subcommittee (SPGS) Chair, and if necessary, the Executive Committee of the Board. This assessment of disclosed relationships for possible COI was based on the relative weight of the financial relationship (i.e., monetary amount) and the relevance of the relationship (i.e., the degree to which an independent observer might reasonably interpret an association as related to the topic or recommendation of consideration).

In the event that a prohibited conflicts are disclosed during the course of the guideline development process, an internal investigation will be conducted to determine if the conflict introduced bias to the process and whether the panelist may continue to serve, provided the conflict can be appropriately managed. The reader of these guidelines should be mindful of this when reviewing the list of disclosures. See the COI summary at the end of the guideline for the disclosures reported to IDSA.

Formulating Clinical Questions

The clinical practice guideline development started in 2019. In line with the National Academy of Medicine standards on trustworthy guidelines,[22] the GRADE approach was used to assess the certainty of evidence and strength of recommendation. Clinical questions were formulated using the “PICO” format (Patient/Population [P]; Intervention/Indicator [I]; Comparator/Control [C]; Outcome [O]) questions.

Determining Outcomes of Interest

All outcomes of interest were identified *a priori* and explicitly rated for their relative importance for decision-making. Each clinical question was assigned to a subgroup of panelists. Ranking of the outcomes by importance for decision-making was determined by consensus within the panel in collaboration with patient representatives for each clinical question question. In situations where a clinical question compared the use of one specific antibiotic regimen to

another (e.g., comparing spectrum of activity, route of administration, or duration of therapy) and the beneficial effects of the two regimens were similar, then the undesirable outcomes could be ranked as critical for decision-making, but several other considerations might have also been taken into account, such as antimicrobial stewardship issues for appropriate use, tolerability, as well as costs and resources.

Panel's judgment on clinical cure versus microbiologic cure: We judged the most important outcome in interventional treatment trials for cUTI to be mortality; however, mortality was rare in the identified clinical trials. In this context, the panel (in collaboration with patient representatives) judged that clinical cure at test-of-cure (TOC) was most critical for decision-making (meaning that this outcome was the main driver of whether an antibiotic should be preferred versus other agents for cUTI).[23] The panel also judged that recurrence of infection at late follow-up, which is influenced by many factors beyond initial antimicrobial therapy (e.g., comorbidities), was important for decision-making about the choice of empiric therapy and the switch from IV to oral therapy. Taking into account the values and preferences of our patient representatives, recurrence of infection was judged to be critical for making decisions about duration of therapy.

In contrast to the FDA guidance on clinical trials for cUTI, the panel decided that clinical cure alone rather than a composite outcome of clinical and microbiologic cure should be the primary metric for comparing antimicrobials for empiric treatment of cUTI. The panel judged that microbiologic eradication was important but not critical for decision-making. The GRADE methodology is outcome-centric and focuses on patient-important outcomes. Clinical cure is the most important outcome to patients and was set as our critical outcome for decision-making when choosing among empiric antibiotics for initial treatment of cUTI. Our analysis placed less emphasis on microbiologic cure for several reasons. First, if the patient no longer has urinary symptoms, from the patient's perspective, the UTI episode is resolved. Second, detecting microbiologic cure or failure requires a urine culture after cessation of antibiotic therapy, and in practice, test of cure urine cultures are discouraged in asymptomatic patients.[15] Third, the majority of the trials that contributed to this cUTI evidence base did not report the outcome of symptomatic recurrence, and very few confirmed whether the organisms causing recurrence were the same as those isolated at microbiologic failure. Another issue is that microbiologic cure or eradication was variably defined as a test-of-cure urine culture with fewer than 104 colony-forming units per mL (CFU/mL) or fewer than 10^3 CFU/mL of the baseline uropathogen, therefore the definition of microbiologic failure also varies. Women with 10^2 CFU/mL can still have symptoms of acute cystitis.

We acknowledge that multiple studies with varying degrees of adjustment for potential confounders show that patients with microbiologic failure might also be more likely to have clinical recurrence.[24,25] This finding is not surprising in light of decades of evidence from uUTI studies that women with asymptomatic bacteriuria (ASB) are also more likely to develop symptomatic UTI. However, the anatomical and immunologic risk factors predisposing to ASB in these patients cannot be corrected by antibiotics and treatment of ASB in multiple populations (e.g., young women, catheterized patients, diabetic patients, renal transplant patients, and nursing home residents) has consistently failed to prevent subsequent bouts of ASB or

symptomatic UTI.[15] Whether or not this finding also applies to the patients in cUTI studies is unknown because, to the knowledge of the panel, no trial has yet randomized patients with microbiologic failure at TOC to antibiotic treatment versus no treatment. That said, we considered microbiologic cure in the decision-making process but did not value it as highly as clinical cure.

Panel's judgment on clinical decision threshold: The panel judged that the clinical decision threshold (or a minimally important difference) would consist of an increase of more than 10% in clinical failure. For example, if the treatment with a specific antibiotic leads to an increase in clinical failure of more than 10% than the comparator antibiotic(s), the panel would judge this difference as clinically unacceptable and important enough to decide to recommend against this antibiotic. Despite not being considered a critical outcome, microbiologic cure was also examined using a decision threshold of 10%. All other outcomes were compared based on statistical difference.

Evidence Review and Synthesis

Search strategy

After scoping the available literature, the panel decided to conduct multiple systematic reviews to summarize and synthesize the evidence for all clinical questions included in the guideline. Two librarians specialized in systematic review designed the literature searches and MeSH terms for Ovid Medline, for EMBASE and Cochrane Central Register of Controlled Trials (CENTRAL), and Cochrane Database of Systematic Reviews. Searches were limited to studies published in English and restricted to two different sets of years of publication: from 2000 up to September 2024 for general concepts (for example, for timing of IV to oral switch or impact of inappropriate empiric antibiotic therapy) or from 2008 up to September 2024 if updating data from the prior IDSA uncomplicated UTI guidelines 2010 (for example, empiric choice of antibiotics).[7] These cut offs (2000 and 2008) were chosen based on concerns about generalizability of older studies to current practice. The initial formal literature search was initially performed in October 2020 and last updated in September 2024. All search strategies are available in the Supplementary materials.

Study selection

A subgroup of panelists screened titles and abstracts of all identified citations. Two panel members independently screened each relevant title and abstract retrieved from the search using Covidence software (Covidence Systematic Review Software VHI, Melbourne, Australia. Available at www.covidence.org). A third reviewer resolved disagreements about relevance. All potentially relevant citations were subjected to a full-text review using predefined inclusion and exclusion criteria tailored to meet the specific population, intervention, and comparator of each clinical question. Abstracts and conference proceedings, letters to the editor, editorials, review articles, and unpublished data were excluded. When acceptable RCTs of effectiveness were found, no additional non-randomized studies or non-comparative evidence (i.e., single-arm case series) were sought. The literature search results were supervised and thoroughly reviewed by

the guideline methodologist for the final selection of the relevant articles. Panel members reviewed the final set of included articles for accuracy.

Data collection and extraction

Once the articles were selected, the guideline methodologist, in conjunction with panelists, extracted the data for patient-important outcomes. Reviewers extracted relevant information into a standardized data extraction form, including study characteristics, study design, participant characteristics, details of the intervention and comparison, outcomes reported and funding source. The numbers of events and total sample sizes were extracted for all dichotomous outcomes, and means with standard deviations were calculated for continuous outcomes.

Data analysis

The quantitative data from RCTs or non-randomized studies with a control arm were combined to obtain a relative risk (RR) for dichotomous outcomes and a mean difference for continuous outcomes and reported with 95% confidence intervals (95% CI). We used the Mantel-Haenszel random-effects model to pool the relative effects unless the number of studies was too small to allow precise estimation of between-study variance, in which case we used the fixed-effects model. The statistical heterogeneity in the pooled estimates was assessed by visual inspection of the forest plots and the I2 statistic. Statistical heterogeneity was deemed substantial if I2 was greater than 50%.[26] When a sufficient number of studies were presented with no substantial heterogeneity, we planned to assess for publication bias using funnel plot asymmetry tests.[27] All quantitative analyses were performed using RevMan.[28]

Risk of bias assessment

In collaboration with the subgroup, the guideline methodologist assessed the risk of bias. The risk of bias was assessed by using the Cochrane risk of bias tool for RCTs, the ROBINS-I[29] for observational studies, QUIPS for prognostic studies,[30] and the QUADAS-2 tool for diagnostic test accuracy studies.[31]

Certainty of evidence assessment

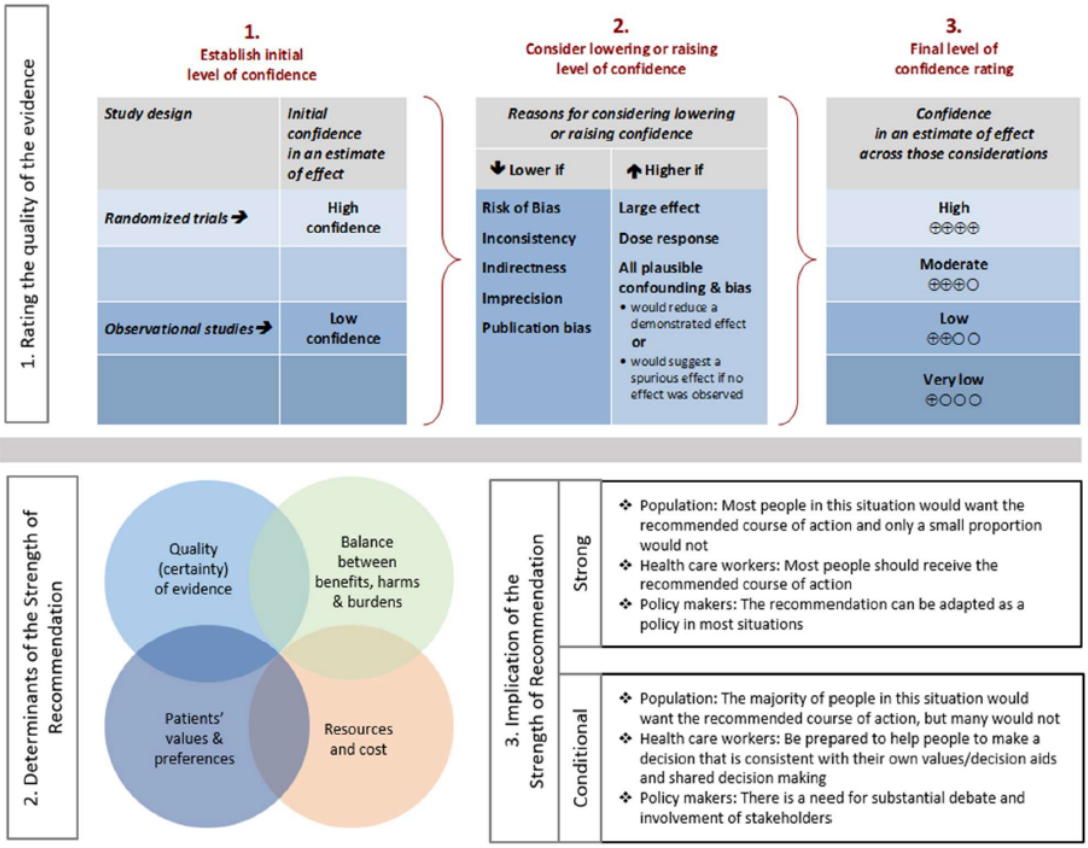
The certainty of the evidence was determined for each critical and important outcome and then for each recommendation using the GRADE approach to rating the confidence in the evidence.[20,32] Within GRADE, the body of evidence across each outcome is assessed for domains that may reduce or increase one's certainty in the evidence. Factors that may reduce one's certainty include risk of bias (study limitations), inconsistency (unexplained heterogeneity across study findings), indirectness (applicability or generalizability to the research question), imprecision (the confidence in the estimate of an effect to support a particular decision) or publication bias (selective publication of studies). One's certainty in the evidence may be strengthened if the following considerations are present: large or very large magnitude of effect, evidence of a dose-response gradient, or opposing residual confounding. GRADE evidence profile and summary of findings tables were developed in GRADEpro Guideline Development Tool.[33] All summary tables were reviewed by subgroup of panelists responsible and edited as appropriate.

Evidence to Recommendations

The panel considered core elements of the GRADE evidence in the decision process, including Certainty of evidence and balance between desirable and undesirable effects. Additional domains were acknowledged where applicable (feasibility, resource use, acceptability). All recommendations were labeled as either “strong” or “conditional” according to the GRADE approach.[19] The words “we recommend” indicate strong recommendations, and the words “we suggest” indicate conditional recommendations. **Figure 2.0** provides the suggested interpretation of strong and conditional recommendations for patients, clinicians, and healthcare policymakers. For recommendations where the comparator treatment or tests are not formally stated, the comparison of interest is implicitly referred to as “not using the intervention” (not using either a specific treatment or a diagnostic test). According to the GRADE guidance on discordant recommendations, strong recommendations in the setting of lower certainty of evidence were assigned only when the panelists believed they conformed to one of the five accepted paradigmatic conditions.[34] According to the GRADE working group, appropriate identification and wording choices were followed for recommendations regarding good practice statements.[35] A good practice statement represents a message the guideline panel perceives as necessary regarding current healthcare practice. It is supported by a large body of indirect evidence that is difficult to summarize and indicates that implementing this recommendation would clearly result in large net positive consequences. “Research Needs” were noted for recommendations as deemed appropriate by the panel.

Final presentation of evidence summaries and the development of the recommendations were conducted through a series of video teleconferences by the whole panel. All members of the panel participated in the preparation of the draft guideline and approved the recommendations. Literature search strategies, PRISMA flow diagram detailing the search results, evidence profiles tables, EtD framework, and additional data, such as quantitative analysis and modeling when appropriate, can be found in the Supplementary Material.

Figure 2.0. Approach and implications to rating the quality of evidence and strength of recommendations using GRADE methodology *(unrestricted use of figure granted by the U.S. GRADE Network)*



Section on oral empiric antibiotics for cUTI

The external review process highlighted the readership's need for more information on oral empiric antibiotics; thus, the section discussing oral empiric antibiotics for cUTI was considerably expanded after completion of the initial formulation of recommendations. The section and associated dosing table are based on a mapping review of clinical studies that compared different oral antibiotics when transitioning from IV in populations either treated for cUTI or Gram-negative bacteremia of predominately urinary origin. The evidence identified from that search was not assessed as per the GRADE methodology and no formal recommendations are made for use of these oral antibiotics as empiric initial therapy cUTI.

Public Patient Involvement

As per the 2011 Institute of Medicine, AGREE II and GIN standards, patients' representatives were involved in the development of this guideline.[36] Public patient involvement was advertised through the MyIDSA website as well as through the social media account of the

UTI Global Alliance. IDSA patient representatives are unpaid. Patient representatives willing to give of their time and effort often are those who suffer most from the disease under study. Patients' representatives were selected if they had a prior history of UTI and were not part of an advocacy group potentially at risk of intellectual or financial COI with the guideline. Patients were involved at different steps of the process, mainly providing feedback for the ranking patient-important outcomes, on patients' values and preferences for the different clinical questions reviewed, and on the final recommendations and potential related implementation issues. A narrative section can be found at the end of the document to highlight concerns expressed by patients experiencing UTIs.

Revision process

Feedback was obtained from three external peer expert reviewers, and involved organizations, i.e. SIDP (Society of Infectious Diseases Pharmacists), AAFP (American Academy of Family Physicians), SHM (Society of Hospital Medicine), AUA (American Urological Association), ASM (American Society of Microbiology), SAEM (Society for Academic Emergency Medicine), ACOG (American College of Obstetricians and Gynecologists), AMMI-CA (Association of Medical Microbiology and the Infectious Disease Canada), European Society of Clinical Microbiology and Infectious Diseases (ESCMID). In addition, the guideline was reviewed by the IDSA Standards and Practice Guidelines Subcommittee (SPGS), the IDSA Quality Subcommittee, and the IDSA Board of Directors. After review and approval by the various organizations and reviewers, the guideline was posted online prior to publication to facilitate a public comment period requesting feedback on the full guideline. The panel reviewed the feedback from the public comment phase and updated the guideline prior to final approval by the IDSA SPGS and Board of Directors.

Notes

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We acknowledge the important contributions of our four patient representatives (Carolyn Turner, Lisa Tecklenburg, Valerie Price, Laura Helgeson) and the two librarians (Elizabeth Kiscaden and Mary Beth McAteer). We also appreciate the efforts of our three external reviewers (Thomas Mac Hooton, Emily Spivak, and Bryan Alexander) and reviews by the Standards and Practice Guidelines Subcommittee. In addition, we would like to acknowledge IDSA staff Senam Attipoe, Genet Demisashi, and Jon Heald for their continual support and guidance developing the guidelines.

Disclaimer

It is important to recognize that guidelines cannot always account for individual variation among patients. They are assessments of current scientific and clinical information provided as an educational service; are not continually updated and may not reflect the most recent

evidence (new evidence may emerge between the time information is drafted and when it is published or read); should not be considered inclusive of all proper methods of care, or as a statement of the standard of care; do not mandate any course of medical care; and are not intended to supplant clinician judgment with respect to particular patients or situations. Whether to follow guidelines and to what extent is voluntary, with the ultimate determination regarding their application to be made by the clinician in the light of each patient's individual circumstances. While IDSA makes every effort to present accurate, complete, and reliable information, these guidelines are presented "as is" without any warranty, either express or implied.

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Possible conflicts of interest

Evaluation of relationships as potential conflicts of interest is determined by a review process. The assessment of disclosed relationships for possible COIs is based on the relative weight of the financial relationship (i.e., monetary amount) and the relevance of the relationship (i.e., the degree to which an association might reasonably be interpreted by an independent observer as related to the topic or recommendation of consideration). The following panelists have reported relationships with indicated companies since 2019, when the guideline work began. **B.T.** served as an advisor for Genentech, GSK, Phiogen Pharma, and Paratek Pharmaceuticals; received research funding from Genentech and Zambon Pharmaceuticals; received grant funding from the MacDonald Research Foundation, Agency for Healthcare Research and Quality, VA HSR&D, CDC, NIH, Mike Hogg Foundation, Neilsen Foundation, and VA RR&D; engaged in activity with Infectious Diseases Board Review Course, Warren Alpert School of Brown University Department of Medicine, Baylor College of Medicine PriMed, and UpToDate; owns

intellectual property for Phiogen Pharma; and owned stocks in Abbott Laboratories, Bristol Meyers Squibb, Abbvie, and Pfizer (all divested when BT learned of the individual stock ownership). **N.C.P.** received funding from Duke University. **K.G.** served as a consultant for Iterum Therapeutics, Tetraphase Pharmaceutical, Paratek Pharmaceutical, First Light Diagnostics, Ocean Spray Inc, Shionogi Inc, Nabriva Therapeutics, Utility Therapeutics, Qiagen Diagnostics, Rebiotix, Spero Therapeutics; serves as a consultant for GlaxoSmithKline, PhenUtest (Innotive), CarbX and PRIME Education; receives renumeration from Up to Date, Inc. Pri-Med and PRIME Education; owned equities in Pfizer Pharmaceutical (past), and Fidelity Managed Investment Account; received research grant funding from VA Health Services Research & Development, VA Cooperative Studies Program, NIAID, and NIH. **E.H.** receives funding from Cystic Fibrosis Foundation; receives honoraria from SIDP; served as an advisor for Nabriva Pharmaceuticals, Merck, GlaxoSmithKlin, Paratek Pharmaceuticals and MeMed; received funding from Merck; engages in activity with SIDP and CLSI. **M.H.** received research grants from AHA, VA QUERI, Veterans Affairs Health Systems Research, National Institute on Aging, Society to Improve Diagnosis in Medicine, and Michael E. DeBakey VA Medical Center; receives funding from Veterans Affairs Health Systems Research. **G.M.** served as an advisor for Light AI and Qiagen; received funding from Light AI, Contrafect, Nabriva, and NIH; engaged in activity with Annals of Emergency Medicine; owns stock in Light AI; receives funding from Abvacc. **R.C.** served as an advisor for the GlaxoSmithKlin, Biofire and Rebiotix. **J.O.** received funding from MITRE Corporation. **M.S.A.** received funding from Merck & Co. Inc and Nebraska DHHS HAI/AR Program (through CDC grants); served as a member and vice chair of PALTmed's Infection Advisory Committee (IAC) and chair of PALTmed's IAC UTI workgroup; served as a member of ASCP's AS and Infection Control committee; served as vice president and president for Nebraska Infection Control Network; served as a member of SHEA's education committee and Learning CE committee; and served as member of IDSA antimicrobial resistance committee; serves as board member and immediate past president for Nebraska Infection Control Network; serves as chair of PALTmed's infection prevention and control committee (previously infection advisory committee) and member of PALTmed's state advocacy committee (previously state based policy and advocacy subcommittee) and clinical practice steering committee; serves as a member of CSTE HAI/AR subcommittee; is member of the SHEA Long-Term Care Special Interest Group, was a member of the COVID-19 State Collaboration Task Force of PALTmed; worked on development of infection control guidance for nursing homes with SHEA and is working on development of infection control guidance for LTACHs and antibiotic initiation guidance for nursing homes with SHEA and also on nursing home infection control educational module development for front-line staff with SHEA; and accepted invitation to serve on HICPAC but onboarding was not completed. **D.D.** received funding from VA CSP, VA Merit Review, VA Cooperative Studies Program, CDC Tuberculosis Trials Consortium, and honoraria from UpToDate. **L.G.** received funding from Antibacterial Resistance Leadership Group, NIH, Zambon Pharmaceuticals, VA Health Services Research & Development, Rebiotix, and AHRQ; receives funding from AHRQ; Neilsen Foundation, VA Health Services Research & Development and Micheal E. DeBakey Veterans Affairs Seed Grant. **S.C.** Received honoraria for the development of sexual and reproductive health educational materials for family physicians from the American Academy of Family Physicians and California Academy of Family Physicians. Received honoraria for the development of training materials on sexual health from Quest Diagnostics. **A.H.** received funding from Swiss National Science Foundation, and serves as the ESCMID representative on the panel. **G.L.** received honoraria from UpToDate; received funding from Lupin Pharmaceutical, NeuMoDx, CDC, NACCHO, NIH, HRSA, DHHS, State Office of Victims Assistance, and IDSOG. **L.N.**

served as an advisor for Utility, Paratek, and GSK; engaged in activity with SHEA, Government of Manitoba, and ESCMID. **A.J.S.** served as an advisor for Spero; received funding from NIH; and engaged in activity with UpToDate, Inc**. S.Y.** received funding from Gilead and Dynamed; received remuneration with Dynamed; served as an advisor for Gilead; received funding from NIH/ACTG, HRSA; received royalties from UpToDate. **V.L.** received funding from Fonds de Recherche du Québec – Santé (FRSQ); received funding from Centre de Recherche du Centre Intégré Universitaire de Santé et de Services Sociaux du Nord-de-l'île-de-Montréal (CIUSSS NIM). No disclosures were reported for all other authors.

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Clinical Practice Guideline by Infectious Diseases Society of America (IDSA): 2025 Guideline on Management and Treatment of Complicated Urinary Tract Infections: Selection of Antibiotic Therapy for Complicated UTI

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A. Initial Selection among Empiric Antibiotic Options for Complicated UTI

B. Stepwise Process to Guide Empiric Antibiotic Choice for Complicated UTI

C. Selection of Definitive Antibiotic Therapy for Complicated UTI

A. Initial Selection among Empiric Antibiotic Options for Complicated UTI

In patients with cUTI, which classes of empiric antibiotic therapy should initially be prioritized?

Recommendations:

I. For patients with **sepsis** due to complicated UTI, we suggest **initially selecting among** the following antibiotics, using the four-step assessment (**Figure 1.1**): third- or fourth-generation cephalosporins, carbapenems, piperacillin-tazobactam, or fluoroquinolones, rather than newer agents (novel beta lactam-beta lactamase inhibitors, cefiderocol, plazomicin) or older aminoglycosides (*conditional recommendation, very low to moderate certainty of evidence*).

Remarks:
-See **Table 1.1** for a more complete list of empiric antibiotic therapy options.
-Please refer to the four-step approach in **Figure 1.1** to choose among these antibiotics for the specific patient (i.e., severity of illness, risk factors for having resistant uropathogen, patient-specific considerations, and antibiogram).
-Agents with broader spectrum of activity against organisms other than Enterobacterales (e.g. *Pseudomonas aeruginosa*, enterococci, or methicillin-resistant *Staphylococcus aureus*) may be considered for patients with sepsis in whom the diagnosis of cUTI is not clear or who are suspected to have cUTI due to these pathogens.

Comments:
-This recommendation places a higher value on providing early, appropriate empiric antibiotic therapy to prevent mortality while deferring stewardship considerations to definitive therapy.
- The certainty of evidence was moderate for all classes of antibiotics, except for third and fourth generation cephalosporins, and older aminoglycosides, for which the certainty of evidence was very low.

II. For patients with suspected complicated UTI without sepsis, we suggest **initially selecting among** the following antibiotics, using the four-step assessment (**Figure 1.1**): third- or fourth-generation cephalosporins, piperacillin-tazobactam, or fluoroquinolones, rather than carbapenems and newer agents (novel beta lactam-beta lactamase inhibitors, cefiderocol, plazomicin) or older aminoglycosides (*conditional recommendation, very low to moderate certainty of evidence*).

Remarks:
-See **Table 1.1** for a more complete list of empiric antibiotic therapy options.
-Please refer to the four-step approach in **Figure 1.1** to choose among these antibiotics for the specific patient (i.e., severity of illness, risk factors for having resistant uropathogen, and patient-specific considerations).
-Other agents (e.g., trimethoprim-sulfamethoxazole, amoxicillin-clavulanate, first or second-generation cephalosporins) are less well studied but may be appropriate in select settings or situations for empiric oral treatment of cUTI.

Comments:
-This recommendation places a higher value on antibiotic stewardship considerations in patients with cUTI who are not septic and in whom the risk of infection-related mortality is

low while also considering costs, resources, and practical aspects of antibiotic administration
-The certainty of evidence was moderate for all classes of antibiotics, except for third and fourth generation cephalosporins and older aminoglycosides, for which the certainty of evidence was very low.

Table 1.1: Potential Empiric Antibiotics for cUTI^ prior to using the four-step approach to choose among these options

Four-Step Approach to choose among these antibiotics: Assess (1) severity of illness, (2) risk factors for resistance, (3) patient-specific considerations, and (4) if septic, consider the antibiogram. See discussion below for details of the four steps.

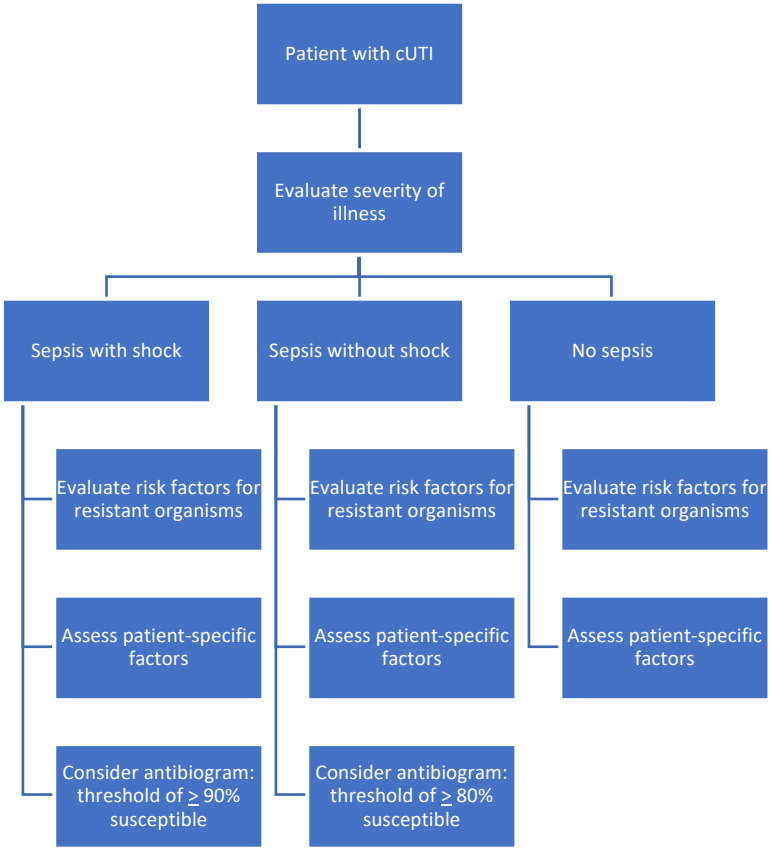
Condition of the Patient	Preferred	Alternative
Sepsis with or without shock**	Third or fourth generation cephalosporins,* carbapenems,# piperacillin-tazobactam, fluoroquinolones&	Novel beta lactam-beta lactamase inhibitors,+ cefiderocol, plazomicin, or older aminoglycosides%
Without sepsis, IV route of therapy	Third or fourth generation cephalosporins,* piperacillin-tazobactam, or fluoroquinolones&	Carbapenems,# newer agents (novel beta lactams-beta lactamase inhibitors,+ cefiderocol, plazomicin), or older aminoglycosides%
Without sepsis, oral route of therapy	Fluoroquinolones& or trimethoprim-sulfamethoxazole	Amoxicillin-clavulanate or oral cephalosporins (see **Table 3.1**)

^Difficult-to-treat resistant pathogens may require use of drugs not listed here (e.g., colistin); refer to IDSA Antimicrobial Resistance guidance.
**Sepsis is life-threatening organ dysfunction related to infection, identified by SOFA score of 2 or higher. Screening tools such as qSOFA or SIRS may be useful for presumptive identification. In sepsis with shock, in step 4 choose an antibiotic for which the susceptibilities of the most relevant organisms are at least 90%. In sepsis without shock, in step 4 choose an antibiotic for which the susceptibilities of the most relevant organisms are at least 80%.
*Third and fourth generation IV cephalosporins include: ceftriaxone, ceftazidime, cefotaxime, and cefepime. (see **Table 2.1 & 3.1**, Dosing of IV and oral antibiotics for cUTI).
&The fluoroquinolones approved for UTI currently include ciprofloxacin and levofloxacin.
#The carbapenems currently include imipenem-cilastatin, doripenem, meropenem, and ertapenem.
+The novel beta lactam-beta lactamase inhibitors currently include ceftolozane-tazobactam, ceftazidime-avibactam, meropenem-vaborbactam, and imipenem-cilastatin-relebactam.
%Older aminoglycosides include gentamicin, amikacin, and tobramycin.

This table was created in 2025; new drugs approved after this date may also be appropriate choices.

Please note that nitrofurantoin and oral fosfomycin are generally not appropriate choices for cUTI because they may not achieve adequate levels in renal parenchyma and blood.

Figure 1.1: Four-step approach to choosing empiric antimicrobial therapy for cUTI



This approach starts with the most important issue—the patient's severity of illness—and then takes into consideration the patient's risk factors for having a pathogen resistant to specific antibiotics or antibiotic classes, as well as practical issues such as antibiotic allergies. Finally, and only for patients with sepsis related to cUTI, the local antibiogram may have a role in helping the provider avoid inappropriate empiric antibiotic therapy if it is recent and relevant to the patient under consideration. The antibiogram is the last of the four recommended steps, as the evidence that using a facility's antibiogram to guide antibiotic prescribing for individual patients improves outcomes is very uncertain. Choosing which organism to focus on in the antibiogram is also a challenge in empiric decision making. The most relevant organism is suggested by the prior urine culture, if available. If not, *E. coli* is the default organism.

Introduction

Complicated UTI (cUTI) is one of the more common reasons for emergency department visits and hospital admission. Selection of appropriate empiric antibiotics for cUTI can be complex because of increasing resistance to antibiotics used to treat UTI, the association between active empiric therapy and improved outcomes in UTI, and the availability of newer and broader spectrum agents. Many new antibiotics have received FDA approval for treating cUTI since the prior IDSA UTI guidelines on cystitis and pyelonephritis were published. The randomized, controlled trials to gain these FDA approvals provide a rich evidence base about the efficacy of various antibiotics for empiric treatment of cUTI. As these trials followed FDA

guidance about cUTI trial design, most are similar in design (non-inferiority) and patient populations. This is, at once, both a strength and a weakness of the evidence base. For example, we can draw comparisons across trials (assuming similar prevalence of resistance), but the inclusion/exclusion criteria were selected for a younger population with few comorbidities and often excluded patients in septic shock. Mortality in the trials that accepted patients without specifying a resistant uropathogen as an inclusion criterion was accordingly low (< 1% in all), which may not reflect real-world survival for patients hospitalized with cUTI. Another common design feature in these randomized, controlled trials is that clinical and microbiological outcomes were assessed at "test of cure," usually 5-10 days after completion of antibiotics. In clinical practice, patients who are feeling better after treatment for cUTI or acute pyelonephritis do not typically return for clinical assessment or repeat urine cultures. In fact, collection of a urine culture in a now asymptomatic patient is discouraged in clinical practice guidelines as asymptomatic bacteriuria (ASB) treatment does not prevent UTI and may predispose to subsequent recurrent UTI.[1-4] Uncertainty over the clinical relevance of ASB at the test of cure study visit influenced the panel's choice of important outcomes, as discussed below.

The scope of this clinical question was limited to empiric choice of antibiotics in suspected cUTI. If the causative organism has already been identified as a difficult-to-treat resistant pathogen, please refer to the IDSA guidance on antimicrobial resistance.[5] For patients with septic shock from a urinary source, in addition to using these cUTI guidelines to guide empiric antimicrobial therapy, please refer to the Sepsis-3 Task Force guidelines for other management strategies.[6] The guidelines below first discuss the evidence for specific antibiotics or classes of antibiotics that can be used as empiric therapy for cUTI. Then, the stepwise approach is discussed, with explanations of the evidence on the potential impact of inappropriate initial antibiotic therapy, the predictive value of specific risk factors for having a resistant uropathogen (including prior urine cultures and prior antibiotic exposure), and modeling approaches for establishing what threshold to use on the antibiogram when choosing empiric antibiotics for a cUTI patient with sepsis.

Background on the trials providing evidence about specific antibiotics for cUTI

All but five of these 15 published trials discussed here were designed as FDA registration trials to gain an FDA indication for a novel agent's use in cUTI and pyelonephritis (See **Table A.1** "Characteristics of Included Studies" in the Supplementary Materials). Thus, most trials followed the FDA-recommended non-inferiority design and largely demonstrated that the newer agents are non-inferior to older antibiotics. Patients enrolled in these randomized, controlled trials may not reflect the real-world population with cUTI; many of the randomized, controlled trials had a low mortality rate (1% or less) compared to the above 5% mortality rates reported in observational studies examining the consequences of inappropriate empiric antimicrobial therapy (IEAT) in cUTI. Another important caveat is that clinical and microbiological outcomes in some trials were affected by high levels of uropathogen resistance to one of the antibiotics under comparison (such as fluoroquinolones or ceftriaxone).

The aim of our systematic review was to assess the balance of benefits and harms for currently used antibiotics (especially in the context of increasing resistance) as well as the drugs newly available in the US since 2008. We found antimicrobial agents were generally comparable in terms of achieving clinical cure as long as the causative pathogen was susceptible to the agent given, and we conclude that other factors, such as patient risk factors for resistance to specific antibiotics, antibiotic stewardship considerations, drug adverse event

profiles, and cost should drive choice among these agents. A decisional strategy for selecting empiric therapy in cUTI will be presented.

Methods for reviewing specific antibiotic classes for empiric treatment of cUTI

To understand the evidence presented, it is important to appreciate the following issues: the choice of antibiotics, the timeframe of the studies, the background level of antibiotic resistance, the prioritization of outcomes, and the clinical decision threshold. The panel's decisions on the choice of antibiotics and years of publication are addressed below. Please see the introduction for a discussion of the prioritization of outcomes and selection of a clinical decision threshold. The background level of antibiotic resistance is addressed under each specific class of antibiotics, in the context of the studies that provided the supporting evidence.

Choice of antibiotics:

The panel established some criteria for which antibiotics to include in these cUTI guidelines. The antibiotic needed to be available in the United States, which implies FDA approval (although not necessarily for a UTI indication). Drugs that sought but did not attain FDA approval for cUTI were not included in our evidence tables, but we discuss several of these drugs below.

Years of publication:

The literature search for randomized, controlled trials of empiric treatment of cUTI started in 2008, updating from prior guidelines. To gather evidence on risk factors for having an organism resistant to specific antibiotics or for more general concepts related to cUTI treatment, literature searches started in 2000. Some older antibiotics have not been tested in a randomized, controlled trial since 2008 but may still have relevance to empiric treatment of cUTI; for such antibiotics, we looked at literature reviews and provided narrative discussion. Ampicillin-sulbactam, cefazolin, and trimethoprim-sulfamethoxazole (TMP-SMX) have not been tested in an RCT for cUTI since 2008; their ongoing relevance to empiric treatment of cUTI is unclear due to the high prevalence of resistance among *Escherichia coli* to these antibiotics. Many antibiotics included in our search strategy were not actually studied in any of the references identified to form the evidence base for this clinical question (e.g. many cephalosporins), and recommendations could be made only via indirect evidence from similar/nearly equivalent antibiotics.

Our literature searches for clinical trials of various antibiotics to treat cUTI extended back to studies published in 2008, picking up where the prior UTI guidelines' literature review stopped. Although the prior UTI guidelines focused on cystitis and pyelonephritis in premenopausal women, the panel judged that literature prior to 2008 would not be relevant to management of cUTI in the current era. In particular, the prevalence of antibiotic resistance among urinary pathogens has changed considerably from 2008 to the present (2023). Additionally, many prior first-line antibiotics for cUTI (e.g., aminoglycosides and ceftriaxone) were rarely studied in randomized, controlled trials from 2008-2023, as much older trials had established their efficacy. The clinical trials gathered to address this clinical question, along with important contextual information such as the main uropathogens isolated and their rates of resistance to the study antibiotics, are given in the Supplemental Materials (Table "Characteristics of the Included Studies").

Antibiotics' effectiveness in these trials related to the prevalence of resistance among uropathogens to each of the two study drugs in the study population at the time. Usually, the

comparator in a given trial was an older agent with a higher prevalence of resistance among the isolated uropathogens. An unconfounded comparison would require the prevalence of resistance to each study drug to be the same. In these guidelines' supporting text and tables, we have reported resistance rates in two different ways: (1) general rate of resistance to a specific antibiotic among all bacteria isolated, tested, and reported in that trial, or (2) the specific rate of resistance to an antibiotic within a treatment group (i.e. the people actually receiving that drug).

Summary of Evidence for Specific Antibiotic Classes for Empiric Treatment of cUTI

Different classes of antibiotics that can be used as empiric therapy for cUTI will be discussed below. Doses of the antibiotics used in these studies are in **Table 2.1** below.

Table 2.1: Dosing of intravenous (IV) antibiotics for complicated UTI used in clinical studies presented in alphabetical order.

Drug	Dosing regimen used in clinical trials for patients with normal renal function
Cefepime	1-2g every 8 to 12 hours[7,8]
Cefepime-enmetazobactam	2g/0.5g (infused over 2 hours) every 8 hours[9]
Cefiderocol	2g (infused over 3 hours) every 8 hours[10,11]
Cefotaxime	1-2g every 8 hours[12]
Ceftazidime	1-2g every 8 hours[13,14]
Ceftazidime-avibactam	2.5g (infused over 2 hours) every 8 hours[15-17]
Ceftolozane-tazobactam	1.5g every 8 hours[18]
Ceftriaxone	1-2g daily[19,20]
Ertapenem	1g daily[20]
Fosfomycin	6g every 8 hours[21]
Imipenem-cilastatin	500mg every 6 hours[17,22] 1g every 8 hours[11]
Imipenem-cilastatin-relebactam	500mg/125mg every 6 hours[22]
Meropenem	1g every 8 hours[19,23]
Meropenem-vaborbactam	2g/2g (infused over 3 hours) every 8 hours[24]
Piperacillin-tazobactam	4.5g every 8 hours[9,21,24]
Plazomicin	10-15mg/kg daily[23,25]

Table 2.1 includes IV dosing for cUTI based on review of randomized controlled trials among patients with complicated UTI.

1) Ceftriaxone / third and fourth generation cephalosporins

Ceftriaxone (and to a lesser extent, other parenteral third and fourth-generation cephalosporins) have a long history of use to treat cUTI including pyelonephritis, including a recommendation in the prior IDSA UTI guidelines of these agents as appropriate treatment for women with acute pyelonephritis.[26] While the increasing prevalence of Enterobacterales producing extended-spectrum beta lactamases threatens to undermine the efficacy of

ceftriaxone and other antibiotics in its class, ceftriaxone remains a frequent choice for empiric therapy of cUTI.

Summary of evidence for empiric use of ceftriaxone to treat cUTI

The key studies establishing the efficacy of ceftriaxone for treatment of cUTI were published prior to our literature search's time frame of 2008-2023. Two randomized, controlled trials compared ceftriaxone and the newer agent ertapenem for cUTI (including acute pyelonephritis), and the findings were summarized in a meta-analysis.[27-29] In both studies patients received the antibiotic IV initially, and then a switch to oral agents was permitted after three days of treatment, to complete 10-14 days total therapy. The most commonly used oral agent was a fluoroquinolone. Of the 850 randomized patients in the combined trials, 480 cases were microbiologically evaluable. The primary efficacy endpoint in these trials was not clinical cure but microbiologic cure at the test of cure visits, 5-9 days after completion of therapy. The primary outcome (microbiologic response) was achieved in 91% in the ceftriaxone arm versus 90% of patients in the ertapenem arm (risk difference (RD): 0.9%; 95% CI, -4.5% to 6.3%). Clinical response was not reported. Baseline resistance to ertapenem and ceftriaxone was not observed in these trials except for a few enterococci and *P. aeruginosa* isolates.[28]

Only one RCT was identified in our present literature search, which reported the results of a multi-center trial from South Korea comparing ceftriaxone to ertapenem for empiric treatment of cUTI (Park 2012).[20] After five days of IV therapy, patients could be switched to oral ciprofloxacin or cefixime to complete 10-14 days of antibiotic therapy. Among the 267 patients enrolled from 2008-2009, uropathogen resistance to ceftriaxone was only 6.2% (including 4.5% ESBL-producing organisms).

Benefits, Harms and Certainty of evidence

The single trial identified in our present literature review reported only a combined endpoint of clinical cure and microbiological response, both assessed at 5-9 days after completion of antibiotic therapy (i.e. data on clinical cure alone were not available).[20] Treatment with ceftriaxone may lead to similar cure rates (cure being a composite of clinical and microbiologic outcomes) in comparison to ertapenem, but the evidence is very uncertain mainly due to serious imprecision (i.e. small sample size). Specifically, the overall combined clinical cure and microbiological response reported in this single trial was 87% for ceftriaxone vs 88% for ertapenem (RD: -0.6%; 95% CI: -11.6% to 10.5% / relative risk (RR): 0.99; 95% CI: 0.88 to 1.13; very low certainty of evidence). For microbiologic cure, ceftriaxone may lead to similar rates of microbiological cure as ertapenem (RD: 0.9%; 95% CI: -9.9% to 11.6% / RR: 1.01; 95%CI: 0.89 to 1.14). Recurrence of infection was not reported in this study.
Patients treated with ceftriaxone may experience fewer non-serious adverse events than the comparator group (4.4% in ceftriaxone group vs 10.6% in ertapenem group), with the ceftriaxone group experiencing fewer drug-related gastrointestinal adverse events (diarrhea and nausea). No serious adverse events were documented, and mortality was not reported.

Other considerations

Ceftriaxone is a practical choice for outpatient antibiotic therapy programs when the once-daily dose is sufficient, both for ease of dosing and for its low cost relative to newer agents. It can also be administered intramuscularly, which can be a particularly useful route in

patients who lack sensation in lower limbs or gluteal areas due to spinal cord injury. Ceftriaxone susceptibility cannot be extrapolated to cefpodoxime or cefepime.

The 2010 publication of the UTI guidelines on cystitis and pyelonephritis recommended a single dose of aminoglycoside or ceftriaxone at the initiation of oral antibiotics to treat acute pyelonephritis, if resistance to the oral agent was a concern.[26] We identified one study of single dose ceftriaxone in non-pregnant adults; this study suggested that a single dose of IV ceftriaxone prior to switching to an oral cephalosporin was an effective strategy for women with pyelonephritis.[30]

Rationale for recommendation and implementation

The panel judged that ceftriaxone, and by extension third and fourth generation cephalosporins remain one of the preferred classes of antibiotics to empirically treat patients with cUTI, particularly in patients without sepsis. If after applying the four empiric antibiotic choice steps (severity of illness, risk factors for resistance, patient-specific factors, and antibiogram), third or fourth generation cephalosporins have not been excluded from consideration, they may be appropriate empiric choices for patients with sepsis related to cUTI.

As an example of applying the four steps, if the patient had an ESBL-producing organism in a recent prior urine culture, ceftriaxone would not be an appropriate choice for cUTI. As another example, if the prevalence of ceftriaxone resistance exceeds 10% or 20%, ceftriaxone should not be used as empiric therapy for patients with cUTI in septic shock or sepsis, respectively.

2) Piperacillin-tazobactam

Piperacillin-tazobactam is an extended-spectrum penicillin and beta-lactamase inhibitor combination with broad activity against Gram-negative organisms, including *Pseudomonas aeruginosa*. Since its approval in 1993, recommended doses of piperacillin-tazobactam have been increased to overcome rising MICs in common pathogens; for the same reason, many sites now employ extended or continuous infusion piperacillin-tazobactam dosing strategies.

Summary of evidence for the empiric use of piperacillin-tazobactam to treat cUTI

Three randomized, controlled, multicenter, international trials included piperacillin-tazobactam as a treatment group for cUTI, including 2,043 evaluable patients with cUTI or acute pyelonephritis (per the FDA definitions).[9,21,24] The aim of these FDA registration trials was to establish the efficacy of novel agents (cefepime-enmetazobactam, meropenem-vaborbactam and IV fosfomycin) versus piperacillin-tazobactam in cUTI/acute pyelonephritis. The TANGO I study (Kaye 2018) compared meropenem-vaborbactam (a novel beta-lactam, beta-lactamase inhibitor, or BLBLI) to piperacillin-tazobactam in 545 patients with cUTI.[24] Patients received an average of 8 days of IV therapy, followed by 2 more days of oral levofloxacin. The ZEUS study (Kaye 2019) compared IV fosfomycin to piperacillin-tazobactam in 464 patients, with 7 days of IV therapy and no oral switch options.[21] In these two trials, resistance to piperacillin-tazobactam was 7% and 13%, while resistance to meropenem was 1% and resistance to the two novel drugs (IV fosfomycin and meropenem-vaborbactam) was not reported. The ALLIUM study (Kaye 2022) compared cefepime-enmetazobactam (a novel BLBLI) to piperacillin-tazobactam in

1,034 with cUTI.[9] Patients also received an average of 8 days of IV therapy and no transition to oral antibiotic was allowed. Patients with resistant uropathogens to either studied drugs were excluded from the analysis. For all three studies, clinical and microbiological outcomes were assessed at test of cure, 7-14 days after end of antibiotics.

Benefits, Harms and Certainty of evidence

Treatment with piperacillin-tazobactam (PT) likely leads to the similar rate of clinical cure at the test of cure timepoint as does treatment with comparators in patients treated for cUTI (overall clinical cure for PT was 88.9% vs 91.5% for comparators; RD: -2.7%; 95% CI: -5.5% to 0.9%/ RR: 0.97; 95% CI: 0.94 to 1.01; moderate certainty of evidence).

The evidence suggests that piperacillin-tazobactam leads to lower microbiological cure at test of cure than the comparators. Overall, microbiological eradication for piperacillin-tazobactam was 60.8% versus 74.2% in the comparator group (RD: -14.1%; 95% CI: -17.8% to -9.6%/ RR: 0.81; 95% CI: 0.76 to 0.87). Recurrence of infection was recorded only in one trial (Kaye 2019) at late follow up. The evidence suggests that treatment with piperacillin-tazobactam leads to similar recurrence of infection rates, as compared to IV fosfomycin (3.9% with piperacillin-tazobactam versus 4.3% with fosfomycin), but this estimate is likely imprecise due to the few events.

The evidence suggests that serious adverse events and mortality were comparable between the two groups (mortality rate was 0.5%). Non-serious adverse events were lower in patients receiving piperacillin-tazobactam versus the comparators (RD: -6.6%; 95% CI: -10.4% to -2.4%/ RR: 0.86; 95% CI: 0.78 to 0.95).

Other considerations

Administration of piperacillin-tazobactam requires an increasingly thoughtful approach, with improved time above the MIC for many pathogens more likely to be achieved through prolonged or continuous infusion. Such dosing strategies create challenges for nursing staff, pharmacists, and patients. Piperacillin-tazobactam is formulated as a salt and can cause hypokalemia or fluid overload. Rising rates of ESBL-producing organisms and multidrug-resistant *Pseudomonas* are challenging the effectiveness of piperacillin-tazobactam.

Rationale for recommendation and implementation

The panel judged that piperacillin-tazobactam remains one of the preferred antibiotics for empiric treatment of patients with cUTI without sepsis. If after applying the four empiric antibiotic choice steps (severity of illness, risk factors for resistance, patient-specific factors, and antibiogram), piperacillin-tazobactam has not been excluded from consideration, it may be an appropriate empiric choice for patients with sepsis related to cUTI.

As an example of applying the four steps, if the patient had an organism resistant to piperacillin-tazobactam in a recent prior urine culture, piperacillin-tazobactam would not be an appropriate choice for cUTI. As another example, if the prevalence of piperacillin-tazobactam resistance exceeds 10% or 20%, piperacillin-tazobactam should not be used as empiric therapy for patients with cUTI in septic shock or sepsis, respectively.

3) Fluoroquinolones

The fluoroquinolones (FQ) have a long history in treatment of urinary tract infections, including pyelonephritis, with the first clinical trials of quinolones to treat UTI appearing nearly 40 years ago.[31,32] While fluoroquinolones are very useful for treating infections of the urinary tract and have excellent oral bioavailability, they also have an important role in treating respiratory infections, bone and joint infections, enteric pathogens, *Pseudomonas aeruginosa*, mycobacteria, and *Neisseria spp*. Unfortunately, widespread use and overuse of fluoroquinolones have led to both dramatic increases in bacterial resistance and a better appreciation of these agents' potential for serious side effects, including collagen-vascular adverse events (tendinitis, tendon, and aortic aneurysm or dissection rupture), peripheral neuropathy, central nervous system effects, hypoglycemia, QT interval prolongation, and *C. difficile* colitis. In 2016 the FDA issued advice that in light of the potentially severe side of effects of fluoroquinolones, these drugs should not be used to treat uncomplicated UTIs in patients who have other treatment options.[33] Given the rising resistance, concern for serious toxicities, and important role quinolones play in more serious infections, many antibiotic stewardship programs have focused on reducing fluoroquinolone use in cUTI.

Summary of evidence for empiric use of fluoroquinolones to treat cUTI

Our literature search from 2008-2023 identified three randomized, controlled trials of empiric treatment of cUTI using a fluoroquinolone (levofloxacin).[18,25,34] These were multicenter, international trials conducted to establish the efficacy of new drugs to treat cUTI (including acute pyelonephritis) and gain FDA approval for cUTI. A total of 1,956 patients were treated for complicated UTI/acute pyelonephritis (as defined by the FDA) in these studies, with enrollment spanning 2003-2013. The prevalence of fluoroquinolone resistance was increasing during this period, and levofloxacin resistance among uropathogens isolated in these trials ranged 15-27% compared to 0.5-6% for the novel comparator agents, which were doripenem (Naber 2009),[34] ceftolozane-tazobactam (ASPECT-cUTI trial, Wagenlehner 2015),[18] and plazomicin (Connolly 2018).[25,35] Patients received IV antibiotics for 5-7 days; one trial permitted additional oral therapy with levofloxacin (Naber 2009). Clinical and microbiologic outcomes were measured at the test of cure (TOC) visit, 5-12 days after the last dose of antibiotics.

Benefits, harms, and certainty in the evidence

Despite the much higher resistance rates to fluoroquinolones than to the comparators, empirical treatment of suspected cUTI/acute pyelonephritis with fluoroquinolones (FQ) likely leads to lower clinical cure rates at test of cure versus the comparators, but this difference was judged clinically unimportant at a decision threshold of 10% (see Methods section on decision threshold). More specifically, clinical cure for FQ was 88.2% vs 91.3% for the comparator group (RD: -3.7%; 95% CI: -6.4% to -0.9%; RR:0.96; 95% CI: 0.93 to 0.99/ moderate certainty of evidence).

Interestingly, fluoroquinolones may lead to similar rates of microbiological cure as the comparators, but the evidence is very uncertain. Microbiological cure in the FQ was 75.9% and was 79.2% in the comparator group (RD: -3.2%; 95% CI: -11.1% to 4.8%/ RR: 0.96; 95% CI: 0.86 to 1.06).

The lower rates of clinical cure with fluoroquinolones are likely explained by the much higher resistance rates to fluoroquinolones versus the new drugs among the isolated uropathogens. One trial performed a post-hoc analysis to analyse the subgroup of patients found to have a fluoroquinolone-resistant pathogen in the ASPECT-cUTI trial (ceftolozane-

tazobactam).[35] A total of 212 patients mainly with acute pyelonephritis and levofloxacin-resistant uropathogens were analysed. In these patients whose baseline organism was resistant to levofloxacin, empirical treatment with levofloxacin was associated with lower rate of clinical cure at TOC (RD: -13.2%; 95% CI: -23.0% to -3.4%). Interestingly, overall response rates in these patients with levofloxacin-resistant uropathogens (i.e. regardless of antibiotic treatment) were lower than in the parent study (83% vs 90% respectively) which suggests that these patients' infections might be more difficult to cure either due to differences in the uropathogens, differences in the patients' risk factors for having UTI, or both. These findings are congruent with multiple observational studies suggesting that patients infected with resistant organisms more often have comorbidities or risk factors for poor outcomes than patients with more susceptible organisms.[36-39]

Recurrence of infection at late follow up was reported only in one study.[18] This evidence suggests that treatment with fluoroquinolones may lead to similar rates of recurrence of infection as the comparators (6% with levofloxacin versus 14% with ceftolozane-tazobactam; RD: -8%; 95% -13.6% to 37.0%),[18] but this estimate is likely imprecise due to the few events and a very small sample size. Serious and non-serious adverse event rates were likely comparable between groups. Mortality was rare (0.1%).

Other considerations

Other considerations specific to fluoroquinolones include their ease of administration, low cost, potential for serious adverse events, and antibiotic stewardship concerns. Fluoroquinolones have excellent bioavailability, with the IV to oral switch being dictated by when the patient can take oral drugs rather than concerns about bioavailability. Antibiotic stewardship concerns are perhaps more relevant to this class of antibiotics than others given that two recent antimicrobial resistance surveillance reports found that approximately 30% of Enterobacterales isolates were resistant to fluoroquinolones in a national study of hospitalized patients,[40] while the SENTRY report on bloodstream isolates 2012-2017 found that 30% of *E. coli* were resistant to ciprofloxacin.[41]

Rationale for recommendation and implementation

The panel judged that fluoroquinolones remain one of the preferred classes of antibiotics to empirically treat patients with cUTI without sepsis, despite the possibility of lower clinical cure rates in patients with fluoroquinolone-resistant uropathogens, because cUTI-related mortality is low in the absence of sepsis. If, after applying the four empiric antibiotic choice steps (severity of illness, risk factors for resistance, practical considerations, and antibiogram), fluoroquinolones have not been excluded from consideration, they may be an appropriate empiric choice for patients with sepsis related to cUTI. Fluoroquinolones as a class can be very effective therapy for cUTI but should be reserved for cases in which resistance is not expected.

As an example of applying the four steps, if the patient had taken a fluoroquinolone within the past 12 months or had a recent urine culture with a fluoroquinolone-resistant organism, a fluoroquinolone would not be an appropriate choice for cUTI. As another example, if the prevalence of fluoroquinolone resistance exceeds 10% or 20%, fluoroquinolones should not be used as empiric therapy for patients with cUTI in septic shock or sepsis, respectively. Caution is advised in applying these recommendations because the studies with supporting evidence last enrolled patients in 2013, and the rate of resistance to fluoroquinolones has increased since that time.

4) Carbapenems (without beta-lactamase inhibitors or BLI)

Carbapenems have a broad spectrum of activity and stability against many beta-lactamases, though carbapenem-resistant organisms are an increasing concern.[42-44] Given carbapenems' value as empiric therapy for suspected Gram-negative bacterial infections in the critically ill, preserving carbapenem effectiveness is a high priority for antibiotic stewardship programs. In this section, we discuss clinical trials of imipenem-cilastatin, ertapenem, meropenem, and doripenem for complicated urinary tract infections (cUTIs). None of these drugs have an oral option. Discussion of tebipenem and sulopenem (neither approved in the US at the time this passage was written) appears elsewhere.

Summary of the evidence for empiric use of carbapenems (without BLI) to treat cUTI

We identified seven randomized, controlled trials published from 2009-2019 (total n=3554 patients) that compared a carbapenem to alternative therapy for cUTI and/or acute pyelonephritis.[11,15-17,20,23,34] In two of the older trials, carbapenems were the new drug of interest, compared to fluoroquinolones (Naber 2009)[34] or ceftriaxone (Park 2012).[20] In five of the trials, carbapenems were the older comparator, and the drug of interest was a novel beta-lactam, beta-lactamase inhibitor combination (Vasquez 2012, Wagenlehner 2016, Carmeli 2016),[15-17] plazomicin (Wagenlehner 2019),[23] or cefiderocol (Portsmouth 2018).[11]

Antimicrobial susceptibilities of the uropathogens in these trials are key context for interpreting their results. For example, resistance to ceftriaxone in the Park 2012 study was 6.2%,[20] and resistance to levofloxacin was 14.8% in the Naber 2009 study;[34] in contrast, resistance to carbapenems in both studies was less than 1%. These high resistance rates to the older comparator would favor carbapenems. On the other hand, in the studies published from 2016 onwards, resistance rates to carbapenems ranged from 3-5%, while resistance to the novel agents was rare.[11,15,16] The average duration of IV therapy in these trials ranged from 5-10 days, and some studies continued with oral therapy based on the susceptibility of the isolated organisms.

Benefits, harms, and certainty of evidence

Treatment with carbapenems likely leads to similar rates of clinical cure as the antibiotic comparators at test of cure across these seven studies. Specifically, clinical cure in patients with cUTI treated with carbapenems was 91.2% vs 89.9% for comparators (RD: 1.8%; 95% CI: -0.9% to 3.6%/ RR: 1.02; 95% CI: 0.99 to 1.04; moderate certainty of evidence). The test of cure visit was typically 7-10 days after the last dose of antibiotics.

The evidence suggests that treatment with carbapenems leads to fewer microbiological cure (at TOC) versus the comparators, but this reduction was judged clinically unimportant at a decision threshold of 10%. More specifically, the microbiologic cure was 72.8% in patients treated with carbapenems than 80.4% with comparators (RD: -8.8%; 95% CI: -13.7% to -2.4%/ RR: 0.89; 95% CI: 0.83 to 0.97), but three of the seven trials accounted for the difference, each comparing carbapenems to a newer agent. Heterogeneity in this estimate is due to the EPIC trial,[23] in which the comparison was between meropenem and plazomicin, and the TOC visit was somewhat later than other trials, at 15-19 days after start of therapy. Another trial driving the microbiologic cure results against carbapenems was REPRISE, in which the comparison was between ceftazidime/avibactam and best available therapy (96% of which was a

carbapenem).[15] Only patients with ceftazidime-resistant Enterobacteriaceae and *Pseudomonas aeruginosa* were included in REPRISE. The third trial compared cefiderocol to imipenem-cilastatin, finding higher microbiologic cure at the TOC in the cefiderocol group than the imipenem group (73% versus 56%).[11] If these three trials are excluded, carbapenems might not have resulted in fewer microbiologic cures than the comparator antibiotics.

Only two of these seven trials reported the outcome of recurrent clinical infection, one of which was the EPIC trial of plazomicin,[23] and the other of which studied cefiderocol as the comparator.[11] In both trials clinical recurrence was measured 3-4 weeks after initiation of IV therapy. The evidence suggests that treatment with carbapenems leads to more recurrence of infection than the comparators in these two trials (8.2% vs 3.4%, respectively; RD: 6.1%; 95% CI: 1.6% to 14.8%/ RR: 2.80; 95% CI: 1.46 to 5.38). Of note, both trials also found a lower microbiologic cure rate in the carbapenem group, potentially implying a connection between microbiologic failure and recurrence of infection; again, however, whether this is a causative relationship (i.e. failing to kill the bacteria in the bladder leads to recurrence) or a linked relationship along a causal pathway (i.e. persistent bacteriuria is a marker for host factors such as urinary retention that also increase the risk of recurrent UTI) is unclear.

Mortality was rare in these randomized, controlled trials (0.4%), as expected given that life-threatening illness, expected imminent death, and serious comorbidities were frequent exclusion criteria. The rates of serious and non-serious adverse events were likely comparable between groups.

Other considerations

Other considerations specific to carbapenems include their crucial role in treating severely ill patients with healthcare-associated infections or risk factors for resistant pathogens. Carbapenems as a class are an important priority of antibiotic stewardship, to preserve their effectiveness.

Rationale for recommendation and implementation

The panel's overall assessment of the evidence was that carbapenems provide adequate clinical cure of cUTI in all studied comparisons. Carbapenems may be inferior to some newer agents (plazomicin, ceftazidime-avibactam, and cefiderocol) for microbiologic cure and prevention of recurrent infection, though the clinical relevance of the discrepancy between clinical and microbiologic cure is unclear.

Carbapenems clearly have a role in empiric treatment of cUTI in patients with sepsis and a higher risk of mortality. In the context of increasing rate of resistance to fluoroquinolones, third and fourth generation cephalosporins, and piperacillin/tazobactam, the panel considered carbapenems to be among the preferred antibiotics to empirically treat patients with sepsis assumed to be caused by cUTI, to avoid inappropriate empiric antibiotic therapy and possible associated excess mortality. However, in patients with cUTI without sepsis, the panel judged that carbapenems are not a first-line empiric antibiotic choice due to stewardship considerations, as initial drug-bug mismatch (causative organism not susceptible to the antibiotic given) in cUTI without sepsis is unlikely to substantially contribute to mortality, and clinical cure is still achieved in the majority of cases.

5) Novel beta-lactam/beta-lactamase inhibitors (BLBLI)

Multidrug-resistant and difficult to treat Gram-negative uropathogens such as carbapenem-resistant Enterobacterales (CRE) uropathogens such as Klebsiella-pneumoniae carbapenemase-producing (KPC) *E. coli,* multidrug resistant *Pseudomonas aeruginosa*, *Acinetobacter spp.*, and *Stenotrophomonas maltophilia* have increased the challenge of treating cUTIs.[45] To meet this challenge, a welcome number of new drugs have been developed, including novel beta-lactams and beta-lactamase inhibitor combinations (BLBLIs). In recent years, ceftazidime-avibactam, ceftolozane-tazobactam, imipenem-cilastatin-relebactam, meropenem-vaborbactam, and cefepime-enmetazobactam have all been approved for use in treating cUTI following one or more FDA registrational trials. In most of these agents the novel beta-lactamase component has been designed to inhibit the activity of beta-lactamases, including certain carbapenemases. In parallel with development of these new combination antibiotics, molecular tests to define the specific resistance genes expressed by the bacterial pathogen are also coming into widespread use. Antibiotic stewardship considerations suggest that the main role of novel BLBLIs currently may be in treating identified multidrug-resistant pathogens, particularly when the resistance gene or mechanism has been established.

Summary of evidence for empiric use of novel beta-lactam/beta-lactamase inhibitors to treat cUTI

The evidence for the use of novel BLBLI in treatment of cUTI include seven randomized, controlled trials, published from 2012-2023 and including 4,432 patients with cUTI (including acute pyelonephritis). All trials evaluated the efficacy of a novel BLBLI versus a standard cUTI comparator, including levofloxacin in the ASPECT-cUTI (Wagenlehner 2015) trial,[18] carbapenems in four trials (Vasquez 2012, Wagenlehner 2016, Carmeli 2016, Sims 2017),[15-17,22] and piperacillin-tazobactam in the TANGO I and ALLIUM trials (Kaye 2018 and Kaye 2022).[9,24] The average duration of IV therapy ranged from 5-10 days, and some trials transitioned patients to oral therapy with a fluoroquinolone or TMP/SMX. Clinical cure and microbiologic cure were usually measured at TOC, 5 to 10 days after end of antibiotic therapy.

Benefits, Harms and Certainty of evidence

Overall, treatment with the novel BLBLIs likely leads to similar clinical cure as the comparator antibiotics at test of cure in patients with cUTI (overall clinical cure for novel BLBLIs was 91.9% vs 89.7% for comparators; RD: 0.9%; 95% CI: -0.9% to 3.6%/ RR: 1.01; 95% CI: 0.99 to 1.04; moderate certainty of evidence).

The evidence suggests that treatment with novel BLBLIs leads to more microbiological cures at test of cure in patients with cUTI versus comparator antibiotics, but this increase was judged clinically unimportant at a decision threshold of 10% (. More specifically, microbiological cure for novel BLBLI was 79.3% and 69.0% for the comparators (RD: 8.3%; 95% CI: 1.4% to 15.9%/ RR: 1.12; 95% CI: 1.02 to 1.23). When stratifying by classes of antibiotics in the comparator group, the difference in microbiological cure was larger when the comparator was piperacillin-tazobactam (RD: 15.0%; 95% CI: 8.1% to 22.4%). Microbiologic cure was more comparable with fluoroquinolones or carbapenems versus BLBLIs. Unfortunately, infection recurrence was not reported in these trials, so the relevance of in the higher microbiological cure rates with novel BLBLIs is unclear.

For harms, serious and non-serious adverse events were likely comparable between groups. Mortality was rare (0.4%), for reasons discussed above related to enrollment criteria for randomized, clinical trials.

Other considerations

Other considerations specific to these novel BLBLI agents are their crucial role in our current armamentarium against multidrug-resistant pathogens, and their higher costs as newer agents. The current role of these antibiotics in management of cUTI may be for use after identification of specific patterns of multidrug resistance. For example, IDSA guidance on antibiotic treatment of antimicrobial resistant Gram-negative infections suggests that the novel BLBLI agents be reserved for treating infections caused by organisms exhibiting carbapenem resistance.[5] Cost also becomes an important consideration with these agents. For example, according to the 2023 Wholesale Acquisition Cost (WAC) database, a dose of piperacillin/tazobactam ranges in cost from $30-$156, and meropenem costs $30-150 per dose, while ceftolozane/tazobactam costs $1443 per dose.[46] Higher costs can lead to inequities in access to these drugs, particularly if hospitals with underserved populations do not include these drugs on their formularies.

Rationale for recommendation and implementation

The novel BLBLIs have been designed to function against specific types of antibiotic resistance, particularly carbapenemases. From an antibiotic stewardship perspective, these drugs are among the few remaining options against multidrug resistant pathogens. As such, they should be reserved for situations in which they are truly needed, such as when susceptibility testing and molecular resistance testing results are available to guide the choice of novel BLBLI. When choosing empirically (prior to culture results) to treat a patient with cUTI, the panel judged that BLBLI are not first choice antibiotics because of these antibiotic stewardship considerations. Empiric use of novel BLBLIs for cUTI should be largely restricted to patients in septic shock for whom prior culture data or risk factors suggest that preferred antimicrobials are likely to be inappropriate.

6) Cefiderocol

Cefiderocol is a novel siderophore cephalosporin brought to the periplasm by bacterial iron transporters through a “Trojan horse” strategy, a unique mechanism of delivery conferring activity against many otherwise highly resistant Gram-negative bacteria.[45,47] Cefiderocol was approved by the FDA in 2019 for treatment of complicated UTI caused by susceptible Gram-negative microorganisms in patients who have limited or no alternative treatment options.[48] Unfortunately, resistance to cefiderocol, related to changes in siderophore receptors or traditional resistance mechanisms to cephalosporins, is already emerging.

Summary of evidence on empiric use of cefiderocol to treat cUTI

Two randomized, controlled trials were included, one of which focused specifically on cUTI caused by Gram-negative pathogens (Portsmouth 2018),[11] and the other which enrolled patients with serious infections caused by carbapenem-resistant Gram-negative bacteria (CREDIBLE-CR trial) (Bassetti 2021).[10] Both trials were multicenter and international. The Portsmouth trial enrolled 452 patients, all with cUTI, while CREDIBLE-CR enrolled 152 patients, of which only 36 had cUTI. The analyses below for clinical and microbiologic cure include 326

evaluable patients with cUTI. The comparator in the Portsmouth trial was imipenem-cilastatin, while the comparator in CREDIBLE-CR was best available therapy, of which the majority were colistin-based regimens. Although CREDIBLE-CR was randomized, the treatment was open label, and the duration of therapy was longer in the cefiderocol group (10.5 days versus 6.5 days).[10] Portsmouth 2018 was randomized and blinded, and duration of IV therapy averaged 9 days.[11] Test of cure in both studies was at 5-9 days after end of treatment.

Many of the causative organisms in these two multicenter, international trials were resistant to antibiotics. In the Portsmouth trial, despite excluding patients who were known to have an organism resistant to carbapenems, 6.5% of organisms were resistant to imipenem, 55% to levofloxacin, 28% to cefepime, and 15% to piperacillin-tazobactam.[11] The top three causative organisms were *Escherichia coli*, *Klebsiella pneumoniae*, and *Pseudomonas aeruginosa*. In the CREDIBLE-CR trial, the organisms causing cUTI were predominantly *Klebsiella pneumoniae* and *Pseudomonas aeruginosa*, and all organisms were resistant to carbapenems.[10] Surprisingly, 8/185 pathogens (4%) had a cefiderocol MIC of greater than or equal to 4 ug/mL at baseline in the main cohort.

Benefits, Harms and Certainty of evidence

Treatment with cefiderocol likely leads to similar clinical cure at test of cure in patients with cUTI versus the comparator antibiotics. . Specifically, clinical cure for cefiderocol was 88.5% vs 86.3% for comparators (RD: 2.6%; 95% CI: -4.3% to 10.4%/ RR: 1.03; 95% CI: 0.95 to 1.12; moderate certainty of evidence).

The evidence suggests that treatment with cefiderocol leads to higher rates of microbiological cures in at test of cure in patients with cUTI versus the comparator antibiotics. More specifically, microbiological cure at TOC for cefiderocol was 72.9% and 54.8% for the comparators (RD: 18.1%; 95% CI: 6.6% to 32.4% / RR: 1.33; 95% CI 1.12 to 1.59). The evidence suggests that cefiderocol leads to similar recurrence of infection at late follow-up (4.8% for cefiderocol vs 9.7% for the comparator group; RD: -4.8%; 95% CI -7.4% to 0.4% / RR: 0.50; 95% CI 0.24 to 1.04), but this estimate is imprecise due to the small number of recurrences documented.

Cefiderocol likely leads to fewer non-serious adverse events than the comparator antibiotics (RD: -11.1%, 95% CI: -18.7% to -2.5% / RR: 0.78; 95% CI: 0.63 to 0.95), but this finding was clearly driven by a high rate of gastrointestinal disturbances in the imipenem-cilastatin group in the Portsmouth 2018 trial. Serious adverse events were frequently reported in these studies. Cefiderocol may lead to comparable rates of serious adverse events (reported in 7.4% in the cefiderocol group and 10.8% in the comparator group) and mortality (1.5% in the cefiderocol group versus 1.3% in the comparator group) versus the comparators. The greater overall mortality with cefiderocol in CREDIBLE-CR appeared to be driven primarily by patients with hospital-acquired/ventilator-associated pneumonia or bloodstream infections and by infections with *Acinetobacter* spp..[10]

Other considerations

Other considerations specific to cefiderocol are that dosing needs to be adjusted carefully according to renal function, with increased frequency required to achieve therapeutic levels for patients on renal replacement therapy with augmented renal clearance (greater than 120 mL/min). As the Portsmouth 2018 trial excluded patients with creatinine clearance less than

20mL/min, the real-world impact of the need for dose adjustment on clinical outcomes remains to be seen.[11] Additionally, resistance to cefiderocol is already emerging. In the CREDIBLE-CR trial, 17% of enrolled patients in the cefiderocol arm received combination therapy, which may have boosted apparent effectiveness.[10] As a newer drug, the costs are expected to be higher than with older antibiotic agents.

Rationale for recommendation and implementation

The panel judged that cefiderocol is an alternative antibiotic to empirically treat patients with cUTI but is not preferred due to stewardship considerations and uncertainties about real-world effectiveness. Care should be taken to ensure adequate dosing in patients receiving renal replacement therapy. Cefiderocol's use may be ideally reserved for highly resistant uropathogens in which the resistance mechanism of the organism is known and should be overcome by cefiderocol (i.e. organisms with metallo-beta-lactamase carbapenemases).

7) Older aminoglycosides (gentamicin, amikacin, and tobramycin)

Aminoglycosides are among the earliest antibiotics to enter clinical use, though their use declined over the past half-century as alternative antibiotics with fewer side effects became available. As a class, aminoglycosides can cause renal impairment and ototoxicity, and these harmful effects often emerge during treatment despite optimized (i.e. once-daily) dosing. With rising rates of antimicrobial resistance, aminoglycosides are receiving renewed interest.

In this section we will discuss the indirect evidence for empiric use of older aminoglycosides (gentamicin, amikacin, and tobramycin) to treat cUTI. Plazomicin, the newest aminoglycoside, will be discussed separately.

No randomized, controlled trials of aminoglycoside monotherapy for empiric treatment of cUTI were published from 2008-2023; these agents were predominately tested in the last century. One systematic review of the literature and meta-analysis included 37 randomized, controlled trials published between 1966 up to 2006, mostly in adults.[49] These trials compared systemically administered aminoglycosides as a single drugs (monotherapy) versus another systemically administered single antibiotic or an antibiotic combination without aminoglycosides. The aminoglycosides studied were gentamicin, amikacin, tobramycin but also netilmicin (not currently available), and dosing was once daily in only three of these trials. The majority of the 26 UTI trials were published between 1981 and 1992 and included both inpatients and outpatients. Many of the UTI trials had very small sample sizes (34-186 patients), and only five reported on organisms' susceptibility to aminoglycosides. Comparator antibiotics were beta-lactam antibiotics (mostly cephalosporins) and quinolones.

Six of the UTI trials reported 30-day mortality, finding a statistically non-significant increase in mortality in the aminoglycoside arm (RR 1.96, 95% CI 0.61-6.29). Overall mortality in these studies were very low, precluding meaningful comparisons.

Twenty of the UTI trials reported on treatment failure, finding no significant difference between groups (RR 1.11, 95% CI 0.94-1.30). Microbiological failure at 5-9 days after end of therapy was higher in the aminoglycoside group in comparison to beta-lactams or quinolones (RR 1.40, 95% CI 1.16-1.69). During the timeframe of these trials, the cephalosporins and

quinolones were the newer drugs. Rates of relapse of infection were not significantly different between groups and were not reported further.

Adverse events were less often reported in the aminoglycoside group compared to the beta-lactams (RR 0.46, 95% CI 0.33-0.63), but the types of adverse events were different. Beta-lactams were associated with rash, phlebitis, gastrointestinal, and hepatic adverse events, while aminoglycosides were associated with nephrotoxicity, which was reported significantly more commonly in patients in the aminoglycoside group (RR 3.61, 95% CI 1.67, 7.80).

In summary, older trials (in which cephalosporins and quinolones were the novel agents without substantial resistance, and in which aminoglycoside resistance frequently went unreported) suggested similar performance of aminoglycosides versus comparators for cUTI, albeit with more nephrotoxicity. Many of these trials were too small to ensure randomization was successful and balanced for important characteristics, and these trials suffered from other methodologic flaws that limit their ability to inform modern clinical practice.

Summary of evidence for empiric use of older aminoglycosides to treat cUTI

More recently, two retrospective studies from Israel compared clinical outcomes of aminoglycosides to non-aminoglycoside antibiotics in patients hospitalized for cUTI between 2014-2019.[50,51] One study evaluated 2,026 patients with pyelonephritis (Elbaz 2020) of which 29% were bacteremic,[51] while the other studied 218 patients with bacteremia of urinary source with ESBL-producing Enterobacterales (Zohar 2019).[50] The rate of ESBL-producing organisms was 30% in the pyelonephritis study and 100% in the bacteremia study (by enrollment criteria). Patients in these two retrospective studies were older than patients in RCTs, with median ages of 79 and 82 years. The non-aminoglycoside antibiotics given were ceftriaxone, piperacillin-tazobactam, carbapenems, and fluoroquinolones. In both studies the primary outcome was 30-day mortality, and multivariate analysis was done to adjust for confounding variables. The mortality rate in the pyelonephritis study was 9.9% and16.6% in the bacteremia study, as expected with real-world data.

Benefits, harms and certainty of evidence

Both retrospective studies found that 30-day mortality was lower in the aminoglycoside group (ranging from RD: -2.4%, 95%CI: -3.9% to -0.6% / adjusted RR: 0.78; 95%CI: 0.65 to 0.95 to RD: -10.3; 95% CI: -21.4% to 0.8% / adjusted OR: 0.51; 95%CI: 0.24 to 1.06, but the certainty in the evidence is very low. Clinical cure was not reported, and microbiological cure was reported only in only one study. The evidence suggests that microbiological cure was comparable between groups (RD; -8.9%; 95%CI: -29.4% to 12.8% / adjusted OR: 0.70; 95%CI: 0.28 to 1.72), but this estimate was imprecise due to a very small sample size.[50]

Acute kidney injury may be comparable between groups in both studies (ranging from RD: -0.1%, 95%CI: -0.1% to 0% / adjusted RR: 0.98; 95%CI: 0.97 to 1.00 and RD: 1.3%; 95%CI: -5.4% to 14.4% / OR: 1.14, 95%CI (0.46 to 2.81) , but these estimates are very uncertain due to the serious risk of bias. Length of hospital stays and rehospitalization at 3 months may favor the aminoglycoside group in one study.[51]

Both studies found that patients who received aminoglycosides had fewer comorbidities, better renal function, and a better functional status than patients who received comparator

antibiotics. Confounding-by-indication with residual confounding remains very likely, despite authors' efforts to adjust for this selection bias. These clinical outcomes should be taken in the context that the patients treated with aminoglycosides were a relatively healthier group.

Other considerations

Aminoglycosides are now given once daily for cUTI,[52] and close consultation with a pharmacist is advised to adjust the dose and interval. Serious and irreversible nephro- and ototoxicity can occur with prolonged courses of therapy. Patients receiving these antibiotics require regular assessment of renal and otic function, and therapeutic drug monitoring to determine trough levels is recommended. Aminoglycosides may not be appropriate for patients with underlying impairment of renal function or hearing loss. Patients receiving IV or IM aminoglycosides as outpatients require close monitoring. These drugs are themselves inexpensive, although the laboratory tests add some costs.

The 2010 publication of the UTI guidelines on cystitis and pyelonephritis recommended a single dose of aminoglycoside or ceftriaxone at the initiation of oral antibiotics to treat acute pyelonephritis, if resistance to the oral agent was a concern.[26] This panel did not find any studies of single dose aminoglycoside as part of a combination treatment for cUTI in adults, but in practice one or two doses of an aminoglycoside are often used as a component of the antibiotic treatment for acute pyelonephritis/cUTI when there is a concern for an ESBL-producing organism.[53]

Rationale for recommendation and implementation

The panel judged that aminoglycosides are an alternative class of antibiotic for empiric treatment of patients with cUTI, especially in populations where the ESBL rate among urinary organisms is increasing. Many patients may have contraindications to receiving aminoglycosides, such as renal insufficiency or advanced hearing loss. The panel judged that aminoglycosides are not a preferred antibiotic for patients with cUTI with or without sepsis due to the relatively greater risk of nephrotoxicity and limited modern evidence base for effectiveness versus alternative agents.

8) Plazomicin

Plazomicin is a "next generation" intravenous aminoglycoside that was designed to evade bacterial aminoglycoside modifying enzymes.[45] These modifications give plazomicin activity against most Enterobacterales, including those resistant to older aminoglycosides. Plazomicin was approved by the FDA for use in complicated UTI in 2018. As with other aminoglycosides, plazomicin can cause nephrotoxicity; the risk of ototoxicity is less clear, but a valid safety concern that needs to be assessed in larger studies.

Summary of evidence for empiric use of plazomicin to treat cUTI

Two randomized, controlled, multicenter international trials compared plazomicin to another antibiotic in empiric treatment of cUTI, for a total of 744 evaluable patients.[23,25] One of these studies, Connolly et al. 2018 was a phase 2 study in which 145 patients were randomized to two different doses of plazomicin or levofloxacin IV for five days, and enrollment occurred between 2010-2012.[25] Resistance among the causative urinary pathogens was high, with 19% of all urinary pathogens resistant to levofloxacin, and 6% of pathogens resistant to plazomicin.

Oral switch at end of therapy was not permitted. Once the trial was underway, enrollment in the lower plazomicin treatment dose group (10mg/kg) was stopped, and all patients were subsequently randomized to the higher (15mg/kg) dose versus levofloxacin. In the EPIC trial (Wagenlehner 2019), 604 patients were randomized to receive IV plazomicin versus meropenem for an average of five days and then were transitioned to oral agents to complete 7-10 days of therapy.[23] The majority of organisms isolated were susceptible to both plazomicin and meropenem. The test of cure visit was at 15-19 days after starting IV therapy for the EPIC trial; the Connolly 2018 trial measured test of cure at 5-12 days after the end of antibiotic treatment. Patients in both trials had to have good renal function (creatinine clearance of >60 ml/min in Connolly and >30 mL/min in the EPIC trial).

Benefits, Harms and Certainty of evidence

Treatment with plazomicin likely leads to similar rates of clinical cure at TOC in patients with cUTI versus treatment with comparator antibiotics. Specifically, the overall clinical cure for plazomicin was 84.3% vs 87.2% for comparators (RD: 0.0%; 95% CI: -6.1% to 6.1%/ RR: 1.00; 95% CI: 0.93 to 1.07; moderate certainty of evidence).

Treatment with plazomicin may lead to higher rates of microbiological cure at test of cure in patients treated for cUTI versus the comparator antibiotics (plazomicin was 81.9% vs comparator group was 72.6%; RD: 12.3%; 95% CI: 5.1% to 21.0% / RR: 1.17; 95% CI: 1.07 to 1.29). The evidence suggests that treatment with plazomicin may lead to similar recurrence of infection at late follow-up (3.2% in the plazomicin group vs 7.0% in the comparator group), but these studies are too heterogeneous to be pooled since these they reported opposite direction of effects.

Although plazomicin likely leads to similar rates of non-serious adverse events versus comparators, but an increase in serum creatinine of 0.5 mg or more per deciliter above baseline was observed in 7.0% in the plazomicin group vs 4.0% in the meropenem group in the EPIC trial,[23] and ototoxicity was reported in two cases (one in each group). Connolly 2018 required enrolled patients to have a creatinine clearance of >60ml/min at baseline, and five plazomicin-treated patients had an increase in serum creatinine of >=0.5 mg/dl during the study, versus one levofloxacin-treated patient.[25] Two patients in the plazomicin group and one in the levofloxacin group developed some signs of ototoxicity. Serious adverse events and mortality were rare (only 1 death among 744 patients). In the context of the relative healthy state of the enrolled subjects compared to the real-world population of patients hospitalized with cUTI, these signals of nephro- and otoxicity are concerning.

Other considerations

Other considerations specific to plazomicin include the need to involve a clinical pharmacist in designing and monitoring the once daily dosing regimen, to avoid nephro- and ototoxicity. Patients with baseline renal impairment and hearing loss may not be good candidates to receive this agent. Patients with creatinine clearance of < 30 mL/min were excluded from the clinical trials of this agent. Plazomicin is generally a more expensive antibiotic than the older agents.

Rationale for recommendation and implementation

The panel judged that plazomicin is an alternative antibiotic to empirically treat patients with cUTI but is not preferred due to concerns with potential serious adverse events associated with aminoglycosides, such as renal insufficiency or ototoxicity.

9) Oral empiric antibiotics for cUTI

Some patients with cUTIs who are seen in an emergency department or urgent care setting with acute pyelonephritis may be candidates for outpatient management. Similarly, patients with CAUTI seen in clinic may often be managed as outpatients. Oral third generation cephalosporins and fluoroquinolones and TMP-SMX may be useful for empiric therapy in such cases (**Table 3.1**).[54,55] High resistance rates to fluoroquinolones in many areas may make these less appealing as empiric options, and adverse effects are a concern. High resistance rates are likewise a concern with TMP-SMX, and prior urine cultures may be helpful in determining if fluoroquinolones or TMP-SMX are likely to be effective.

Although robust clinical trials of oral cephalosporins such as cefpodoxime as initial treatment for cUTI in adults are lacking, in practice cephalosporins are used in many settings as step-down therapy, when ESBL-production is not a major concern.[55-57] When choosing an oral cephalosporin for cUTI, both oral absorption and urinary excretion may be relevant parameters (See dosing **Table 3.1**) for consideration. Observational studies suggest that third generation oral cephalosporins may be comparable to oral fluoroquinolones or TMP-SMX as step down therapy in patients with cUTI and gram-negative bacteremia.[55-57] However, such studies are conflicting on whether earlier generation cephalosporins (e.g. cephalexin), oral beta-lactams (e.g. amoxicillin and amoxicillin clavulanate), and cephalosporins with low bioavailability (e.g. cefdinir) are as efficacious as alternatives; these should be used cautiously and with optimized dosing.[58-61]

As an example, in one retrospective study that included patients who received cefdinir (which has low urinary excretion of only 13-23% and low oral absorption of only 25%) and a lower dose of cephalexin (500 mg every 8 hours), readmissions for UTI were higher in the beta-lactam group compared to those who received fluoroquinolones or TMP-SMX.[59]

Amoxicillin-clavulanate and cephalexin have potentially lower efficacy as demonstrated in multiple studies.[58,60] Additionally, we did not find substantial data supporting the use of ampicillin, cefadroxil, cefaclor, or cefdinir for cUTI. Ideally, a patient who receives any of these oral options as their initial empiric therapy would have a urine culture from a prior episode showing susceptibility to the agent chosen.[62]

Furthermore, trials of three days of beta-lactam antibiotics for acute cystitis in women (cefpodoxime, amoxicillin-clavulanate, cefadroxil, and amoxicillin) consistently found lower clinical and microbiologic cure in the beta-lactam recipients, in comparison to three days of ciprofloxacin or trimethoprim-sulfamethoxazole.[63,64] These trials provide indirect evidence that beta-lactams are not as effective for acute cystitis when used for the same duration as other classes of antibiotics; whether these results are generalizable to empiric use of beta-lactam antibiotics to treat complicated UTI is unknown. Another concern with treating cUTI with oral beta-lactam antibiotics is that standard dosing may not achieve adequate levels in the urine. For example, a retrospective cohort study found that 7 days of IV or highly bioavailable antibiotics was as effective as 14 days of antibiotic therapy for bacteremic cUTI; of note, the doses of beta-

lactams considered to be bioavailable were the following: amoxicillin 1000 mg orally every 8 hours, amoxicillin-clavulanate 875–1000 mg orally every 8 hours, or cephalexin 1000 mg orally every 6 hours.[65] Increasingly institutions are using higher dose regimens for oral beta-lactams and cephalosporins as step down therapy for Gram-negative bacteremia of urinary origin.[54,66]

The 2010 publication of the UTI guidelines on cystitis and pyelonephritis recommended a single dose of aminoglycoside or ceftriaxone at the initiation of oral antibiotics to treat acute pyelonephritis, if resistance to the oral agent was a concern.[26] This panel did not find any studies of single dose aminoglycoside as part of a combination treatment for cUTI in adults, but in practice one or two doses of an aminoglycoside are often used as a component of the antibiotic treatment for acute pyelonephritis when there is a concern for an ESBL-producing organism.[53] We identified one study of single dose ceftriaxone in non-pregnant adults; this study suggested that a single dose of IV ceftriaxone prior to switching to an oral cephalosporin was an effective strategy for women with pyelonephritis.[30]

Nitrofurantoin and oral fosfomycin are generally not appropriate choices for cUTI due to inadequate levels in tissue/bloodstream. If treating suspected bacteremia, the oral agent needs to achieve therapeutic levels in the bloodstream, which would preclude using nitrofurantoin and oral fosfomycin. Likewise, if treating pyelonephritis, nitrofurantoin does not reach adequate drug levels in the renal parenchyma and would not be an appropriate agent for an oral switch. Oral fosfomycin has been used in small studies to treat cUTI (including pyelonephritis), but multi-dose regimens are generally used and are inconsistent across the literature; the effectiveness and dosing of fosfomycin for cUTI needs additional study.[67,68] Oral fosfomycin has been used to treat chronic bacterial prostatitis but has not formally been evaluated in acute bacterial prostatitis trials.[69,70]

Table 3.1: Dosing of oral antibiotics for complicated UTI in alphabetical order

Drug	Oral absorption (%)	Urinary excretion (%)	Dose for patients with normal renal function
Amoxicillin-clavulanate	80 (amoxicillin)[71] variable (clavulanate)[72]	50-70 (amoxicillin)[71] 25-40% (clavulanate)[71]	875mg-125mg every 8 to 12 hours[58-61,66,73-76] Other regimens may be more effective[a]
Cefixime	50[77]	50[77]	400mg once daily[30]
Cefpodoxime	50[77]	80[77]	200mg to 400mg every 12 hours[55,61,78]
Ceftibuten	75-90[77]	73[77]	9mg/kg daily (children)[b] 400mg daily or 200mg every 12 hours (adults)[79,80]
Cefuroxime	52[77,81]	90[77,81]	500mg every 12 hours[61,82]
Cephalexin	90[77]	90[77]	500mg to 1000mg every 6 hours[54,58-60,66,73-75,83] Other regimens may be more effective[a]
Ciprofloxacin	70[84]	40-50[84]	500mg to 750mg every 12 hours [54,61,66,85,86]
Levofloxacin	99[87]	64-100[87]	500mg to 750mg daily[25,54,78,86]

Other oral beta-lactams (e.g. amoxicillin, cefadroxil, cefaclor, cefdinir)	Comparative clinical outcomes data vs highly bioavailable oral alternatives are more limited and/or discouraging; consider use with infectious disease pharmacist consultation if alternatives are not available.		
Trimethoprim-sulfamethoxazole	70-90[88]	84 (sulfamethoxazole), 66 (trimethoprim)[88]	800mg-160mg every 12 hours[61,85]
[a]Despite routine use of optimized dosing, the majority of studies comparing switch to oral beta-lactams versus fluoroquinolones or trimethoprim-sulfamethoxazole for cUTI have found inferior outcomes with oral beta-lactams when amoxicillin-clavulanate or cephalexin were the predominant oral beta-lactams being used. [b]Ceftibuten is the sole oral beta-lactam in this table with modern randomized, controlled trial data for cUTI in both children in adults; however, while it produced comparable clinical outcomes versus trimethoprim-sulfamethoxazole in children, in adults relapses were higher with ceftibuten versus norfloxacin.			

10) Other antibiotics relevant to treatment of cUTI

IV Fosfomycin

Fosfomycin is an older antibiotic agent whose susceptibility testing is complicated, not easily amenable to automated methods, and rarely performed. In the U.S., fosfomycin susceptibility breakpoints for fosfomycin are established only for *E. coli* and for *Enterococcus faecalis*, and only for the oral formulation of fosfomycin. In practice, clinicians are often asked to determine whether fosfomycin would be an acceptable treatment for other urinary pathogens, such as *Klebsiella spp.*, but MIC breakpoints do not currently exist for non-*E. coli* organisms. Resistance to fosfomycin is increasing globally,[89] but as this antibiotic is infrequently used in the United States, many urinary organisms retain susceptibility.

Since intravenous fosfomycin is not available in the United States, fosfomycin is not an option for empiric therapy for most patients hospitalized for cUTI. Oral fosfomycin (fosfomycin trometamol) is a white powder administered as a sachet, to be dissolved in in water and then swallowed. This oral formation is available in the US and Europe and is used to treat uncomplicated UTI, or as step down therapy for patients with cUTI who are switching from IV medications to an oral agent and whose organism is known to be susceptible to fosfomycin. Fosfomycin's IV formulation is a calcium salt; this IV formulation is approved for cUTI for adults in Europe and Asia.

Summary of evidence for empiric use of IV fosfomycin to treat cUTI

The evidence on empiric use of IV fosfomycin for cUTI comes from two randomized, controlled trials which enrolled 607 evaluable patients. The open-label FOREST trial was conducted in 22 hospitals in Spain and enrolled patients with bacteremic UTIs caused by MDR *E. coli* (Sojo-Dorado 2022).[19] Although this trial was published in 2022, the 143 evaluable patients were enrolled from 2014-2018. The comparator in the FOREST trial was ceftriaxone or meropenem (if the causative organism was resistant to ceftriaxone), and the majority of patients transitioned to oral therapy after 5.5 days of IV therapy. Resistance to ceftriaxone among the

MDR *E. coli* was high, 57%. The ZEUS trial, published in 2019, enrolled 464 evaluable patients between 2016-2017.[21] Patients were randomized to seven days of IV fosfomycin versus piperacillin-tazobactam in a double-blind design, with no oral switch after IV therapy. All uropathogens were included in the ZEUS trial, and resistance to piperacillin/tazobactam was only 6.9%. Resistance to fosfomycin was not reported in the ZEUS trial and was an exclusion factor for the FOREST trial.

Benefits, Harms and Certainty of evidence

Treatment with IV fosfomycin likely leads to similar rates of clinical cure at test of cure in patients with cUTI versus treatment with comparator antibiotics. The overall clinical cure for fosfomycin was 92.2% vs 91.2% for comparators (RD: 0.9%; 95% CI: -3.6% to 5.5% / RR: 1.01; 95% CI: 0.96 to 1.06; moderate certainty of evidence).

Similarly, treatment with IV fosfomycin likely leads to similar rates of microbiological cure at test of cure in patients with cUTI versus treatment with the comparator antibiotics (RD: 6.4%; 95% CI: -1.9 % to 15.4% / RR: 1.10; 95% CI: 0.97 to 1.24). The evidence suggests that recurrence of infection at late follow-up may be comparable between groups, but this estimate is likely imprecise due to the small number of recurrences of infection documented. Of note, in the FOREST study, both relapses and reinfections were recorded, with reinfections defined as symptomatic UTI caused by a different strain of bacteria. Among the 132 patients in this analysis, 14 had relapse versus 8 with reinfection, demonstrating the very important principle that patients who are hospitalized for cUTI are at risk for another cUTI, regardless of initial treatment given.

Harms are an important consideration with IV fosfomycin, which is a salt and delivers 330 mg of sodium per gram of IV fosfomycin disodium, in comparison to the 65 mg of sodium per gram of piperacillin in the combination product piperacillin-tazobactam.[90] The IV formulation of fosfomycin contains significant sodium content that can exacerbate cardiac failure among individuals unable to tolerate large increases in volume. The evidence suggests that treatment with IV fosfomycin may lead to more non-serious adverse events than treatment with the comparator antibiotics (RD: 10.6%, 95% CI: 1.3% to 22.1% / RR: 1.33; 95% CI: 1.04 to 1.69), although this estimate only reflects the ZEUS trial.[21] In the ZEUS trial the main non-serious adverse events were mild and transient hypokalemia and elevated serum aminotransferases, both more frequently documented in the IV fosfomycin group. In the smaller FOREST trial, non-serious adverse events cannot be quantified between the two arms, but fosfomycin was discontinued among 6 patients due to adverse events (4 for heart failure); heart failure was reported among 6 patients treated with fosfomycin, all older than 80 years of age.[19]

Other considerations

Other considerations specific to IV fosfomycin include that the IV formulation is not available in the United States, and susceptibility testing of fosfomycin is not routinely conducted nor reported.

Rationale for recommendation and implementation

The panel judged that IV fosfomycin is not a first line antibiotic for empiric treatment of cUTI due to lack of availability, difficulty in conducting susceptibility testing, and concerns about adverse events, particularly sodium overload and hypokalemia. However, if available, this drug

can be an alternative antibiotic to empirically treat patients with cUTI without risk factors for resistance to fosfomycin. Caution should be taken in treating patients with known, chronic heart failure with IV fosfomycin.

Colistin

No randomized clinical trials exist to support the empirical use of colistin in cUTI. However, colistin (the active form of the commercially available parenteral prodrug colistimethate sodium, also polymyxin E) is suggested by the IDSA guidance document on managing antibiotic resistant pathogens as an alternative agent for treating uncomplicated cystitis caused by carbapenem-resistant Enterobacterales or *Pseudomonas aeruginosa* with difficult-to-treat resistance.[5] Colistin is also mentioned as a potential treatment for UTIs caused by carbapenem-resistant *Acinetobacter baumannii*. Colistin can cause nephrotoxicity and should rarely be used as an empiric initial treatment for cUTI.. Polymyxin B should not be used to treat UTI because of its predominantly non-renal clearance.

Drugs that do not currently have FDA approval for empiric treatment of cUTI

Three antibiotics that were tested in randomized, controlled trials seeking FDA approval for a UTI indication did not succeed in receiving this approval. These three agents merit brief discussion, as it is possible that additional trials or worsening resistance will lead to their use in the future for cUTI. These three antibiotics are omadacycline, tebipenem, and sulopenem.

Omadacycline is a modified tetracycline that has increased activity against some tetracycline-resistant organisms.[91] Omadacycline is available both orally and intravenously and is FDA approved for treatment of skin and soft tissue infections and pneumonia. Although active against *E. coli*, omadacycline is not active in vitro against the cUTI pathogens *Morganella* spp., *Proteus* spp., and *Providencia* spp. Overcash et al. conducted a phase 1 trial of omadacycline IV and/or orally to treat acute pyelonephritis in women; 201 patients were randomized, and omadacycline was “not non-inferior” to levofloxacin (did not meet non-inferiority criteria).[92] Another trial of omadacycline (IV and oral) for uncomplicated cystitis in 31 women found that 65% reported nausea, with 31% reporting vomiting.[93]

Tebipenem pivoxil hydrobromide is an orally bioavailable carbapenem prodrug that is converted to the active moiety, tebipenem, by enterocytes. Tebipenem has broad-spectrum activity against multidrug-resistant gram-negative pathogens, including ESBL-producing Enterobacterales. Tebipenem was studied in a phase 3 trial of treatment of cUTI, with IV ertapenem as the comparator.[94] Both groups were treated for 7 to 10 days (up to 14 days if bacteremic), and the primary outcome was a composite “overall response” that required both clinical and microbiologic response. Overall 1,372 patients were randomized, and 868 were included in the microbiological intent to treat group assessed for the primary outcome. The composite success outcome was reached by 59% in the tebipenem group versus 62% in the ertapenem group, meaning that tebipenem was non-inferior to ertapenem. Clinical cure was high in both arms of the study, at 93% and 94%, for tebipenem and ertapenem respectively. However, the FDA did not issue an approval for tebipenem for treatment of UTI.

Sulopenem is a thiopenem (sulfur fused to 5-member beta-lactam ring) which is active against ESBL-producing and Amp-C beta-lactamase producing Enterobacterales (organisms resistant to third generation cephalosporins such as ceftriaxone).[95] Sulopenem has an IV and an

oral formulation. The oral prodrug, sulopenem etzadroxil, is co-formulated with probenecid to reduce renal clearance and increase systemic levels of sulopenem. A phase 3 trial in 1395 patients with cUTI (including pyelonephritis) randomized participants to received IV followed by oral sulopenem, versus IV ertapenem followed by oral ciprofloxacin or amoxicillin/clavulanate.[96] The primary outcome was a composite of clinical and microbiologic cure, and microbiologic cure was stringently defined as reduction of the original pathogen to < 10^3 colony forming units (CFU)/mL (in contrast to other trials using a <10^4 CFU/mL threshold for microbiologic cure). Overall, in the modified microbiologic intention to treat or mMITT population (patients with a baseline urine culture with $\geq 10^5$ of an Enterobacterales pathogen susceptible to sulopenem and ertapenem), sulopenem was not non-inferior to ertapenem, with composite outcome rates of 67.8% and 73.9%, respectively. Although clinical cure was 88% or higher in both arms of the study, the microbiologic response was lower in the sulopenem arm than the ertapenem arm (71.2% versus 78.0%). The difference in outcomes between the two treatment arms was driven primarily by presence of asymptomatic bacteriuria in the sulopenem-treated group at test of cure (day 21). Closer examination of the microbiologic failures revealed that oral ciprofloxacin was better at eliminating Enterobacterales from the urine than oral sulopenem, provided that the organisms were susceptible to fluoroquinolones (4.7% microbiologic failure with ciprofloxacin versus 21.8% with sulopenem). The results of this trial highlight the potential dichotomy between clinical cure and microbiologic response and raise the question of the clinical significance of finding bacteriuria at the test of cure visit.

Other Considerations

Antibiotic stewardship considerations

In light of antibiotic stewardship principles (i.e., “coordinated interventions designed to improve and measure the appropriate use of [antibiotic] agents by promoting the selection of the optimal [antibiotic] drug regimen including dosing, duration of therapy, and route of administration” [per IDSA guidelines]),[97] we advocate for the appropriate use of more narrow-spectrum antibiotics in patients without specific risk factors for infection caused by resistant pathogens. One meta-analysis reported that the incidence of *C. difficile* infection could be reduced by lowering exposure to ‘high-risk’ antibiotics, defined as clindamycin, fluoroquinolones, and cephalosporins, monobactams, and carbapenems.[98] For empiric treatment of cUTI, avoidance of antibiotics with a broad spectrum of activity when an agent with narrower spectrum of activity may be appropriate is aligned with principles of antibiotic stewardship. Empiric antibiotic choice always involves weighing antibiotic stewardship concerns versus the risk of inappropriate initial antibiotic choice.

Patients’ values and preferences

This guideline recommendation focuses on which antibiotics to choose at that critical point at which the patient with cUTI presents for care and the causative organism has not yet been identified (empiric antibiotic choice). Empiric antibiotics typically are continued for up to 72 hours before being replaced with tailored antibiotics based on culture results and other emerging data. In that context, avoiding mortality by choosing initially appropriate antibiotic therapy is the most important outcome. When expected mortality is low, consultation with the patient representatives participating in this guidelines panel further supported that treatment (whatever the choice of empirical therapy) should mainly focus on achieving clinical cure. If clinical cure is expected to be similar between different treatments, additional considerations

include antibiotic-associated adverse events, decreasing the risk of recurrence of infection, and avoiding readmission to hospital. Reducing the length of hospitalization and facilitating the ease of administration were considered important, but the choice of antibiotics by itself was not a driving factor in their decision-making process.

Costs, Resources, Feasibility and Equity

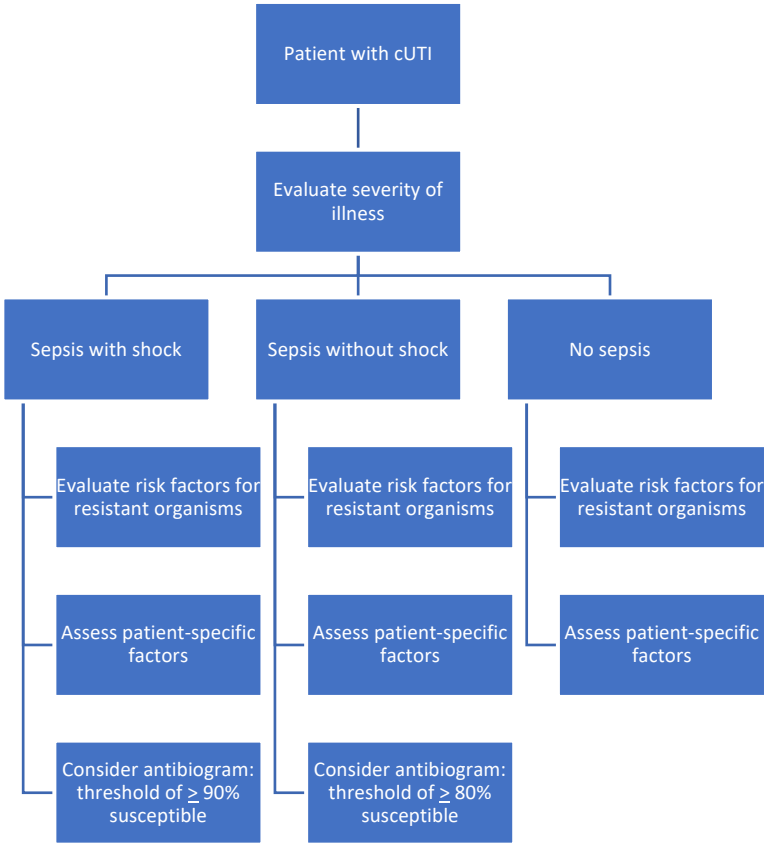
It is not possible for the guidelines panel to offer nationally generalizable direct comparisons of cUTI antibiotic costs because (at least in the United States) these costs vary widely based on the drug wholesaler and their contracts with individual pharmacies and institutions. That said, at the time of development of these recommendations, the average wholesale prices reported by the drug cost analysis tool Medi-Span (https://www.wolterskluwer.com/en/solutions/medi-span) suggests the antibiotics studied for cUTI can be categorized into three cost groups: low, medium, and high. Levofloxacin and ceftriaxone can be considered low-cost, with daily costs ranging from about $1 to about $50. Piperacillin-tazobactam and the carbapenems can be considered medium cost, with daily costs ranging from about $15 to about $150. Plazomicin, cefiderocol, and the novel cephalosporin and carbapenem beta-lactamase inhibitor combinations can be considered high-cost, with daily costs ranging from about $500 to $1500.

Thus, the potential excess cost of a 7-day course of cUTI treatment with agents other than levofloxacin or ceftriaxone is on the scale of a few hundred to a thousand dollars for piperacillin-tazobactam or the carbapenems, or several thousand to ten thousand dollars for the novel agents. Additionally, we consider that all of these antibiotic agents are given IV except for levofloxacin and ertapenem (which have oral and IM formulations, respectively), and thus would at minimum incur additional costs in the hundreds to thousands-dollar range for administration of outpatient parenteral antibiotic therapy (OPAT). Finally, we note that all of these agents other than levofloxacin, ceftriaxone, ertapenem, and plazomicin have every six hour or every eight hour dosing schedules, and so if given with on-label dosing could require the excess costs of extended hospitalization or nursing facility stay, likely in the several thousands to ten thousands of dollars range.

B. Process to Guide Empiric Antibiotic Choice for Complicated UTI

To optimize the selection of empiric antibiotic therapy for patients with suspected complicated UTI, we propose the following four step approach (**Figure 1.1**): 1) assess the severity of illness (for initial prioritization of empiric antibiotic therapy), 2) consider patient-specific risk factors for resistant uropathogens (for optimization of coverage), 3) evaluate other patient-specific considerations (to reduce the risk of adverse events), and 4) for patients with sepsis, consult a relevant local antibiogram if available (to further improve the likelihood of giving appropriate empiric therapy in septic patients).

Figure 1: Four-step approach to choosing empiric antimicrobial therapy for cUTI



This approach starts with the most important issue—the patient's severity of illness—and then takes into consideration the patient's risk factors for having a pathogen resistant to specific antibiotics or antibiotic classes, as well as practical issues such as antibiotic allergies. Finally, and only for patients with sepsis related to cUTI, the local antibiogram may have a role in helping the provider avoid inappropriate empiric antibiotic therapy if it is recent and relevant to the patient under consideration. The antibiogram is the last of the four recommended steps, as the evidence that using a facility's antibiogram to guide antibiotic prescribing for individual patients improves outcomes is very uncertain. Choosing which organism to focus on in the antibiogram is also a challenge in empiric decision making. The most relevant organism is suggested by the prior urine culture, if available. If not, *E. coli* is the default organism.

STEP 1: SEVERITY OF ILLNESS (initial prioritization of empiric antibiotic therapy)

In patients with suspected cUTI (including pyelonephritis), should selection of empiric antibiotic therapy be guided by severity of illness?

Recommendation:

I. For patients with suspected complicated UTI (including pyelonephritis), we suggest that the selection of empiric antibiotic therapy be initially guided by the severity of illness, specifically by whether the patient is in sepsis or not (*conditional recommendation, very low certainty of evidence*).

Remarks:

-Sepsis is defined per the Sepsis-3 Task Force as life-threatening organ dysfunction caused by a dysregulated host response to infection. These patients can be identified by SOFA score increase of 2 points or more, reflecting an in-hospital mortality greater than 10%, or presumptively identified with screening tools such as qSOFA or SIRS.[6,99]

The panel judged this clinical question to be the most important to guiding cUTI recommendations, especially after acknowledging that populations enrolled in randomized, controlled trials of empiric antibiotic choice have lower baseline mortality than general cUTI patient populations. Clinical practice is to use broader spectrum antibiotics in sicker patients (with the assumption that giving broader antibiotics means the empiric antibiotics are more likely to be effective against the pathogen). Quantifying the downstream impacts of choosing the wrong initial empiric antibiotic was necessary to provide stronger evidence for this practice. To identify subpopulations in which inappropriate antibiotic therapy has a significant impact on mortality is critical as it is one of a few factors that can be modified. A literature search was performed to define the prognostic impact of inappropriate empiric antibiotic therapy (IEAT) in patients with cUTI.

Prognostic Impact of Inappropriate Empiric Antibiotic Therapy (IEAT) in cUTI

First, we explored the question of quantifying the impact of choosing inappropriate empiric antibiotic therapy for cUTI, stratifying by severity of illness. To summarize the findings, an increase in mortality was observed in patients at high risk of mortality with cUTI receiving IEAT, with a pooled adjusted OR of 1.56 (0.99 to 2.46), as compared to patients receiving appropriate empiric antibiotic therapy. Below we will detail how this estimate was achieved.

Methods

To estimate the prognostic impact of IEAT in patients with cUTI, our systematic review of the literature included studies published since 2000 and studying adults with cUTI, most of whom were admitted to a hospital. Studies meeting these criteria were included if they reported the effect of IEAT on 30-day mortality or in-hospital mortality using a multivariate analysis. These studies were necessarily observational, as ethical considerations prohibit randomizing patients to appropriate versus inappropriate initial antibiotic therapy.

Empiric antibiotic therapy was defined as appropriate if the recovered pathogen in the urine culture (and/or blood culture) was susceptible in vitro to the antibiotics given before those culture results were available. For the analysis of the impact of IEAT, we focused on the outcome of mortality for two reasons: (1) mortality is the most patient-important outcome for decision making, and (2) mortality was the only outcome that was defined consistently across these studies. Note that mortality in these observational studies was much higher (above 5%) than in the randomized, controlled trials presented above which did not require a resistant pathogen as an inclusion criterion (less than 1%). Therefore, our models assessed the impact of IEAT on mortality, which was not possible using the RCT data.

Summary of evidence

Our systematic literature review identified eight observational studies reporting the impact of IEAT on mortality from cUTI after adjusting for other factors associated with mortality (Babich 2017, Esparcia 2014, Holmbom 2022, Korkmaz 2020, Ortega 2013, Righolt 2020, Rodriguez-Gomez 2019, Wiggers 2019).[100-107] (See **Supplementary Table B1.a** in Supplemental Materials). Three studies only included patients with bacteremia from suspected cUTI (Holmbom 2022, Ortega 2013, Wiggers 2019).[102,104,107] Two studies focused specifically on catheter-associated UTI (CAUTI) (Babich 2017, Ortega 2013).[100,104] These eight studies included a total 3,802 patients hospitalized with cUTI of whom 3,593 were further analysed. In most studies the average age of the patients was over 70. The eight included studies compared the effect on mortality of using an inappropriate versus appropriate empiric antibiotic therapy (IEAT versus appropriate empiric antibiotic therapy, or AEAT) based on subsequent urine culture in vitro susceptibility testing. IEAT was frequent, occurring in a mean of 27.5% across studies and ranging from 10.3%[102] to 50.8%.[100] All these studies were observational, and all but one[100] were retrospective. Clinicians' initial choice of empiric antibiotic therapy introduced the bias of confounding by indication, which was either partially or not accounted for at all in most studies. Confounding by indication in this context means that sicker patients likely received different antibiotics than less sick patients.

Estimate of the prognostic Impact of IEAT on mortality

To understand the impact of IEAT on mortality, we first looked at mortality in the patients that received AEAT, to provide an understanding of these patients' health outcomes under optimal antibiotic treatment conditions. The baseline mortality rate in patients receiving AEAT from these 8 studies averaged 14% (ranging from 5.8%[101] up to 34%). [106] Seven studies reported adjusted odds ratios for risk factors for mortality, including IEAT as one of the variables studied. (all except Rodriguez-Gomez 2019[106]). Combining these results, a statistically non-significant increase in mortality was observed in patients with cUTI receiving IEAT, with a pooled adjusted OR of 1.56 (0.99 to 2.46), as compared to patients receiving AEAT. Similarly, the only study reporting an adjusted hazard ratio (HR) for risk factors of mortality showed an adjusted HR of 1.99 (0.94 to 4.21) for mortality in patients with cUTI receiving IEAT.[106] Only one study utilized a propensity score to account for confounding-by-indication, and this study did not show an impact of IEAT on mortality (adjusted OR of 0.72 (0.39-1.32).[100] However, this study was exclusively in catheterized patients, overall mortality was very high (33%), and the diagnosis of CAUTI is often inaccurate.

Certainty in the evidence

The panel recognised that these estimates of the prognostic impact of IEAT on mortality in patients admitted for cUTI are very uncertain for various reasons, the most important being the serious risk of bias due to confounding by indication and residual confounding. Specifically, all cUTI studies of inappropriate antibiotic therapy were observational. Studies included in our estimate did not stratify for type of population, type of pathogen, or type of antibiotics. Reporting bias was also judged to be very likely. We limited the evidence base to studies that included IEAT in adjusted (multivariate) analysis of mortality, meaning that studies that found IEAT to be statistically insignificant in the unadjusted analysis and thus did not study IEAT in multivariate analysis were excluded from our evidence base.[108-110] This reporting bias likely caused an overestimation of the effect of IEAT on mortality. These studies were heterogeneous in terms of populations studied and statistical analyses performed. The multivariate analyses used different variables to adjust the odds ratio from one study to another. Thus, our estimated odds ratio is very uncertain.

Uncertainty in the diagnosis of cUTI in these retrospective studies also undermines the certainty about how much IEAT contributes to mortality. If a patient actually had another condition (particularly a non-infectious condition), the choice of antibiotics would not impact mortality. Alternatively, if patients are already very frail with high expected mortality or are presenting in later stages of sepsis, even appropriate empiric antibiotics may do little to improve their prognosis. In line with this theory, Esparcia 2014 reported that of the older adults admitted with cUTI, IEAT had a negative impact on 30-day survival only among those with a lower APACHE II score (less than 15).[101]

Finally, the panel was concerned about whether these findings are generalizable to cUTI populations at low risk of mortality (such as patients without sepsis, especially since studies not reporting mortality or low mortality rates were not included in this analysis. The panel acknowledges that when risk of mortality from cUTI is low, clinical failure may be more likely with inappropriate initial empiric antibiotic therapy. [35,111]

Other supporting evidence

Our landscape analysis of the literature found indirect evidence that supported our estimated odds ratio of 1.56 for mortality related to IEAT. Two meta-analyses addressed the question of whether inappropriate antibiotics (IEAT) were associated with mortality.[112,113] One of these studies included patients with UTI (Marquet 2015),[112] while the other did not specifically include UTI as a diagnosis, but many of the patients with gram-negative bacteremia may have had a urinary source (Paul 2010).[113] In both, IEAT was associated with an increase in mortality, and the pooled, adjusted odds ratios for mortality were 1.41 (95%CI: 1.22 to 1.61) and 1.60 (95%CI: 1.37 to 1.86) respectively. However, the studies that went into the meta-analyses were themselves observational and subject to the same biases discussed above for the studies of IEAT in patients with cUTI. Despite potential limitations in generalizing from studies including non-UTI patients, these meta-analyses further support our findings in patients with sepsis secondary to cUTI.

Other considerations

The urgency to provide appropriate initial empiric antibiotic therapy for patients with cUTI and sepsis or severe illness must be balanced against the potential harms of administering unnecessarily broad antibiotics. On the one hand, if every patient admitted to the hospital for

cUTI received antibiotics that would cover ESBL-producing Enterobacterales, multidrug-resistant *Pseudomonas aeruginosa*, and vancomycin-resistant enterococci, the rate of IEAT would initially be very low. However, such non-discriminant overuse of antibiotics would lead to more antibiotic side effects for the individual patients, more collateral damage to patients' microbiomes, increased risk of *Clostridioides difficile*-associated diarrhea, and most critically, inevitable emergence of resistant organisms rendering the broad-spectrum regimens less effective and leading to excess mortality from infection in future episodes of care. The tolerable level of risk of choosing a regimen that may prove to be ineffective thus must be determined by the patient's severity of illness at presentation.

Rationale for recommendation

Based on this analysis, the panel judged that inappropriate empiric antibiotic therapy has differential impact on mortality in different subpopulations. In patients with cUTI at high baseline risk of mortality (with sepsis or septic shock), inappropriate empiric antibiotic therapy may increase mortality, but available data are very uncertain and likely overestimate the negative impact of IEAT. Nevertheless, indirect evidence also supports that selecting AEAT reduces mortality in patients with severe infections (sepsis with or without septic shock). In patients with cUTI with low baseline risk of mortality (without sepsis), choice of empiric antibiotics might be more likely to impact clinical failure rather than mortality (as excess mortality is unlikely), but our systematic review of the literature could not estimate excess clinical failure due to wide variation in definitions across studies.

Consequently, the panel judged that the selection of empiric antibiotic therapy be initially guided by the severity of illness (specifically by stratifying whether the patient is in sepsis or not). In other words, severity of illness can be used to guide choice of empiric antibiotic therapy for cUTI.

More specifically patients with sepsis (especially with septic shock) receiving early AEAT (which may require broad-spectrum antibiotics) may prevent (or avoid excess) mortality, and stewardship considerations may be deferred to definitive therapy. In patients without sepsis, antibiotic stewardship considerations may favor choosing the narrowest spectrum agent likely to provide an effective therapy (even if appropriateness of the empiric therapy remains uncertain).

Implementation issues (taking it to the bedside)

Defining sepsis at the bedside can be clinically challenging. The patient's severity of illness should be determined using the established SOFA score when possible. However, data to calculate SOFA score may not be readily available in some clinical settings. Screening tools such as SIRS criteria or qSOFA score may assist with presumptive identification of possible sepsis, though they have limitations of sensitivity and specificity.[6,114,115]

Note that much of the present evidence base for treatment is based on older definitions of sepsis such as SIRS criteria. SIRS criteria as a screening tool are more sensitive than specific for sepsis and do not correlate as well with severity of illness or outcomes as the Sepsis-3 SOFA criteria.[115] As such, applying this corpus of evidence to patient care in patients identified as 'septic' today requires translation.

STEP 2: PATIENT-SPECIFIC RISK FACTORS FOR RESISTANT UROPATHOGENS (optimization of coverage)

In patients with cUTI (including pyelonephritis), should selection of empiric antibiotic therapy be guided by patient-specific prior urine culture results and patient-specific risk factors for resistant uropathogens to optimize selection)?

Recommendations:

I. In patients with complicated UTI (including acute pyelonephritis), we suggest avoiding antibiotics to which the patient has had a resistant pathogen isolated from the urine previously (*conditional recommendation, very low certainty of evidence*)

> **Remarks:**
> **-**More recent urine cultures may be a better guide than more distant urine cultures.
> **-**The time frame for paired cultures (urine samples collected from the same patient at different occasions) varied, but the median was 3-6 months.

II. In patients with complicated UTI (including acute pyelonephritis), we suggest avoiding fluoroquinolones if the patient has been exposed to that class of antibiotic in the past 12 months (*conditional recommendation, very low certainty of evidence*).

> **Remarks:** More recent antibiotic exposure may be a better guide than more distant antibiotic exposure.

In clinical practice, clinicians' choices of empiric antibiotic therapy for cUTI are often influenced by perceived risk factors for having a resistant pathogen, or by results from a prior urine culture. The aim of this section is to identify patient-specific risk factors for having resistant uropathogens.

STEP 2-A: Prior urine cultures as a guide to empiric antibiotic therapy

In absence of evidence assessing the direct impact of selecting antibiotics based on prior urine culture results on clinical outcomes, our literature search initially focused on studies evaluating prior urine cultures' impact on appropriateness of antibiotic therapy in patients being treated for a current episode of UTI.

Prior urine cultures' impact on appropriateness of empiric antibiotic therapy

First, we explored the question of whether information on prior urine culture(s) should guide selection of appropriate empiric therapy for the current episode of UTI. We found that when empiric antibiotic therapy for the current UTI was concordant with the prior urine culture microbiologic data, the likelihood that the treatment would be effective against the uropathogen increased by at least 7-fold.

Methods

We included studies that had been published between 2000 and present (2023) based on patients presenting with any type of UTI and from any geographic location. These studies

needed to report on the impact of prior urine culture on appropriateness of empiric antibiotic therapy in patients treated for a current episode of UTI.

Summary of evidence

A total of 2 observational retrospective studies reported on the impact of prior urine culture on appropriateness of empiric antibiotic therapy in patients treated for a current episode of UTI.[62,116] Linsenmeyer at al. (Linsenmeyer 2015) studied a total of 101 patients corresponding to 126 episodes of multidrug-resistant organisms UTIs from three VA medical centers (from 2010 to 2013).[62] In the 95 episodes of MDR UTI (resistant to three or more classes of antibiotics) in which a prior urine culture was available within two years (of which 73% within 6 months), the same pathogen was identified in 92% of episodes. Similarly, Almomani et al. performed a retrospective study (Almomani 2020) in 483 patients corresponding to 693 episodes of UTI with paired urine cultures within 12 months (median interval between paired isolates was 3 months); in this study the first urine culture had to have an ESBL-producing organism as an inclusion criterion.[116]

Concordance between empiric antibiotic therapy and prior microbiological data was defined slightly differently in the 2 studies: Lisenmeyer 2015 defined concordance if EAT was active against all isolated Gram-negative bacterial pathogens of the index episode,[62] while Almomani 2020 defined concordance if the patient received proper therapy as per guidelines and previous microbiological data (i.e., as per the in vitro susceptibility of empiric therapy in ESBL-UTI).[116] In both studies, when there were numerous previous cultures, the culture with a profile with the most resistance was used to determine the classification of concordant EAT. Both studies stratified the association for different time frames, while Lisenmeyer 2015 also provided a stratification per classes of antibiotics.

Estimate on prior urine cultures' impact on appropriateness of empiric antibiotic therapy

When the choice of empiric antibiotic therapy for the index (current) infection was concordant with the prior microbiologic data, the likelihood of appropriateness of treatment against the uropathogen was increased by at least 7-fold (low certainty in the evidence). In summary, these two studies suggest that a prior urine culture can be helpful to optimize the appropriateness of empiric antibiotic therapy. One of the 2 studies also showed that a shorter duration between index and prior cultures was significantly associated with increased chance of ESBL-positive result in the current culture.

Certainty in the evidence

The panel recognised that these estimates on the impact of using prior urine culture on appropriateness of empirical therapy are uncertain due to the high risk of bias due to study design and due to serious concerns regarding residual confounding. Furthermore, this relationship is indirect and can be influenced by other factors that are difficult to measure, such as local practices in antibiotic choice. Lastly, whether appropriate empiric therapy improves clinical outcomes also remains unclear.[35,111]

Other supporting evidence

Our literature search also focused on other lines of supporting evidence: (1) predictive values of prior urine cultures in patients with paired urine cultures, and (2) prior uropathogen resistance to a specific antibiotic as a risk factor for current resistance. Refer to Supplementary Materials for methods and more detailed results. A summary of results follows.

1) Predictive values of prior urine cultures for current susceptibility or resistance

A total of 4 observational studies reported on the diagnostic test accuracy of prior urine culture to predict susceptibility or resistance to various antibiotics in patients who had at least two urine cultures available for review (MacFadden 2014, Dickstein 2016, Vellinga 2010 and Valentine-King 2023).[117-120] The timeframe of the prior cultures varied from within four weeks to more than 32 weeks. All these studies compared susceptibility phenotypes in the paired urine cultures, while one also reported on the likelihood of identifying the same organism.[118] All included studies reported on positive predictive value (PPV) and negative predictive value (NPV). In the context of paired urine cultures, PPV refers to the probability of a prior resistant culture to accurately predict future resistance. NPV refers to the probability of a prior susceptible urine culture to accurately predict future susceptibility.

Overall, having a prior uropathogen that showed susceptibility to a given antibiotic was better at predicting current susceptibility than a prior uropathogen with resistance to a given antibiotic was at predicting current resistance. NPV was higher than PPV across these studies (**Supplementary Table B2B.1**, Supplemental Materials). When looking at the predictive values of urine culture collected within the last 12 months, NPVs were generally above 80% (with a median NPV of 95% for fluoroquinolone, 72% for 3rd gen cephalosporins, 86% for TMP-SMX, and 98% for carbapenems), while PPVs varied greatly, ranging between 40 to 85% (with a median PPV of 76% for fluoroquinolones, 56% for 3rd gen cephalosporins, 59% for TMP-SMX, and 48% for carbapenems). When stratifying for specific antibiotics, NPVs were higher than PPVs in most cases.

We identified four factors influencing PPV and NPV across studies: prevalence of resistance,[117-120] time since prior culture,[117,118,120] intervening negative culture,[117,118] and antibiotic exposure between cultures.[118] To summarize, a prior urine culture can predict the susceptibility of the organism in the present urine culture, with the caveat that higher baseline prevalence of resistance, longer time between cultures, an intervening negative culture, and intervening receipt of antibiotics diminish the predictive value of prior cultures.

2) Prior uropathogen resistance to a specific antibiotic as a risk factor for current resistance

Three observational studies (all from the United States) reported the predictive value of identifying uropathogen resistance to a specific antibiotic in prior urine culture, after adjusting for other risk factors of resistance.[121-123] Only 20-30% of the patients in these study populations had a prior urine culture.[121,122] These studies only evaluated prior resistance as a predictive factor (rather than both prior susceptibility and resistance). The timeframe of the prior cultures varied from within 12 months[123] to within 6 years.[121,122]

Across studies and after stratifying for specific antibiotics, having a prior resistant uropathogen was an independent risk factor for identifying a uropathogen resistant to the same antibiotic in a UTI patient's present urine culture (**Table 4.1**). More specifically, adjusted ORs ranged 5.5-12.8 for fluoroquinolones,[121,122] 4.7-8.6 for TMP-SMX,[121-123] and 21.7 for third generation

cephalosporins.[122] Interestingly, two of these studies also showed that having more than one prior culture with the same resistance further increased the odds of identifying a uropathogen resistant to the same antibiotic.[121,122]

Table 4.1: Estimates of prior uropathogen resistance as a risk factor for current resistance

Antibiotics	**aORs of resistance (range)**	**Interval between cultures**	**Prevalence of resistance**
Fluoroquinolones[121,122]	-If one prior culture Cipro-R: 5.51 (3.33-9.16) to 12.8 (8.5-19.0) -If 2 or more prior culture Cipro-R: 6.1 (2.73-14.08) to 28.4 (13.2-60.7)	Up to 6 years	Cipro-R: 10.3% to 19.1%
Third generation cephalosporins [122]	-If one prior culture C3-R: 21.7 (7-69.2) -If 2 or more prior culture C3: 32.5 (5.06-126.4)	Up to 6 years	C3-R 6.9%
TMP/SMX[121-123]	-If one prior culture in the last 12 months TMP/SMX-R: 8.58 (3.92-18.81)	Last 12 months	TMP/SMX-R 20.3%
	-If one prior culture TMP/SMX-R: 4.7 (3.5-6.5) to 4.78 (2.87-8.07) -If 2 or more prior culture TMP/SMX-R: 5.4 (3.1-9.4) to 6.66 (2.85-17)	Up to 6 years	TMP/SMX-R 19.4%to 25.6%
aOR: adjusted odds ratio; R: resistant; Cipro: ciprofloxacin; C3: third generation cephalosporins; TMP/SMX: trimethoprim/sulfamethoxazole			
Limitations: For studies reporting on the predictive value of prior uropathogen resistance as a risk factor, many of the patient did not have a prior urine culture. The subpopulation of patients who had a prior culture (with or without a specific resistance) might be different from the rest of the cohort who did not have a prior urine culture (i.e. population with recurrent UTI vs first episode of UTI), which could have introduced a selection bias. Furthermore, most studies were aiming at creating pragmatic algorithms which included and adjusted for a limited number of risk factors in analyses, making these estimates more prone to residual confounding.			

Other considerations

The paired culture studies showed that NPV (susceptibility in the first culture) was predictive of susceptibility, much more strongly than PPV (resistance in the first culture) was predictive of resistance. However, the larger body of literature looked at risk factors from the patient's past that predict current resistance, not susceptibility. When a clinician is choosing an antibiotic, the cognitive choice is most commonly structured as which antibiotics to avoid because of potential lack of effectiveness. Therefore, these recommendations concerning use of prior cultures to guide therapy are structured around prior resistance, to make them more feasible to implement.

Rationale for the recommendation

Based on this evidence, the panel suggests avoiding antibiotics to which the patient has had a resistant pathogen isolated from the urine previously, but also recognized that other factors may influence this decision; for example, a recent culture may provide stronger evidence to support this decision, particularly when results are discordant between more distant cultures and a more recent culture. We did not find literature on predictive value of prior blood culture for current urine culture susceptibility profile, but the panel believes that susceptibility could be extrapolated from a prior blood culture collected during a cUTI.

Implementation issues (taking it to the bedside)

Many patients visit more than one healthcare system. Therefore, prior urine culture reports may not be available to the clinician treating the current episode of cUTI. In some cases, a discharge summary or verbal report upon transfer of the patient to the emergency department may simply note that the patient had a prior ESBL producing organism, without providing specific culture results. Such information can probably be given the same weight as a prior urine culture with an ESBL-producing organism. Verbal reports of "ESBL positivity" or "KPC" can be a clue to the likely pattern of resistance to multiple antibiotics. Urine cultures in real-life scenarios can have multiple organisms or contaminants, potentially obscuring the true pathogen. Patients may have serial urine cultures that show different organisms with different resistance patterns. In such cases the overall resistance phenotype, or conversely, the overall susceptibility phenotype, is probably a reasonable guide for empiric antibiotic choice. Unfortunately, if the prior organisms have been treated with antibiotics, the possibility of a new organism emerging is real. Some patients, due to prior antibiotic exposure, may develop infections with less common pathogens (e.g., *Pseudomonas aeruginosa* or *Enterococcus spp.*), though predicting which patients will develop UTI with these pathogens is beyond the scope of these guidelines. Patients with indwelling urinary catheters (transurethral or suprapubic) and/or neurogenic bladders have often been heavily treated for Gram-negative urinary organisms in the past and are likely to be carrying resistant Gram-negative organisms or non-Enterobacterales organisms.[124]

STEP 2-B: Risk factors of resistance to a specific antibiotic class as a guide to empiric antibiotic therapy

To identify patient-specific risk factors for resistant uropathogens that could help guide the selection of appropriate empiric therapy, we performed a systematic review of the literature aiming at identifying the strongest predictor(s) of specific resistance(s) or resistance pattern(s), such as FQ, ESBL or TMP-SMX.

Methods

Our criteria for inclusion were that the study had to have been published between 2000 and present (2023) and reported on North American populations (United States, Canada, and Mexico), as risk factors of antibiotic resistance will vary depending on the local epidemiology. Included studies had to report on adults with cUTI, meaning that studies that were based on laboratory data only (i.e. without a confirmed clinical diagnosis of UTI) were excluded. Many of the studies also included adults with uncomplicated UTI, but the study was included as long as

some of the patients had been diagnosed with cUTI. Studies needed to report on risk factors for resistance among common Gram-negative uropathogens. Finally, studies meeting these criteria were included only if they reported adjusted relative risks using a multivariate analysis.

For selected risk factors to be useful in a clinical practice (i.e. develop actionable recommendations based on these risk factors), these needed to be **specific to an antibiotic class (e.g. FQ-R, ESBL and TMP/SMX-R)** rather than general (**e.g. MDR**). For example, if residing in a nursing home is associated with an increased risk of resistant uropathogens but not specifically to an antibiotic class, then this risk factor is descriptive and too general to recommend avoiding a specific antibiotic class when selecting empiric antibiotic therapy. Many factors (e.g. nursing home residence, presence of indwelling urinary catheters) are associated with risk of having a multidrug resistant organism. These factors may all relate to the common pathway of healthcare exposure and thus more antibiotic exposure. Nevertheless, knowing that the patient is at higher risk for having a multidrug resistant organism does not guide the clinician at the moment of choosing specific empiric antibiotics.

Due to the expected limitations intrinsic to the identified literature (see below), the panel judged that only risk factors that were **strong predictors** should be reported in our final analysis. We defined strong predictor as an independent risk factor (reported in at least 2 included studies) with an adjusted odds ratio (aOR) consistently greater than five across the included studies. We chose this threshold as one at which knowing a risk factor was likely to meaningfully alter post-test probabilities of resistance to a specific antibiotic and usefully inform antibiotic selection. All risk factors reported as independently associated with resistance to a specific antibiotic were considered for further analysis (see Supplemental Materials for additional details).

Summary of evidence

Our systematic review of the literature identified a total of 16 observational studies reporting on the various risk factors for having a urinary pathogen resistant to a specific antibiotic class in patients with cUTI.[39,123,125-138] Of these risk factors, only one was considered a strong predictor for a resistance to a specific antibiotic class: prior exposure to the same antibiotic (fluoroquinolones) in the last 12 months (**Table 5.1**).

Table 5.1: Independent risk factors of resistance to a specific antibiotic class

Risk factors	aORs of resistance	Timing	References	Strength of association
Risk factors of fluoroquinolones resistance				
Prior exposure to fluoroquinolones (in the last year)	4.62 (1.09-19.61) 15.73 (6.15-40.26) 30.35 (5.82-158.42)	Prior month	Khawcharoenporn 2012[130] Rattanaumpawan 2010[133] Killgore 2004[131]	Strong Predictor
	23.35 (8.20-76.85)	Prior 3 months	Shah 2017[135]	
	21.8 (3.7 – 127.1)	Prior 6 months	Cohen 2006[126]	
	7.6 (2.1-27.5) 13.16 (3.11-68.43)	Prior 12 months	Johnson 2008[39] Shah 2017[135]	
	1.95 (1.66 – 2.28)	Unclear	Rich 2022[134]	
Healthcare exposure (hospital or nursing	Nursing home 1.93 (1.22 – 3.07)	Current	Faine 2022	Weak Predictor

home in the last 3 months)	2.80 (1.02-7.25) 4.41 (1.79-10.88)	Current Current	Shah 2017[135] Rattanaumpawan 2010[133]	
	Hospitalisation 2.0 (1.0-3.9) 2.19 (1.31-3.64) 3.99 (2.38-16.30) 0.97 (0.87 – 1.09)	Each prior week of hospitalisation Prior 2 weeks Prior 3 months Past year	Johnson 2008[39] Kratochwill 2015 Rattanaumpawan 2010[133] Rich 2022[134]	
	Nosocomial 2.56 (1.31-5.02)	Prior 3 months	Khawcharoenporn 2012[130]	
Risk factors of resistance to TMP/SMX				
Prior exposure to TMP/SMX	2.36 (1.94-2.88) 2.58 (1.13-5.89)	Unclear Prior 12 months	Rich 2022[134] DeMarsh 2020[123]	Weak Predictor
aOR: adjusted odds ratio Strong predictor: independent risk factor (reported in at least 2 included studies) with an aOR consistently greater than five across the included studies; Weak predictor: independent risk factor (reported in at least 2 included studies) with an adjusted odds ratio aOR not consistently greater than five across the included studies				

Effect of prior fluoroquinolone (FQ) exposure on risk of FQ resistance

A total of 7 observational studies reported on prior fluoroquinolone use as an independent risk factor for identifying fluoroquinolone resistant uropathogens in patients with cUTI (Killgore 2004, Cohen 2006, Johnson 2008, Rattanaumpawan 2010, Khawcharoenporn 2012, Shah 2017, and Rich 2022).[39,126,130,131,133-135] Data collection in these published studies occurred between 1998 to 2019. The risk factor studied was fluoroquinolone exposure versus no prior fluoroquinolone exposure. Unfortunately, the time frame for prior exposure varied from 1 week to 12 months, and the nature of the exposure was not defined further in terms of dose or duration. See Supplemental Materials for additional details on the studied populations.

Overall, the adjusted odds ratio for prior FQ exposure ranged from 2.0 (1.7 – 2.3)[134] to 30.4 (5.8 – 158.4)[39,126,130,131,133-135] The pooled, adjusted odds ratio was 13.7 (8.4-22.4) (**Table 6.1**).[131] A dose response gradient was seen within one study that looked at two different time periods for fluoroquinolone exposure (prior 3 months: 23.4 (8.2-76.9) vs prior 12 months: 13.2 (3.1-68.4).[135] The literature suggested a strong relationship between prior fluoroquinolone exposure and having a fluoroquinolone-resistant organism when presenting with cUTI.

Table 6.1: Impact of prior fluoroquinolone exposure on the adjusted odds ratio for fluoroquinolone resistance UTI

Time frame for exposure	Adjusted odds ratio (aOR)	References
Prior month	4.6 (1.1-19.6) 15.7 (6.2-40.3) 30.4 (5.8-158 .4)	Khawcharoenporn 2012,[130] Rattanaumpawan 2010,[133] Killgore 2004[131]
Prior 3 months	23.4 (8.2-76.9)	Shah 2017[135]
Prior 6 months	21.8 (3.7-127.1)	Cohen 2006[126]
Prior 12 months	7.6 (2.1-27.5) to 13.2 (3.1-68.4)	Johnson 2008,[39] Shah 2017[135]

Other weak predictors of resistance to specific antibiotics

The literature also suggested a weak relationship between prior exposure to TMP/SMX and having an organism resistant to TMP/SMX in the current episode of cUTI.[123,134] However, TMP/SMX is not a preferred agent for initial empiric treatment of cUTI. We did not find any other predictive relationships between prior exposure to specific classes of antibiotics and having an organism resistant to those classes of antibiotics in a subsequent cUTI. The resistance risk factor literature may have had a bias towards detecting fluoroquinolone resistance, as more studies included fluoroquinolone exposure as a potential risk factor than to other antibiotic classes, and because fluoroquinolone resistance is more prevalent than resistance to some other classes (e.g. carbapenems).

Healthcare exposure to a hospital or nursing home within the prior three months was identified as a weak predictor of having a urinary organism resistant to fluoroquinolones.[39,127,130,132-135]However, healthcare exposure is likely linked to other risk factors for having a resistant organism (e.g. prior antibiotic exposure, comorbidities, presence of indwelling devices, and procedures).

Our systematic review of the literature (focused on studies from North America) did not identify an association between exposure to third generation cephalosporins and having an ESBL-producing uropathogen.[125,127-129,136-138]

Certainty in the evidence

The panel recognized that these estimates on the relative risk of prior exposure of fluoroquinolones on current fluoroquinolone resistance in patients with cUTI are very uncertain due to the high risk of bias due to study design as well as potential residual confounding. Interestingly, a dose-response gradient (i.e. an incremental increase in the risk of fluoroquinolone resistance with more recent exposure to fluoroquinolones) was observed in one study, which supports the biologic plausibility of prior fluoroquinolone exposure as a true risk factor. Nevertheless, our certainty in the evidence remains very low.

Of note, the value of a predictive factor depends on the baseline resistance rate, thus in a setting in which the prevalence of resistance to a specific antibiotic class is very low or very high, these predictors would not be as discriminative. Thus, these results seem generalizable to current North American practice but may vary with local epidemiology.

Rationale for recommendation

Based on this evidence, the panel suggests avoiding fluoroquinolones if the patient has been exposed to that class of antibiotic in the past 12 months, but also recognized that recent exposure (e.g. less than 3 months) may provide stronger evidence to support this decision.

Implementation issues (taking it to the bedside)

The treating clinician who has just diagnosed a patient with cUTI may not have information about the patient's prior antibiotic exposures, particularly if the patient visits more than one healthcare system or is coming from a long-term care facility. Even if the patient is within the same healthcare system, searching the electronic health record for prior antibiotics

can be time-consuming. Patients and their caregivers may not be able to identify specific antibiotics or the timeframe during which they were prescribed. Another important consideration is that the recommendation to consider fluoroquinolone exposure as a risk factor is drawn from literature in which the prevalence of fluoroquinolone resistance ranged from 10 to 45%. If local prevalence of fluoroquinolone resistance among urinary *E. coli* is higher than 45%, then the absence of prior exposure to fluoroquinolones cannot be taken as evidence that the patient's current urinary organism will be susceptible to fluoroquinolones.

STEP 3: OTHER PATIENT-SPECIFIC CONSIDERATIONS (prevention of possible undesirable events)

In patients with cUTI (including pyelonephritis), should selection of empiric antibiotic therapy be further guided by patient-specific considerations?

Recommendation:

I. In patients suspected of cUTI, empiric antibiotic therapy selection should account for patient-specific considerations (e.g. risk of allergic reaction, contraindications, or drug-drug interactions) to avoid preventable adverse events (*good practice statement*).

ΩzGuidelines cannot fully capture individual considerations. As a best practice, clinicians should take into account an individual patient's specific considerations. Such considerations should consider the patient's antibiotic allergy history, contraindications to receiving specific antibiotics, and potential drugs-drug interactions when selecting among the preferred and/or alternative options for empiric antibiotic therapy. Common sense suggests avoiding an antibiotic which the patient was taking when the current symptoms developed, or one which the patient took very recently.

STEP 4: ANTIBIOGRAM (tailoring empiric antibiotic therapy in septic patients)

In patients with cUTI (including pyelonephritis), should selection of empiric antibiotic therapy be further tailored by consulting an antibiogram?

Recommendations:

I. In patients with **sepsis** assumed to be caused by complicated UTI (including acute pyelonephritis), we suggest using an antibiogram to further tailor empiric antibiotic choice **only if** the antibiogram is local, recent, and relevant to the patient (*conditional recommendation, very low certainty of evidence*).

Remarks:

-An antibiogram is considered local if derived from the same healthcare facility, recent if based on data from the prior 12 months and relevant to the patient if based on organisms from a similar patient population.

-If an antibiogram is being used to further tailor empirical antibiotic choice, consider selecting an antibiotic for which 90% or more of the most relevant organism(s) are susceptible in patients in septic shock, or for which 80% or more of the most relevant organism(s) are susceptible in patients with sepsis without shock. These cutoffs are

based on modeling of increased mortality risk associated with inappropriate empiric antibiotics in sepsis and septic shock.
-Septic shock is defined by the Sepsis-3 Task Force as a subset of sepsis in which despite volume resuscitation, vasopressors are required to maintain blood pressure and serum lactate level is greater than 2 mmol/L, reflecting an in-hospital mortality greater than 40%.[6,99]

II. For patients with suspected complicated UTI without sepsis (including acute pyelonephritis), we make no specific recommendation about using an antibiogram to further tailor empiric antibiotic choice (*no recommendation, knowledge gap*).

Remarks:
-Patients who are not septic have a lower risk of mortality from cUTI (less than or equal to 5%) and initial inappropriate empiric antibiotic choice has little impact on mortality. Routine use of broader-spectrum agents in suspected complicated UTI without sepsis may drive antimicrobial resistance without substantial patient benefit.

Based on the review of the literature, the panel identified a lack of evidence about applicability of antibiograms to individual patients, particularly those with cUTI, and whether use of an antibiogram improves clinical outcomes. Therefore, modeling was performed to determine the impact of applying an antibiogram-based threshold for predicted susceptibility of a uropathogen to an empiric antibiotic choice, with the outcome of the modeling being excess mortality due to IEAT.

The panel judged that the use of an antibiogram for choosing empirical therapy for an individual patient with cUTI should be restricted to patients with sepsis for the following reasons: (1) The antibiogram's value in guiding antibiotic choice for an individual patient is unproven, particularly when the organism itself is not known. (2) Modeling for use of the antibiogram to avoid excess mortality due to inappropriate empiric antibiotic therapy (IEAT) was relevant only to patients with an expected mortality of 10% or higher. (3) Antibiograms need to meet certain standards to be useful, and the general practitioner is unlikely to know how the facility's antibiogram was constructed.

Modeling for impact of using antibiogram thresholds on excess mortality

Our analysis of eight studies that compared mortality outcomes with IEAT and AEAT found a pooled odds ratio of 1.56 increased mortality with IEAT (see IEAT section above). This estimate of 1.56 is likely an overestimate because studies that did not find IEAT to be a significant factor in mortality were not included in our modeling. The panel judged that a reasonable susceptibility threshold for choosing an antibiotic to use for a patient with cUTI and sepsis or septic shock should confer no more than a 1% risk of excess mortality due to IEAT. In other words, the panel judged that the tolerable level of excess mortality risk should be less than one per 100 patients treated. Using this odds ratio of 1.56, the threshold for avoiding 1% increased mortality in septic shock is to choose an antibiotic for which at least 90% of the relevant organism are predicted to be susceptible. If the patient is septic but not in shock, the threshold to avoid a 1% increase in mortality is to choose an antibiotic for which at least 80% of the relevant organism are predicted to be susceptible. See **Table S7** in the supplemental materials.

For patients with cUTI with a low baseline mortality (5%), the choice of empiric antibiotics might be influenced more by clinical failure than mortality (as excess mortality is so unlikely). Unfortunately, we judged that modeling for excess clinical failure would not be sound

because both the definition of clinical failure and the study populations varied substantially across studies. For example, one trial included mortality in the composite outcome of clinical failure, while another included microbiologic persistence in the outcome of clinical failure.

Other considerations when using an antibiogram to guide empiric antibiotic choice in individual patients

While patient-specific risk factors help define if a given patient may have a urinary organism that is resistant to one or more antibiotics, the local antibiogram may not necessarily apply to an individual patient.[139] For example, if a patient has recently been living in a different state, or recently traveled extensively in another country, the local hospital's antibiogram may not apply. If the patient is undergoing chemotherapy and has been hospitalized multiple times recently, the outpatient antibiogram may not be relevant, even if the patient is presenting for care in an outpatient setting. Antibiograms derived from hospitalized patients may not be relevant to primary care settings. Application of an antibiogram at the point of care can be difficult when the causative organism is not known. In a patient with recurrent infections or another recent infection, the patient's microbiological history and antibiotic exposure may provide more relevant information than a facility-wide antibiogram.[139] As an illustration of these concerns, a recent study in 127 VA facilities involving 2.2 million isolates of *E. coli* and *Klebsiella* species in 1 million patients found that a hospital-level antibiogram had limited ability to predict whether the patient's organisms would be resistant to specific antibiotics. [140] Another caveat is that evidence is lacking about whether use of an antibiogram in making empiric antibiotic choices improves clinical outcomes.[97]

Quality standards for antibiograms

The Clinical and Laboratory Standards Institute (CLSI) establishes standards for analysis and presentation of cumulative antibiotic susceptibility data.[141] Antibiogram data should be updated at least annually, and that only information from diagnostic cultures should be presented (rather than from surveillance cultures). At least 30 isolates of a species should be available during the time period covered if the organism will be presented in the antibiogram. Going beyond the CLSI basic standards, many facilities create antibiograms that are stratified by patient characteristics, such as sex and age, and/or are location-specific, such as antibiograms for the emergency department or intensive care unit.[139] The 2016 IDSA guidelines on implementing antimicrobial stewardship programs recommend developing a stratified antibiogram (by location, age, etc.). However, this recommendation is based on the idea that stratified antibiograms can expose important differences in susceptibility to guide facility-wide antibiotic recommendations rather than necessarily improving empiric antibiotic choice for individual patients.[97] We anticipate that the expansion of individualized risk prediction models may eventually supplant the use of facility-wide antibiogram to guide empiric antibiotic choice for a specific patient.[139]

Rationale for recommendation

Despite all the challenges in application of an antibiogram to an individual patient, the panel judged that a local, recent, and relevant antibiogram might help avoid excess mortality if used to guide empiric antibiotic choice for cUTI patients with sepsis. In patients with septic shock, the threshold for avoiding a 1% increase in mortality due to inappropriate empiric antibiotic therapy is to choose an antibiotic for which at least 90% of the isolates of the relevant organism are predicted to be susceptible. If the patient is septic but not in shock, the threshold to avoid a 1% increase in mortality is to choose an antibiotic for which at least 80% of the

isolates of relevant organism are predicted to be susceptible. These thresholds were established based on the current literature on inappropriate empiric antibiotic therapy and mortality, which yielded an odds ratio of 1.56 (95% CI of 0.99-2.46), indicating uncertainty related to the imprecision of this estimate. Newer literature and changes in this estimate may require recalculation of the thresholds; these guidelines establish a method for doing such calculations. (See **Table S7** Supplemental Materials). The increased mortality threshold judged by the panel as the acceptable upper limit (1% excess mortality, or 1 per 100 patients) was selected by consensus and is not informed by direct research on patient or clinician values. The applicability of these thresholds to real-world antibiotic appropriateness and their impact on clinical outcomes remains to be established and would be a suitable topic for future research.

For patients with suspected complicated UTI without sepsis (including acute pyelonephritis), the panel does not make a specific recommendation about using an antibiogram to further tailor empiric antibiotic choice. In patients whose risk of mortality from cUTI is less than or equal to 5%, initial inappropriate empiric antibiotic choice may delay recovery, lengthen hospitalization, or reduce the likelihood of clinical cure but has little predicted impact on mortality through our modeling.

Implementation issues (taking it to the bedside)

To guide antibiotic choice for a patient with sepsis from cUTI, the clinician would ideally be able to use a locally developed antibiogram that addresses the relevant patient population (e.g. ICU) and/or has a focus on urinary pathogens. Determining which organism to focus on in the antibiogram represents an additional challenge. The most relevant organism is suggested by the prior urine culture, if available. For example, if the prior urine culture had an ESBL-producing *Klebsiella* species, consult the antibiogram for effective therapy for ESBL-producing *Klebsiella*. In the absence of prior culture information, the panel suggests defaulting to the susceptibilities of *E. coli* when making an empiric choice of antibiotics for cUTI. The advent of rapid tests for antibiotic susceptibilities directly from urine may greatly reduce the role of the antibiogram in empiric antibiotic choice.

These guidelines are designed to avoid excess mortality through inappropriate empiric antibiotic therapy, but not to provide 100% coverage of all organisms that could possibly grow from the urine of a patient with cUTI or bacteremia of a urinary source. The healthcare clinician should apply individual judgment to considerations of whether the patient is likely to have a non-Enterobacterales organism (e.g. *Enterococcus spp.* or *Pseudomonas aeruginosa*) or a highly resistant urinary pathogen based on prior cultures and antibiotic exposures.

C. Selection of Definitive Antibiotic Therapy for Complicated UTI

In patients with microbiologically confirmed cUTI, should definitive effective antibiotic therapy be targeted based on the results of urine culture rather than continuing empiric broad-spectrum antibiotics?

Recommendation:

I. In patients with confirmed complicated UTI, we suggest selecting a definitive effective antibiotic with a targeted spectrum based on the results of urine culture (identification and susceptibility) as soon as these are available, rather than continuing empiric broad-spectrum antibiotics for the complete duration of treatment (*conditional recommendation, low certainty of the evidence*).

> **Comment:**
> -This recommendation places a high value on de-escalating antibiotic therapy based on culture results (stewardship considerations) while optimizing the effectiveness of therapy (improving clinical cure and reducing recurrence of infection). De-escalation may be less practical in cases of cUTI managed in the outpatient setting.

Balance of Benefits and Harms

Assuming that the balance of benefits (clinical cure, recurrence of infection) and harms (serious and non-serious events) as well as other considerations regarding costs/resources between targeted spectrum and broad spectrum-antibiotics are equivalent, then the remaining considerations mainly relate to stewardship issues.

Other considerations and stewardship issues

As mentioned above, the selection of definitive therapy should involve a consideration of the appropriate route of therapy (oral versus intravenous), the costs of different antibiotic treatment options, and the resources required to administer various antibiotic regimens (e.g. two IV antibiotics can cost the same per dose, but one may require administration three times per day, while the other is given once daily). Some antibiotics require therapeutic drug monitoring or pharmacist involvement to adjust dosing. Burden on patients and hospital staff are additional considerations. The antibiotic stewardship goal is usually to choose a more targeted antibiotic with the fewest collateral effects on the intestinal microbiota, when the susceptibility profile supports such a choice.

De-escalation (switching from broad to narrow agents) may not be practical in some situations, particularly when a patient with cUTI has been discharged with oral antibiotics to complete their treatment course. For example, the benefit of changing an oral third generation cephalosporin to a narrow-spectrum, first-generation cephalosporin for a few days of remaining therapy may be marginal. Also, calling patients to make this switch would require additional effort from the healthcare team, could confuse and inconvenience the patient, and might leave the patient with leftover antibiotics at home.

Rationale for recommendation

In patients with confirmed complicated UTI, we suggest selecting a definitive effective antibiotic with a targeted spectrum based on the results of urine culture as soon as these are available, rather than continuing empiric broad-spectrum antibiotics for the complete duration of treatment (conditional recommendation, low certainty of the evidence). This recommendation places a high value on deescalating antibiotic therapy based on culture results (stewardship considerations) while optimizing the effectiveness of therapy (improving clinical cure and reducing recurrence of infection). Whether a few days of a more targeted agent in comparison to a broad-spectrum regimen confer individual or global benefits in terms of antibiotic resistance is unclear, but tailoring the antibiotic regimen is a cornerstone principle of antibiotic stewardship.[142]

Implementation issues (taking it to the bedside)

Following up on the urine culture results is particularly important for patients managed for cUTI in the outpatient setting, to confirm susceptibility and switch to an effective antibiotic if needed.[143] A patient who has been diagnosed with and treated for cUTI may subsequently have a negative urine culture. When the clinician lacks culture evidence to guide antibiotic tailoring, the patient's clinical picture can be a guide. Patients who are improving on broad-spectrum, intravenous antibiotic therapy may be good candidates for transitioning to a more focused treatment regimen and for consideration of switch to an oral agent if resistant pathogens are not isolated.

Conclusions and research needs

Conclusions

Many of the classes of antibiotics approved to treat cUTI demonstrate similar efficacy as other classes of antibiotics in randomized, controlled trials. Bacterial resistance prevalence is always changing, and new antibiotics will be developed. In this context, the panel developed a four-step process for choosing empiric antibiotic therapy for cUTI that will potentially outlast the table of specific antibiotic choices. These steps are: 1) assess the severity of illness (for initial prioritization of antibiotics), 2) consider patient-specific risk factors for resistant uropathogens (for optimization of coverage), 3) evaluate other patient-specific considerations (to reduce the risk of adverse events), and 4) for patients with sepsis, consult a relevant local antibiogram if available (to further improve the likelihood of giving appropriate empiric therapy). The panel also developed a modeling approach to predict the impact of inappropriate empiric antibiotic therapy on mortality in cUTI that can be updated with new data as this emerges. However, the panel believes that the clinical principles of assessing severity of illness, considering the patient's individual risk factors, and evaluating which drug may suit the patient best will continue to be relevant to guiding treatment of cUTI.

Research needs

Whether or not microbiologic cure is a patient-relevant outcome is unclear. For the patients that have clinical cure but microbiologic failure (persistence of greater than 10^4 count-forming-unites/ml of bacteria in the urine at the test of cure time point), we know that such patients are at higher risk of clinical UTI by long term follow up. These patients presumably have asymptomatic bacteriuria at the TOC time point, since their clinical symptoms have resolved. What we do not know is whether treating the bacteriuria will reduce their risk of subsequent UTI.

To address this issue, a randomized, placebo-controlled, blinded trial would need to enroll patients who have been treated for cUTI/AP and who have clinical cure but microbiologic failure, and to randomize them to antibiotic treatment versus no treatment. Ideally the follow up period for recurrent UTI would be extended beyond 30 days after the last dose of antibiotics, out to one year, as antibiotic treatment could paradoxically increase the risk of recurrent UTI (rUTI) by disrupting the microbiome. Ideally this study would also perform molecular typing of the urinary organisms to see if rUTI is caused by the same or different organisms.

Little attention is given in the literature to determining whether men with febrile UTI have acute prostatitis. A systematic study of the signs and symptoms of prostatic involvement in men with UTI could help better define prostatitis as a clinical entity and thus facilitate detection of this condition in cUTI trials.

Another key research need is how to manage UTI in transgender adults. There is insufficient research on transgender and gender diverse individuals to inform guidelines and an urgent need for more research in this area. Nearly 1.6 million people ages 13 and up in the United States identify as transgender or have gender diverse experience. No clinical data exists to describe the epidemiology and risk factors for UTI in transgender and gender diverse individuals. We do not know if gender affirming therapy with hormonal treatment increases or decreases the risk of UTI. In addition to these epidemiology studies, future clinical trials should include gender-diverse samples and collect appropriately inclusive gender demographics on participants.[144]

The 2010 UTI guidelines on cystitis and pyelonephritis recommended one dose of aminoglycosides (or ceftriaxone) in patients with acute pyelonephritis who would otherwise be treated with entirely oral therapy, if there was concern for possibly having an organism resistant to the oral agent. This strategy should be explored in a RCT comparing a single dose or short course of aminoglycoside to placebo at the start of outpatient therapy for cUTI.

Finally, we need more data to define the role of the antibiogram in choosing empiric therapy for cUTI. Ideally this topic would be addressed in an RCT in which clinicians made empiric choices guided by an antibiogram or not guided by an antibiogram. The question of whether an antibiogram is relevant and helpful to choosing an empiric antibiotic treatment regimen for an individual patient with cUTI is an open one. The modeling strategy used in determining the antibiogram susceptibility thresholds to guide empiric antibiotic choice in cUTI patients with sepsis could be tested in a real-world study to determine the value of such thresholds at improving clinical outcomes. Research on how to use rapid molecular testing to identify urinary organisms and susceptibility at the point of care is needed, as injudicious use of such tests may drive overtreatment and overly broad antibiotic use.

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Clinical Practice Guideline by Infectious Diseases Society of America (IDSA): 2025 Guideline on Management and Treatment of Complicated Urinary Tract Infections: Timing of Intravenous to Oral Antibiotics Transition for Complicated UTI

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In patients who are being treated parenterally for cUTI, are clinically improving, can take an oral medication and for whom an oral option is available, should parenteral therapy be transitioned to oral rather than continued for the complete duration of therapy?

Recommendations

I. In patients with complicated UTI (including acute pyelonephritis) treated initially with parenteral therapy who are clinically improving, able to take oral medication, and for whom an effective oral option is available, we suggest transitioning to oral antibiotics rather than continuing parenteral therapy for the remaining treatment duration (*conditional recommendation, low certainty of the evidence*)

Comments:

- This recommendation places a high value on reducing avoidable intravenous catheter-related adverse events, costs, and resources, as well as taking into account practical aspects of antibiotic administration.

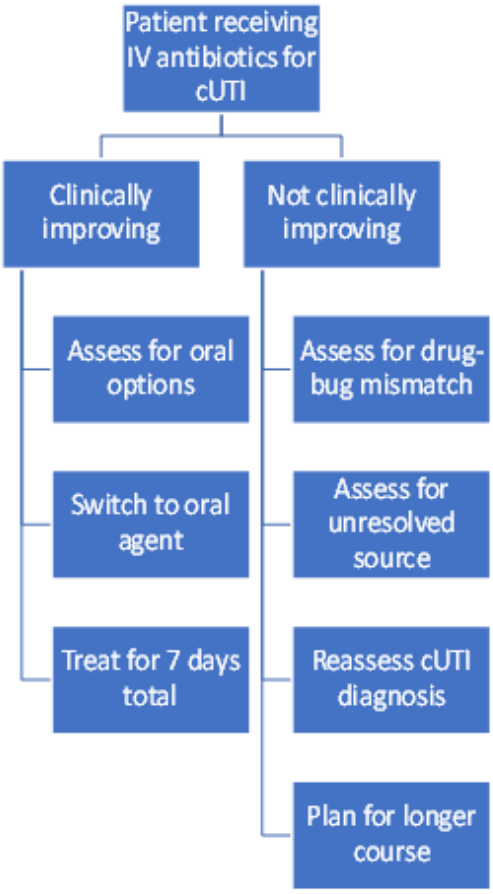
- The trials supporting this recommendation mostly excluded patients with indwelling urinary catheters, sepsis or septic shock, immunocompromised states, severe renal insufficiency, and functional or structural abnormalities of the urinary tract. Some patients in these subpopulations may need an individualized plan of therapy.
- An effective antimicrobial agent means that the antibiotic achieves therapeutic levels in the urine and relevant tissue and is active against the causative pathogen.
- Refer to **Figure 1.2 for a stepwise assessment of the intravenous to oral switch and the duration of antibiotic therapy.**

II. In patients presenting with complicated UTI (including acute pyelonephritis) and associated Gram-negative bacteremia treated initially with parenteral therapy who are clinically improving, able to take oral medication, and for whom an effective oral option is available, we suggest transitioning to oral antibiotics rather than continuing parenteral therapy for the remaining treatment duration (conditional recommendation, very low certainty of the evidence).

Comments*:*

- The trials supporting this recommendation mostly included patients who were afebrile, hemodynamically stable, and had achieved source control (relief of any urinary obstruction) before transitioning to oral antibiotics.
- An effective antimicrobial agent for bacteremic patients means that the antibiotic achieves therapeutic levels in the bloodstream, urine, and relevant tissue and is active against the causative pathogen.
- Refer to **Figure 1.2** for a stepwise assessment of the intravenous to oral switch and the duration of antibiotic therapy.

Figure 1.2: Stepwise assessment of IV to oral switch and duration of antibiotic therapy



Abbreviations: IV=intravenous, cUTI=complicated UTI. Drug-bug mismatch means that the causative organism is not susceptible to the antibiotic prescribed.

Introduction

Antibiotic therapy is typically given for the shortest effective duration, and administered orally rather than intravenously when appropriate, in order to minimize adverse events related to therapy. An increasing number of clinical trials support early IV treatment with transition to oral therapy for infectious syndromes.[1,2] From a pharmacological point of view, antibiotic efficacy depends on the levels of the antibiotic obtained in serum and tissue, not the route of administration.[2] In practice, providers often switch from intravenous to oral antibiotics during the course of therapy for complicated UTI, once a patient is improving. We sought evidence about whether such transition impacts clinical outcomes of cUTI treatment versus continuing intravenous therapy. We also looked for information from clinical trials to guide the timing of this switch and appropriate circumstances for switching to the oral route.

From a practical perspective, an IV to oral switch, if equivalent in terms of outcomes, would be desirable because switching can reduce the need for intravenous access, complications from intravenous devices, nursing time and effort to administer the medication, volume of fluid and sodium given to the patient, duration of hospitalization, healthcare costs, and inconvenience to the patient.

Some patients with cUTI can be managed entirely with oral antibiotics in the outpatient setting. Please see a discussion of this topic in clinical question 1, under the section on "**Oral antibiotics for cUTI.**" Also see **Table 1.2** Dosing of oral antibiotics for complicated UTI.

Summary of the evidence

Our systematic review of the literature, from January 2000 up to September 2024, identified four randomized, controlled trials (RCT) comparing transitioning to oral therapy to continuing parenteral therapy for the total duration of antimicrobial therapy for adults with complicated UTI (So-Ngern 2023, Monmaturapoj 2012, Malaisri 2017, and Concia 2006).[3-6]

Studied population: These four trials included 186 adult inpatients and outpatients from Asia and Europe, mostly female (82%). Although all enrolled patients had a presumptive or confirmed diagnosis of cUTI, the definition of cUTI varied between trials. Three trials included only patients with acute pyelonephritis,[3,4] while the third included only patients with cUTI associated with confirmed or presumed sepsis.[5] The Malaisri 2017 trial excluded bacteremic patients with UTI,[3] while bacteremia was present in 32% (n=15) of the patients in Concia 2006[5], 21% (n=17) of the patients in Monmaturapoj 2012[4], and 14% (n=3) of the patients in So-Ngern 2023.[6] The Malaisri 2017 trial[3] and So-Ngerm 2023[6] only included patients with UTI caused by extended spectrum beta lactamase (ESBL)-producing organisms, while the 2 other trials included UTI caused by all significant uropathogens.[4,5] The oral agents studied were fluoroquinolones (levofloxacin, prulifloxacin, and sitafloxacin) and cephalosporins (cefditoren pivoxil), which were compared to either ertapenem, piperacillin-tazobactam, or ceftriaxone. The IV to oral switch was performed on day three or four of antibiotic therapy in patients whose clinical symptoms and laboratory parameters were improving and who could take oral medications. In three studies patients were no longer febrile at the time of switch, while the third study did not specify.

Pregnant and lactating women, as well as patients with sepsis or septic shock, immunosuppression, severe renal impairment, recurrent UTI within one month, structural abnormalities of the urinary tract (not further defined), or indwelling urinary catheters were excluded from most of these studies.

Studied comparison: The included trials compared transitioning to oral therapy against continuing parenteral antibiotics for the total duration of antimicrobial therapy.[3-6] Three studies used oral fluoroquinolones, and the total duration of therapy varied from 10 to 14 days. In these trials the patients enrolled had an effective oral option, meaning that the drug in the oral switch arm had good oral bioavailability, was excreted in the urine, and was active against the causative pathogen. Three of the four studies screened for susceptibility to antibiotic given in the oral arm, while one did not but was published before fluoroquinolone resistance became widespread.[5]

Although neither cefditoren pivoxil, prulifloxacin or sitafloxacin are currently available in North America, these trials have relevant information that bears on the IV to oral switch question and thus were included. Sitafloxacin and prulifloxacin are broad-spectrum oral fluoroquinolones, active against many Gram-positive, Gram-negative, and anaerobic bacteria, including strains resistant to other fluoroquinolones. Cefditoren pivoxil is a broad-spectrum, oral, third-generation cephalosporin. Cefditoren has a high volume of distribution, and 20-30% of the drug is excreted unchanged in the urine. The antimicrobial therapy varied among studies: (1) IV ceftriaxone followed by oral cefditoren pivoxil versus IV ceftriaxone;[4] (2) IV carbapenems followed by oral sitafloxacin versus IV ertapenem;[3] (3) IV empirical antibiotics followed by oral prulifloxacin versus IV ertapenem[6], and (3) IV levofloxacin followed by oral levofloxacin versus IV piperacillin-tazobactam, and in both arms patients also received amikacin for at least 3 days.[5] See the supplementary material (Characteristics of the studies).

Study design and risk of bias**:** All studies were judged at “unclear” risk of selection bias due to either: 1) lack of reporting of the method used to generate randomization, or 2) randomization of such a small sample that meaningful comparison of groups at baseline was not possible; therefore, whether randomization was successful at balancing important characteristics was unclear.

Three trials were open-label studies, meaning that participants, healthcare workers, and outcome assessors were not blinded to the treatment arms.[3,5] Unblinded studies can affect the outcomes that require subjective judgment, such as how clinical improvement or adverse events are measured and interpreted, thus potentially introducing detection and/or performance bias. All four studies were funded by industry, potentially introducing bias due to financial conflict of interest. See the supplementary material (Cochrane Risk of Bias).

Studied outcomes**:** The only patient-important outcome considered critical for decision-making was clinical cure (at end of therapy, EOT). Other outcomes considered important for decision-making included recurrence of infection, length of hospital stay, serious adverse events, IV catheter associated adverse events, and non-serious adverse events. No studies reported readmission rate or microbiological cure.

Benefits, Harms, and Certainty of the Evidence (CoE)

Benefits and harms: Overall, transitioning from IV to oral therapy in the course of treatment for cUTI does not appear to reduce clinical cure or increase recurrence of infection, and transitioning may lead to potentially fewer intravenous catheter-related harms.

The evidence suggests that transition to oral therapy in patients with cUTI does not reduce clinical cure at EOT or TOC (risk difference or RD: 1.8%; 95%CI: -3.6% to 7.2%/ relative risk or RR: 1.02; 95% CI: 0.96 to 1.08; low CoE) as compared to patients continued on parenteral therapy for the full duration of treatment.

Transition to oral therapy may not increase recurrence of infection (RD: -2.0%; 95%CI: -2.9% to 6.1% / RR: 0.33; 95% CI: 0.04 to 3.05; low CoE), but this estimate is imprecise due to very few events and small sample size.

In the one study examining duration of hospitalization (Concia 2006),[5] transition to oral therapy might have reduced length of hospital stay (median 10.9 versus 17.2 days; absolute reduction of 6.3 days, 95% CI: 11.8 to 0.8 days fewer; very low CoE). This evidence is very uncertain due to imprecision (small sample size of 47 patients), risk of bias due to unblinded study design, and indirectness or lack of generalizability (i.e. the length of hospitalization reported in this study was directly influenced by the route of administration of antimicrobials since all patients received parenteral antibiotics in hospital). Despite this uncertainty, transitioning patients to oral antibiotics is very likely to shorten the duration of hospitalizations if receipt of IV antibiotics will delay hospital discharge.

Despite the available evidence being very uncertain, transition to oral therapy may reduce IV catheter related adverse events (RD: -4.9%; 95%CI: -11.5% to 1.7% / RR: 0.20; 95% CI: 0.01 to 4.04; very low CoE).

Each day of IV catheterization confers risks of adverse events related to the catheter. However, as IV treatment of cUTI is typically 7 days in duration or less, a switch between 3-7 days to oral therapy may not have appreciable benefit in terms of avoiding adverse events of catheterization in the individual patient. However, benefits may be realized in prevention of adverse events (such as infections) over a larger number of patients. An IV to oral switch can also reduce the volume of fluid and sodium given to the patient, but these outcomes were not studied in the included trials.

Transition to oral therapy may have little to no effect on serious adverse events (RD: -0.8%; 95%CI: -1.9% to 6.3% / RR: 0.65; 95%CI 0.11 to 3.88; low CoE) and on non-serious adverse events (RD: 1.0%; 95%CI: -2.1% to 16.7%/ RR: 1.35; 95% CI: 0.27 to 6.67; very low CoE). Imprecision due to few events and small sample size make these assessments uncertain.

Certainty of Evidence**:** The panel recognized that transitioning to oral antimicrobial treatment may provide the same potential benefits as continuing parenteral therapy (no reduction in clinical efficacy and no increase in recurrence of infection), may reduce length of hospitalization

in certain clinical contexts, and possibly reduce adverse events (IV catheter associated adverse events, the benefit of which was judged by the panel to be small, especially with shorter duration of therapy). The panel agreed the overall certainty of evidence for transitioning to oral antimicrobial treatment compared to continuing parenteral therapy for the duration of treatment is low, mainly due to concerns with the risk of bias and imprecision in the estimates. However, IV to oral switch is common practice. See the supplementary material (Evidence Profile Table).

Other supporting evidence

Supporting evidence from trials studying similar treatment strategies: Two additional trials evaluated similar treatment strategies aimed at restricting the use of parenteral therapy in pre-selected populations. One of these trials looked at early switch versus very early switch (single dose IV), while the other looked at oral therapy throughout versus early switch. The very early switch trial was in women with uncomplicated pyelonephritis and compared the efficacy of a single dose of IV ceftriaxone followed by oral cefixime to standard treatment of IV ceftriaxone while awaiting culture results. Clinical response on day three or four of therapy was excellent and comparable between the two strategies (Sanchez 2002).[7] Another trial compared oral sitafloxacin therapy (throughout the course) to parenteral ceftriaxone with early transition to oral cephalosporins for 7-14 days in adults with complicated UTI or pyelonephritis. Oral therapy from the start was non-inferior to early transitioning to oral therapy, but these results might have been influenced by the difference in resistance rate to the study antibiotic within each arm (6.4% to sitafloxacin vs 30.4% to either ceftriaxone or cefdinir, respectively).[8] In conclusion, both of these studies support the idea that switching to oral antibiotics is effective in treating cUTI.

Supporting evidence from trials studying other aspects of cUTI treatment: The long-held belief that IV antibiotics are more effective than oral has been undermined by data from randomized, controlled trials of treatment for numerous serious infections, including osteomyelitis, bacteremia, and endocarditis (Davar 2022).[1] Transitioning to an effective oral antibiotic has become common practice for patients with cUTI showing clinical improvement. Most modern trials designed to either optimize the choice of empirical therapy or the duration of treatment for cUTI have permitted early transition to oral therapy. These trials reported excellent clinical outcomes similar to those in studies restricting treatment to parenteral therapy.

From trials studying optimal duration of therapy for cUTI: Our systematic review of the literature for clinical question 3 (duration of therapy for cUTI) looked at shorter treatment (5 to 7 days) versus longer treatment durations (10-14 days) for cUTI. All 10 trial protocols included clinical question 3 started treatment with either oral therapy or parenteral therapy but permitted transitioning to oral therapy (either as per protocol or as decided by the physician in charge). This analysis showed that shorter duration of effective antimicrobial therapy was not associated with worse outcomes in patients with cUTI, even when transitioning to oral therapy during the course of treatment. See summary of evidence for the clinical question on duration of therapy for cUTI.

From trials studying optimal choice of definitive antibiotic therapy for cUTI when transitioning to oral antibiotics: A randomized, controlled trial enrolling 97 women with cUTI

which evaluated switch to oral fosfomycin versus oral ciprofloxacin after five days of IV therapy found clinical cure rates of 75% in both arms at 30-35 days post end of therapy.[9] Likewise, an oral switch study in 51 adults with cUTI (mostly pyelonephritis) who were switched to oral fosfomycin versus oral levofloxacin reported clinical cure rates of 84% and 86%, respectively.[10] In a study of IV fosfomycin versus beta-lactam antibiotics for bacteremic cUTI, 61 patients in the IV fosfomycin group were switched to oral fosfomycin, of whom 57 (93%) achieved clinical cure at the test of cure endpoint.[11] Of note, oral fosfomycin is not appropriate as an initial empiric treatment for cUTI, due to inadequate levels in tissue/bloodstream. Additionally, an observational cohort study of patients with gram-negative bacteremia from cUTI found that switch to oral fluoroquinolone or TMP/SMX had similar rates of recurrence within 60 days as completion of the full course of therapy intravenously.[12]

Supporting evidence from pediatric population: Two prior systematic reviews touched on this topic, although neither directly addressed the question in an adult patient population. A Cochrane review of routes of administration of antibiotics for severe UTI, published in 2007, included 15 studies, of which 9 focused on children, and 1 on pregnant women.[13] Only three of the six studies in adults were published after 2000. Studies were small and heterogeneous; only six addressed a specific comparison of switch (IV to oral) versus continuing IV therapy for the duration of treatment. Overall, no evidence was found that one route of antibiotic administration was less effective for the treatment of cUTI. However, patients who could not tolerate oral therapy were excluded from these trials. Voloumanou et al. published a systematic review in 2008 of early switch to oral versus intravenous therapy or late switch for patients hospitalized with pyelonephritis.[14] Of these eight studies, only two enrolled adults (rather than a pediatric population; both were conducted prior to 2000). Early switch was defined as occurring on days 1-4 of therapy; late switch was after 7-10 days of therapy. The antibiotics studied in these two trials were gentamicin, ceftriaxone, and cefixime. Overall, early switch to oral therapy was as effective as intravenous therapy for clinical and microbiologic cure rates, as well as preventing renal scarring in a pediatric population. While the populations included in these meta-analyses are not directly generalizable to the adult population, the concept that outcomes did not differ with route of administration is relevant.

Special Populations and Special Situations

Presence of bacteremia

As our systematic review of the literature identified only four randomized, controlled trials of IV to oral antibiotic switch, this small number of trials did not permit post-hoc analyses. Across these four trials only 35 patients had bacteremia, so we were unable to formally stratify the analyses for presence or absence of bacteremia in cUTI.

A recent meta-analysis pooled the results of post-hoc analyses of 3 randomized, controlled trials (Yahav 2019, von Dach 2020, Molina 2022)[15-17] comparing 7 versus 14 days of antibiotics to treat uncomplicated Gram-negative bacteremia for cUTI patients who had become

afebrile, were hemodynamically stable, and had appropriate source control by the time of randomization (Turjeman 2022).[18] All trials permitted step-down oral therapy as per the physician in charge. This analysis showed that shorter duration of effective therapy was not associated with worse outcomes in bacteremic patients with cUTI, even when many (or most) were transitioned to oral therapy. One of these 3 trials showed that transitioning from IV to oral was not associated with treatment failure in bacteremic cUTI patients, regardless of duration of antibiotic therapy (fixed 7-day course, fixed 14-day course, or duration guided by C-reactive protein) (von Dach 2020, personal communication). [16] In these three bacteremia treatment trials, the percentage of patients enrolled who had an ESBL-producing organism ranged from 6.9% to 18.7% (Von Dach 2020 and Yahav 2019).[15,16] The BALANCE trial further supports oral switch for patients with bacteremia in the context of a shorter overall duration of therapy.[19]

The evidence found did not specifically address the clinical scenario of a patient with cUTI or pyelonephritis who is treated with oral therapy in the emergency department or clinic and later discovered to have been bacteremic with a Gram-negative organism at the time of presentation. However, evidence suggests that such patients, if improving clinically, would not have to be switched to parenteral therapy simply because they were originally bacteremic. (Talan 2000, Von Dach 2020)[16,20,21]

Males with cUTI

We are unable to perform stratification for male patients only, as we could not access the original data from the majority of trials that included men and women. However, we believe that men with cUTI are equally eligible for IV to oral switch as women, with the caveat that an oral drug should be chosen that can penetrate the prostate in men with febrile cUTI. The panel is not aware of a validated approach to determine whether the prostate is involved in men with febrile UTI, so choosing a treatment that does penetrate the prostate is reasonable. Classes of UTI-relevant antimicrobials that have adequate prostatic penetration include fluoroquinolones and sulfonamides.[22] Some beta-lactam antibiotics have poor penetration into the prostate and prostatic fluid, although many cephalosporins do achieve therapeutic levels in the prostate. Nitrofurantoin does not appear to reach therapeutic levels in prostatic fluid and should not be used to treat acute or chronic bacterial prostatitis.[23,24] Evidence for effectiveness of oral fosfomycin as a treatment for acute bacterial prostatitis is sparse.[25]

Resistant pathogens

The four trials mainly focused on transitioning from IV to PO when the oral therapy was considered to be effective for the infecting organism. Therefore, it is important to consider resistance rates to antibiotics among pathogens isolated in these trials. In the 2006 Concia study, no information was provided on resistance, but resistance to the two agents tested (piperacillin-tazobactam and levofloxacin) was not high at the time of that study.[5] In the 2017 Malaisri trial of sitafloxacin versus ertapenem, a urine culture with an ESBL-producing *E. coli* was required for enrollment.[3] All causative organisms isolated in the enrolled patients in this trial

from Thailand were susceptible to carbapenems, and 94% were susceptible to sitafloxacin; only 25% were susceptible to ciprofloxacin or levofloxacin. Similarly, patients enrolled in the So-Ngern 2023 trial (prulifloxacin versus ertapenem) had to present with a UTI caused by an ESBL-producing organism which was susceptible to both studied drugs (with 76% of them being susceptible to ciprofloxacin and 100% to carbapenems).[6] The Monmaturapoj (2012) trial, also in Thailand, excluded patients with ESBL-producing *E. coli* or *Pseudomonas aeruginosa*, and having an organism susceptible to the study drugs (ceftriaxone and cefditoren pivoxil) was a requirement for inclusion.[4] In this study, 32% of *E. coli* strains were resistant to standard fluoroquinolones. In summary, 3 of these 4 trials provided effective therapy in both arms of the trial, supporting that IV to oral switch generally requires use of oral agent to which the causative pathogen is sensitive.

Pharmacologic issues and potential choices for oral switch

Consider the following criteria when choosing an oral route of therapy for cUTI: (1) patient is clinically improving, (2) if applicable, source control has been achieved, (3) the patient can absorb oral antibiotics, (4) an oral regimen is available for the target pathogen that achieves adequate levels where needed (e.g., bloodstream if bacteremia present), and (5) there are no patient-level psychosocial or economic factors that would favor the IV route.[1] From a pharmacological point of view, the extent of tissue penetration is not necessarily determined by the route of delivery. Instead, an oral dose should be chosen that will achieve levels in plasma similar to those achieved through the IV route, which may require a higher oral dose. Gastrointestinal disorders that could preclude IV to oral switch include malabsorption, short bowel syndrome, ileus, severe diarrhea, motility disorders, vomiting, delayed gastric emptying, or reduced gut perfusion due to shock (Landersdorfer 2023).[2] Care should also be taken to avoid concomitant administration of certain medications, such as supplements (calcium, iron, magnesium) or sucralfate, that can reduce bioavailability of certain antibiotics (e.g. fluoroquinolones); some antibiotics' absorption may also be affected by food.

Nitrofurantoin and oral fosfomycin are generally not appropriate choices for cUTI and/or suspected bacteremia due to inadequate levels in tissue/bloodstream. Oral fosfomycin has been used in small studies to treat cUTI (including pyelonephritis), but its effectiveness needs to be confirmed in a larger study.[10,26] Oral fosfomycin has been used to treat chronic bacterial prostatitis but has not formally been evaluated in acute or chronic bacterial prostatitis trials.[27]

Patients with severe renal insufficiency may have either delayed clearance of some antibiotics or heightened clearance due to renal replacement therapy. Consultation with a pharmacist would be advisable in patients with severe renal insufficiency when planning an IV to oral switch for treatment of cUTI.

Ideally, the choice of oral step-down therapy can be guided by susceptibility testing of the causative pathogen, but often the organism has not been identified. Oral switch therapy in such cases is usually guided by the suspected urinary organism(s) and the patient's response to the empirical agent given. In other words, if the patient has improved while on ceftriaxone, switching to an oral third generation cephalosporin would be a reasonable choice. Commonly

used oral switch options for cUTI include fluoroquinolones,TMP-SMX, and third generation cephalosporins (**Table 1.2**).[28,29]

Although robust clinical trials of oral cephalosporins such as cefpodoxime as initial treatment for cUTI in adults are lacking, in practice cephalosporins are used in many settings as step-down therapy, when ESBL-production is not a major concern.[29-31] When choosing an oral cephalosporin for cUTI, both oral absorption and urinary excretion may be relevant parameters (See dosing **Table 1.2**) for consideration. Observational studies suggest that third generation oral cephalosporins may be comparable to oral fluoroquinolones or TMP-SMX as step down therapy in patients with cUTI and gram-negative bacteremia.[29-31] However, such studies are conflicting on whether earlier generation cephalosporins (e.g. cephalexin), oral beta-lactams (e.g. amoxicillin and amoxicillin clavulanate), and cephalosporins with low bioavailability (e.g. cefdinir) are as efficacious as alternatives; these should be used cautiously and with optimized dosing.[12,32-34]

As an example, in one retrospective study that included patients who received cefdinir (which has low urinary excretion of only 13-23% and low oral absorption of only 25%) and a lower dose of cephalexin (500 mg every 8 hours), readmissions for UTI were higher in the beta-lactam group compared to those who received fluoroquinolones or TMP-SMX.[32]

Amoxicillin-clavulanate and cephalexin have potentially lower efficacy as demonstrated in multiple studies.[12,33] Additionally, we did not find substantial data supporting the use of ampicillin, cefadroxil, cefaclor, or cefdinir for cUTI. Ideally, a patient who receives any of these oral options as their initial empiric therapy would have a urine culture from a prior episode showing susceptibility to the agent chosen.[35]

Furthermore, trials of three days of beta-lactam antibiotics for acute cystitis in women (cefpodoxime, amoxicillin-clavulanate, cefadroxil, and amoxicillin) consistently found lower clinical and microbiologic cure in the beta-lactam recipients, in comparison to three days of ciprofloxacin or trimethoprim-sulfamethoxazole.[36,37] These trials provide indirect evidence that beta-lactams are not as effective for acute cystitis when used for the same duration as other classes of antibiotics; whether these results are generalizable to using beta-lactam antibiotics as oral switch therapy to treat complicated UTI is unknown. Another concern with treating cUTI with oral beta-lactam antibiotics is that standard dosing may not achieve adequate levels in the urine. For example, a retrospective cohort study found that 7 days of IV or highly bioavailable antibiotics was as effective as 14 days of antibiotic therapy for bacteremic cUTI; of note, the doses of beta-lactams considered to be bioavailable were the following: amoxicillin 1000 mg orally every 8 hours, amoxicillin-clavulanate 875–1000 mg orally every 8 hours, or cephalexin 1000 mg orally every 6 hours.[38] Increasingly institutions are using higher dose regimens for oral beta-lactams and cephalosporins as step down therapy for Gram-negative bacteremia of urinary origin.[28,39]

Table 1.2: Dosing of oral antibiotics for complicated UTI (in alphabetical order)

Drugs	Oral absorption (%)	Urinary excretion (%)	Dose for patients with normal renal function
Amoxicillin-clavulanate	80 (amoxicillin)[40] variable (clavulanate)[41]	50-70 (amoxicillin)[40] 25-40% (clavulanate)[40]	875mg-125mg every 8 to 12 hours[12,32-34,39,42-45] Other regimens may be more effective[a]
Cefixime	50[46]	50[46]	400mg once daily[7]
Cefpodoxime	50[46]	80[46]	200mg to 400mg every 12 hours[29,34,47]
Ceftibuten	75-90[46]	73[46]	9mg/kg daily (children)[b] 400mg daily or 200mg every 12 hours (adults)[48,49]
Cefuroxime	52[46,50]	90[46,50]	500mg every 12 hours[34,51]
Cephalexin	90[46]	90[46]	500mg to 1000mg every 6 hours[12,28,32,33,39,42-44,52] Other regimens may be more effective[a]
Ciprofloxacin	70[53]	40-50[53]	500mg to 750mg every 12 hours [20,28,34,39,54]
Levofloxacin	99[55]	64-100[55]	500mg to 750mg daily[28,47,54,56]
Other oral beta-lactams (e.g. amoxicillin, cefadroxil, cefaclor, cefdinir)	Comparative clinical outcomes data vs highly bioavailable oral alternatives are more limited and/or discouraging; consider use with infectious disease pharmacist consultation if alternatives are not available.		
Trimethoprim-sulfamethoxazole	70-90[57]	84 (sulfamethoxazole), 66 (trimethoprim)[57]	800mg-160mg every 12 hours[20,34]

[a]Despite routine use of optimized dosing, the majority of studies comparing switch to oral beta-lactams versus fluoroquinolones or trimethoprim-sulfamethoxazole for cUTI have found inferior outcomes with oral beta-lactams when amoxicillin-clavulanate or cephalexin were the predominant oral beta-lactams being used.

[b]Ceftibuten is the sole oral beta-lactam in this table with modern randomized, controlled trial data for cUTI in both children in adults; however, while it produced comparable clinical outcomes versus trimethoprim-sulfamethoxazole in children, in adults relapses were higher with ceftibuten versus norfloxacin.

Other considerations

Stewardship considerations

Transitioning to oral therapy may permit earlier discharge, reducing potential exposure to nosocomially-acquired pathogens such as *C. difficile*, and may avoid placement of a central or midline catheter, reducing the likelihood of central-line (or midline) associated bloodstream infection. Evidence is not sufficient to mandate an IV to oral switch from a stewardship perspective.

Patients' values and preferences

The route of administration of antibiotics for cUTI needs to be individualized by patient preference. Our patient representatives commented that side effects of some oral antibiotics can be worse than the side effects of some IV antibiotics, and that responses are individualized.

Preference for receiving treatment at home and the perceived ease of taking oral antibiotics favor oral treatment. IV devices can be painful and limit mobility. Although some patients (and physicians) erroneously believe that IV antibiotics may be better or stronger, patients are likely to prefer oral antibiotics if efficacy is equivalent to that of IV antibiotics. [58-60]

Consultation with patient representatives participating in this guideline further supported that treatment (whatever the route of administration) should mainly focus on achieving clinical cure without increasing the risk of recurrence of infection and readmission to hospital. Reducing the length of hospitalization and facilitating ease of administration were considered important, but the route of administration alone was not a driving factor in their decision-making process.

Costs, Resources, Feasibility and Equity

While no specific studies evaluate the cost effectiveness of transitioning to oral therapy rather than continuing parenteral therapy for cUTI, the costs of administering oral antibiotics are significantly lower than either outpatient parenteral antibiotic therapy (OPAT) or inpatient administration of IV therapy. OPAT is associated with decreased costs compared to prolonging hospitalization for administration of those agents,[61] but OPAT costs are significantly higher than oral therapy in several infections.[62] Switching to oral antibiotics for bone and joint infections rather than OPAT can provide reductions in length of stay and costs.[63] In endocarditis, transitioning to oral therapy was also associated with a reduction in IV catheter complications.[64] Thus, the panel judged that moderate to large savings favor transitioning to oral therapy rather than continuing parenteral therapy for the completion of the treatment for cUTI.

The panel could not identify a scenario in which transitioning to oral therapy would not be more feasible or would not increase equity as compared to continuing parenteral treatment (either in hospital or through OPAT).

Conclusions and research needs

The guideline panel suggests transitioning to oral antimicrobial treatment rather than continuing parenteral therapy in most patients with cUTI, including acute pyelonephritis, for those who are clinically improving, can take an oral medication, and for whom an effective oral option is available. The panel notes that a majority of the patients included in the studies supporting this recommendation were female and without indwelling urinary catheters. The oral antibiotics in three of the four studies were fluoroquinolones (including two not available in the United States, sitafloxacin and prulifloxacin). The evidence base included patients with both pyelonephritis and cUTI, although these are different infectious entities.

For patients with Gram-negative bacteremia associated with cUTI, the panel suggests transitioning to oral therapy in patients who are clinically improving, have adequate source control, who can take an oral medication, and for whom an oral option is available. Source control in this context primarily meant relief of urinary obstructions; patients with abscesses in the genitourinary tract were generally not included in these trials.

Further clinical trials are needed to confirm the efficacy of the widely used strategy of giving one dose of an IV antimicrobial agent to patients with acute pyelonephritis, in addition to a course of oral antibiotics, in comparison to treating entirely with oral antibiotics. Very little is known about shorter course therapy for cUTI that does not involve fluoroquinolones, and fluoroquinolone therapy is becoming increasingly less relevant for cUTI as resistance rates increase.

Additional research into the safety of transitioning to oral therapy for certain subpopulations at higher risk of treatment failure or complications, such as patients with indwelling urinary catheters, sepsis or septic shock, immunocompromised status, severe renal insufficiency, and functional or structural abnormalities of the urinary tract is needed to ascertain whether this transition is safe and effective in these scenarios.

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Clinical Practice Guideline by Infectious Diseases Society of America (IDSA): 2025 Guideline on Management and Treatment of Complicated Urinary Tract Infections: Duration of Antibiotics for Complicated UTI

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In patients presenting with complicated UTI (cUTI) with a clinical response to therapy, should total duration of antibiotics be prolonged to >7 days rather than shorter (<=7 days)?

Recommendations:

I. In patients presenting with complicated UTI (including acute pyelonephritis) and who are improving clinically on effective therapy, we suggest treating with a shorter course of antimicrobials, using either 5-7 days of a fluoroquinolone (conditional recommendation, moderate certainty of evidence) or 7 days of a non-fluoroquinolone antibiotic (*conditional recommendation, very low certainty of evidence*), rather than a longer course (10-14 days).

Definitions:

- An effective antimicrobial agent achieves therapeutic levels in the urine and relevant tissue and is active against the causative pathogen.
- The duration of therapy is counted from the first day of effective antibiotic therapy.

Comments:

- Most studies supporting this recommendation excluded patients with indwelling urinary catheters, severe sepsis, immunocompromising conditions, abscesses in the urinary tract, chronic kidney disease, bacterial prostatitis, complete urinary obstruction, or undergoing urologic surgical procedures. Some patients in these subpopulations may be at higher risk for complications or treatment failure and may need an individualized duration of therapy.
- Men with febrile UTI in whom acute bacterial prostatitis is suspected may benefit from a longer treatment duration (i.e., 10-14 days), although evidence to guide the optimal duration in this subgroup is lacking.
- This recommendation is driven by evidence from trials that primarily studied fluoroquinolones during a time when fluroquinolone resistance was less common. Evidence for short courses of oral beta lactams in cUTI is more limited, and higher doses may be required for efficacy.
- Consider evaluation for an ongoing nidus of infection requiring source control in patients who do not have prompt clinical improvement.
- This recommendation places a high value on antibiotic stewardship considerations as well as reducing the burden of antimicrobial administration from a healthcare perspective and reducing the burden of taking antibiotics from a patient perspective.
- Refer to **Figure 1.3 for a stepwise assessment of the intravenous to oral switch and the duration of antibiotic therapy.**

II. In patients presenting with complicated UTI with associated Gram-negative bacteremia and who are improving clinically on effective therapy, we suggest treating with a shorter course (7 days) of antimicrobial therapy rather than a longer course (14 days) (*conditional recommendation, low certainty in the evidence*).

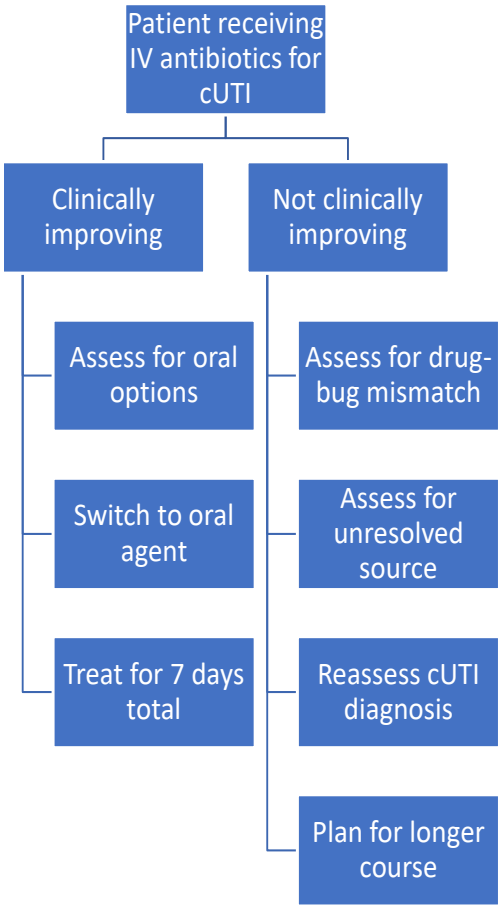
Definitions:

- An effective antimicrobial agent for bacteremic patients means that the antibiotic achieves therapeutic levels in the bloodstream, urine, and relevant tissue and is active against the causative pathogen.
- The duration of therapy is counted from the first day of effective antibiotic therapy.

Comments:

- Men with febrile, bacteremic UTI in whom acute bacterial prostatitis is suspected may benefit from a longer treatment duration (i.e., 10-14 days), although evidence to guide the optimal duration in this subgroup is lacking.
- Consider evaluation for an ongoing nidus of infection requiring source control in patients who do not have prompt clinical improvement.
- This recommendation places a high value on antibiotic stewardship considerations as well as reducing the burden of antimicrobial administration from a healthcare perspective and reducing the burden of taking antibiotics from a patient perspective.
- Refer to **Figure 1.3** for a stepwise assessment of the intravenous to oral switch and the duration of antibiotic therapy.

Figure 1.3: Stepwise assessment of IV to oral switch and duration of antibiotic therapy



Abbreviations: IV=intravenous, cUTI=complicated UTI. Drug-bug mismatch means that the causative organism is not susceptible to the antibiotic prescribed.

Introduction

The duration of therapy is important for antibiotic stewardship and has relevance to patients' wellbeing. Although it is unclear whether shortening an antibiotic course from 14 to 7 days has a meaningful impact on the individual's microbiome,[1] broader societal concern to reduce overall antibiotic use remains a priority.[2] Many studies show that infections are often treated for longer than guidelines recommend, and evidence demonstrates that each additional day of antibiotics increases the risk of patient harm, particularly in terms of adverse effects, and can negatively impact patient well-being.[3-5] These risks to individuals are further compounded by the public health concern of increased selection for and transmission of multidrug resistant organisms. Both IDSA guidelines on establishing antibiotic stewardship programs[6] and the CDC Core Elements of Hospital Antibiotic Stewardship Programs[7] recommend implementing strategies to reduce antibiotic therapy to the shortest effective duration.

Summary of evidence

Our systematic review of the literature (spanning from January 2000 up to September 2023) identified ten randomized, controlled trials comparing treatment with short duration (less than or equal to seven days) with longer duration of antimicrobial therapy (greater than seven days) for adults with complicated UTI.[8-17]

Studied populations and clinical settings: These ten trials enrolled and randomized a total of 2,945 patients, of which 2,681 were further described and analyzed. The timeframe for enrollment spanned from October 1994 up to April 2019. Trials were performed in North America (Darouiche 2014, Peterson 2008, Talan 2000),[8-10] Europe (Dinh 2017, Sandberg 2012, van Niewkoop 2017, Wagenlehner 2018, Lafaurie 2023),[11-15] and Asia (Ren 2017, Rubrabhatla 2018).[16,17] The settings for enrollment varied, with some trials enrolling only outpatients who presented to a primary care clinic or the emergency department,[10,14,15] some enrolling both outpatients and hospitalized patients,[9,11,13,17] and others exclusively enrolling patients who had been hospitalized for cUTI.[12,16] One of the ten trials enrolled hospitalized patients with spinal cord injury and an indwelling urinary catheter.[8]

Most studies enrolled both adult males and females,[8,9,12,13,16,17] while three trials included only adult women,[10,12,14,15] and one only included adult men.[11] Across all ten trials, a total of 909 males (33.9%) and 1,772 females (66.1%) were studied.

The clinical distinctions between uncomplicated pyelonephritis, complicated pyelonephritis, febrile UTI, and complicated UTI are not standardized or clearly understood from a pathogenesis point of view. Not surprisingly, several studies that included patients with both pyelonephritis and other types of cUTI did not specify how many patients fell into each category of infection. All studies included patients with a presumptive or confirmed diagnosis of cUTI as defined by these guidelines (which includes acute uncomplicated pyelonephritis). Two trials included only women with acute uncomplicated pyelonephritis, excluding other forms of complicated UTI.[10,15] Definitions of cUTI varied widely between trials and were not always clearly specified. For example, van Nieuwkoop et al. required symptoms of UTI with fever and a positive urine nitrite test or pyuria,[13] while Sandberg et al. required only presence of a urinary

symptom plus fever.[14] Peterson et al. enrolled patients with a diagnosis of either acute pyelonephritis or complicated UTI, but the definitions for these two conditions were not clear.[9]

The organism causing the cUTI was *Escherichia coli* in the majority of cases. Of the nine trials reporting on causal uropathogens,[9-17] the organism most frequently identified was *Escherichia coli* (1,202/1,604 (74.9%), median of 84.2%, ranging from 36.9% to 97.7% between studies).

Exclusion criteria varied greatly between studies. Studies excluded patients with indwelling urinary catheters, with the exception of three trials.[8,9,13] The subpopulation of patients with urinary catheters comprised only 5-7% of the overall population included in our meta-analysis. Eight of the studies excluded patients who presented with severe sepsis or septic shock, while two trials included this sicker population.[9,13] Immunosuppression was a reason for exclusion, except for two trials.[9,13] Features used to define immunosuppression included neutropenia, transplant recipient, advanced cirrhosis, and receiving >20 mg/day of corticosteroids. Patients with severe chronic kidney disease, on renal replacement therapy or with acute prostatitis were excluded from most studies (except van Niewkoop 2017 and Lafaurie 2023).[11,13] Overall, only a small number of patients within the total cohort had indwelling catheters, severe sepsis on initial presentation, immunocompromised status, chronic kidney disease, or acute bacterial prostatitis.

The majority of studies formally excluded patients with suppurative or chronic infections of the urinary tract, such as renal abscesses, chronic pyelonephritis and chronic bacterial prostatitis. Patients with complete urinary obstruction or recent urinary instrumentation, surgery, or lithotripsy were excluded. Additionally, pregnant and lactating women were excluded. Overall, the evidence identified to address the question of duration of treatment for cUTI did not include these subpopulations.

Comparisons Studied: The trials included in this evidence base compared either a 5-day course of antibiotics against 10-day course[8,9,12,15] or a 7-day course of antibiotics against 14-day course,[10,11,13,14,16] except for one study which compared 5 to 7-14 days of two different doses of levofloxacin.[17]

The majority of included trials addressed duration of fluoroquinolone therapy. Five of these trials examined the effect of a shorter versus longer duration of fluoroquinolone therapy,[9,12,14,15,17] while two trials permitted the use of a non-fluoroquinolone regimen in hospitalized patients before switching to an effective fluoroquinolone as soon as the results of urine culture became available, for 7 versus 14 days total therapy.[11,13] Talan et al. (2000) compared 7 days of ciprofloxacin to 14 days of trimethoprim-sulfamethoxazole in 255 women with acute pyelonephritis.[10] Only one small trial studied duration of therapy with two non-fluoroquinolone regimens, in the context of a very high fluoroquinolone resistance rate (about 78% of *E. coli* isolates were resistant to ciprofloxacin).[16] Ceftriaxone and amikacin were the antibiotics used most often in this trial. The Darouiche et al. trial (2014) studied 5 versus 10 days of antibiotics in 61 persons with spinal cord injury and indwelling catheters.[8] Antibiotic regimens were not specified by the study protocol, and 30% of patients received a

fluoroquinolone. In summary, 8 of the 10 trials specified fluoroquinolone use in one or both arms of the trial.

Many of these trials were conducted before widespread emergence of resistance to fluoroquinolones. Most trials studying the optimal duration of therapy using fluoroquinolones either avoided randomizing patients with fluoroquinolone-resistant uropathogens[11,13] or withdrew them from the analysis after randomization.[14,15] Two of the three trials that did not systematically exclude this subpopulation reported fluoroquinolone resistance rates of uropathogens varying from 9% to 16%.[9,12] Enrollment was completed in the most recent of these trials in 2014. The trial which compared a fluoroquinolone to trimethoprim-sulfamethoxazole was performed in 1994-1997 (Talan et al., 2000).[10] *Escherichia coli* was the causative pathogen in 93% of these cases of acute pyelonephritis, with resistance rates of 0% to ciprofloxacin and 18% to trimethoprim-sulfamethoxazole.

Two of these trials provided only indirect evidence on the effect of shorter duration of therapy since their results may have been influenced by another simultaneous intervention in addition to different durations of therapy. The Darouiche et al. study (2014) compared a 5-day course therapy plus urinary catheter exchange vs a 10-day course of therapy without catheter exchange in patients with spinal cord injury and long-term indwelling urinary catheters, so the two arms differed in catheter exchange in addition to duration of therapy.[8] Talan et al. (2000) compared a 7-day course of a fluoroquinolone to a 14-day course of trimethoprim-sulfamethoxazole in women with acute uncomplicated pyelonephritis, so the two arms differed in terms of antibiotic given as well as duration.[10] See the supplementary material (Characteristics of the studies).

Study design and risk of bias: All studies but one[12] consisted of non-inferiority trials, with a predefined non-inferiority margin varying from 10% to 15%. Most trials randomized patients at diagnosis (prior to any clinical assessment) except for Rhudrabhatla et al. (2018) in which patients needed to have improved clinically to be randomized (eligible patients should have clinically improved following empirical or culture-guided antibiotic treatment and should be afebrile for >48 hours at the time of randomization).[16] Most trials simply assessed clinical improvement as part of their primary outcome for clinical cure.

Four trials were open-label studies, meaning that participants, healthcare workers, and outcome assessors were not blinded to the treatment arms.[8,15-17] Unblinded studies can affect the outcomes that require judgment, such as how clinical improvement or adverse events are measured and interpreted, thus potentially introducing detection and/or performance bias.

Results of multiple trials may have been influenced by incomplete outcome data due to exclusion after randomization (e.g., early withdrawal secondary to the lack of a confirmatory positive urine culture or presence of an uropathogen showing resistance to the investigational drug).[8-10,14,15] Unfortunately, the full extent of this potential attrition was not completely assessable in most studies, and attrition could reduce sample size without introducing bias. The three studies funded by industry may also have been biased due to potential financial conflict of interest.[9,10] Two studies showed evidence of potentially failed randomization.[8,11] See the supplementary material (Cochrane Risk of Bias).

Studied outcomes: The patient-important outcomes which were considered critical for decision-making included clinical cure at end of therapy (EOT) or at the first follow up after EOT and recurrence of infection (up to a maximum of 180 days). Other outcomes that were considered important for decision-making were microbiologic cure, serious adverse events, non-serious adverse events, length of hospital stay, and readmission rate. While most trials reported on these outcomes, only two studies reported on readmission/rehospitalization rate,[13,16] and only one reported on length of hospital stay.[16]

Microbiological failure (post-treatment asymptomatic bacteriuria) is frequently considered a surrogate marker for recurrence of infection. Since recurrence of infection was already included in our analysis as a critical outcome for decision-making, the panelists and the patients considered microbiologic cure as important but not critical for decision-making.

Benefits, Harms, and Certainty of the Evidence (CoE)

Benefits and harms: Overall, a shorter course of treatment likely does not reduce clinical efficacy or increase recurrence of infection, but a shorter course may not reduce adverse events.

Treatment with a shorter course of antimicrobial therapy in patients with cUTI likely does not reduce clinical cure (at TOC) (risk difference or RD: 0%; 95% CI: -2.6% to 3.5% / relative risk or RR:1.00; 95% CI: 0.97 to 1.04; moderate CoE) and may not reduce microbiological cure (at TOC) (risk difference or RD: -0.8%; 95% CI: -5.1% to 4.2% / relative risk or RR:0.99; 95% CI: 0.94 to 1.05; low CoE) as compared to patients treated with a longer course.

Shorter duration of therapy likely does not increase recurrence of infection (up to 180 days of follow-up) (RD: 0.5%; 95% CI: -2.1% to 4.5% / RR:1.07; 95% CI: 0.69 to 1.65; moderate CoE). The evidence suggests that shorter therapy may not increase readmission/rehospitalization during follow-up (RD: 0%; 95% CI -0.7% to 6.9% / RR: 0.99; 95% CI: 0.10 to 9.33; low CoE).

A shorter course of antimicrobial therapy might have reduced the length of hospital stay in the only study that looked at this outcome (median 8 days vs. 14 days, absolute reduction of 6 days, $p<0.001$; very low CoE). However, length of hospitalization was likely influenced by the route of administration of antimicrobials, as all patients received parenteral antibiotics throughout this study for the assigned duration in the hospital, without switching to an oral option.[16]

The evidence provided by the ten included trials suggests that shorter course of antimicrobial therapy results in little to no reduction in serious and non-serious adverse events in these trial populations during follow- (RD: -0.6%; 95% CI: -1.6% to 0.9% / RR: 0.82; 95%CI: 0.54 to 1.25; low CoE, and RD: -2.3%; 95% CI: -6.0% to 2.0% / RR: 0.92; 95% CI: 0.79 to 1.07; low CoE, respectively).

Other supporting evidence on harms: A recently published review included 71 randomized, controlled trials published between 1972 to 2016 on various fixed duration of antibiotics

(between 3 and 15 days) for different types of infections (of which 29 studies looked at UTI more specifically).[3] This meta-analysis found that each additional day of antibiotic therapy is associated with an increased odds of experiencing adverse events (OR 1.04; 95% CI: 1.02 to 1.07) and severe adverse events (OR 1.09; 95% CI: 1.00 to 1.19), but not of superinfection (such as *C. difficile* and candidiasis) and antimicrobial resistance.

Certainty of Evidence: The panel agreed on the overall certainty of evidence for treatment with a short course compared to a longer course of treatment as being moderate, mainly due to concerns with the risk of bias. Given that shorter course of treatment: (1) likely does not reduce clinical efficacy, (2) probably does not increase recurrence of infection or readmission/rehospitalization, and (3) may reduce length of hospitalization, the panel recognized the benefit of a shorter course of treatment. The shorter course of treatment may not reduce adverse events compared to the longer course. See the supplementary material (Evidence Profile Table).

Special Populations and Special Situations

Choice of antibiotics

Fluoroquinolone antibiotics: Seven out of 10 studies compared shorter to longer course of fluoroquinolone therapy in patients with cUTI (including pyelonephritis).[9,11-15,17] Our stratified analysis focusing on fluoroquinolone therapy showed that treatment with a shorter course of fluoroquinolone likely does not reduce clinical cure (at TOC) (RD: -1.8%; 95% CI: -3.5% to 0.9% / RR:0.98; 95% CI: 0.96 to 1.01; moderate CoE) and may not reduce microbiological cure (at TOC) (RD: -1.7%; 95% CI: -5.9% to 2.5% / RR:0.98; 95% CI: 0.93 to 1.03; low CoE). Furthermore, shorter therapy likely does not increase recurrence of infection (up to 90 days of follow-up) (RD: -0.5%; 95% CI: -2.8% to 3.2% / RR:0.93; 95% CI: 0.58 to 1.49; moderate CoE) and may not increase readmission/ rehospitalization during follow-up (RD: -0.5%; 95% CI: -1.4 to 0.5% / RR: 3.00; 95%CI: 0.1 to 72.7; low CoE). See the supplementary material (Evidence Profile Table).

Non-fluoroquinolone antibiotics: One trial directly compared shorter versus longer duration of non-fluoroquinolone antibiotics. The evidence from this one trial suggests that shorter duration of antibiotics may not lead to reduced clinical or microbiological cure, or to more recurrence of infection or readmission, but these estimates are very uncertain (very low CoE) due to the very small number of patients enrolled in this trial (54 patients).[16] Patients who received the shorter course of antibiotics possibly had a shorter length of hospitalization as compared to those treated with longer duration. Patients in this trial were primarily treated with aminoglycosides; 78% of the *E. coli* isolates collected from patients in this trial were resistant to fluoroquinolones. See the supplementary material (Forest plots).

Furthermore, trials of three days of beta-lactam antibiotics for acute cystitis in women (cefpodoxime, amoxicillin-clavulanate, cefadroxil, and amoxicillin) consistently found lower clinical and microbiologic cure in the beta-lactam recipients, in comparison to three days of ciprofloxacin or trimethoprim-sulfamethoxazole.[18,19] These trials provide indirect evidence that

beta-lactams are not as effective for acute cystitis when used for the same duration as other classes of antibiotics; whether these results are generalizable to the necessary duration of beta-lactam antibiotic to treat complicated UTI is unknown. Another concern with treating cUTI with oral beta-lactam antibiotics is that standard dosing may not achieve adequate levels in the urine. For example, a retrospective cohort study found that 7 days of IV or highly bioavailable antibiotics was as effective as 14 days of antibiotic therapy for bacteremic cUTI; of note, the doses of beta-lactams considered to be bioavailable were the following: amoxicillin 1000 mg orally every 8 hours, amoxicillin-clavulanate 875–1000 mg orally every 8 hours, or cephalexin 1000 mg orally every 6 hours.[20] Increasingly institutions are using higher dose regimens for oral beta-lactams and cephalosporins as step down therapy for Gram-negative bacteremia of urinary origin.[21,22]

Presence of bacteremia

Complicated UTI is frequently accompanied by bacteremia, although the prevalence of bacteremia is difficult to specify as populations and definitions of cUTI vary across studies. Our systematic review of the literature identified three trials that performed post-hoc analyses stratifying for subgroup of patients presenting with cUTI with or without bacteremia.[10,13,14] Although the analysis of these post-hoc analyses did not show a reduction in clinical cure at TOC between shorter and longer course of treatment, this analysis only included a very small subgroup of each of the initial three cohorts, which thus brought serious concerns regarding risk of bias (selection bias) and imprecision (very small sample size, n=91). These issues undermine the certainty of this finding when making inferences on efficacy of short course of antibiotics among bacteremic patients. Similar results were reported in a previously published meta-analysis including randomized, controlled trials published between 1988-2012 on duration of therapy for pyelonephritis (only including 86 of 2,515 patients with bacteremia).[23]

We found additional evidence about how to manage bacteremic cUTI from randomized, controlled trials of duration of therapy for Gram-negative bacteremia arising from multiple infectious etiologies. Three recently published randomized, controlled trials compared 7 versus 14 days of antibiotics to treat uncomplicated Gram-negative bacteremia in patients who were on antibiotic therapy, afebrile, hemodynamically stable, and with appropriate source control.[24-26] Source control in the context of cUTI was defined as relief of urinary obstruction, if present. None of the studies were focused on cUTI, but between 55% to 68% of patients in these studies had a urinary source of bacteremia. Antibiotics given in these studies were drawn from multiple classes. A recent meta-analysis of these three trials pooled the results of post-hoc analyses for cUTI patients.[24] The meta-analysis showed that shorter duration was not associated with an increase in relapse of bacteremia, mortality at 30 or 90-days, or readmission in patients with cUTI. Unfortunately, some uncertainty in these estimates remains due to the inherent nature of these post-hoc analyses and the lack of blinding.

Post hoc analysis of one of these three RCTs showed inferiority of the 7-day treatment in the 173 males with UTI and bacteremia included in this cohort, although the trial excluded men with known prostatitis.[24,27] In these 173 men, 24% of the 7-day treatment group, and 37% of the 14-day treatment group had an indwelling urinary device (urinary catheter, nephrostomy tube, or double J catheter). The difference in the primary composite outcome (which included all-cause

mortality; relapse, suppurative, or distant complications; and readmission or extended hospitalization over 14 days) between the two groups (RD: 6.2%; 95%CI: -8.6% to 21.2%) was driven primarily by a higher rehospitalization rate in the 7-day treatment group (RD: 4.5%; 95%CI: -10.2% to 19.3%). Whether this post hoc analysis limited by a small sample size from a single trial should guide management of men with cUTI and bacteremia is unclear.

A recent trial added evidence in support of our recommendation. The BALANCE trial randomized 3,608 hospitalized patients with bacteremia to 7 versus 14 days of antibiotics (IV or oral); the majority of the patients had a Gram-negative organism, and 42% had a urinary source of bacteremia. The shorter duration of therapy was non-inferior to the longer duration of antibiotic therapy for the primary outcome of death within 90 days, both overall and in the subgroup of patients with bacteremia from a urinary source (risk difference -1.9% with 95% CI -5.2 to 1.4), which provides support for our recommendation.[28]

Males with cUTI

Across all 10 trials, seven enrolled men, including a total of 909 men with cUTI or pyelonephritis. Of these seven trials, only two (van Nieuwkoop FUTIRST 2017 and Lafaurie PROSTASHORT 2023) could provide data for a stratified analysis of outcomes in men with cUTI, covering 326 men (36% of the men included in these seven trials).[11,13] In both of these studies, men were not systematically screened with rectal examination and *were not excluded if suspected of having acute prostatitis*. Of note, this analysis does not include 583 men who are represented in other studies presented here, because we could not obtain stratified information from the authors of these studies based on male sex/gender. The stratified analysis that follows is thus based on these two trials but does not include the larger population of men who were enrolled in the other five trials; in these five trials the overall clinical outcomes of shorter and longer duration of therapy for cUTI were comparable.

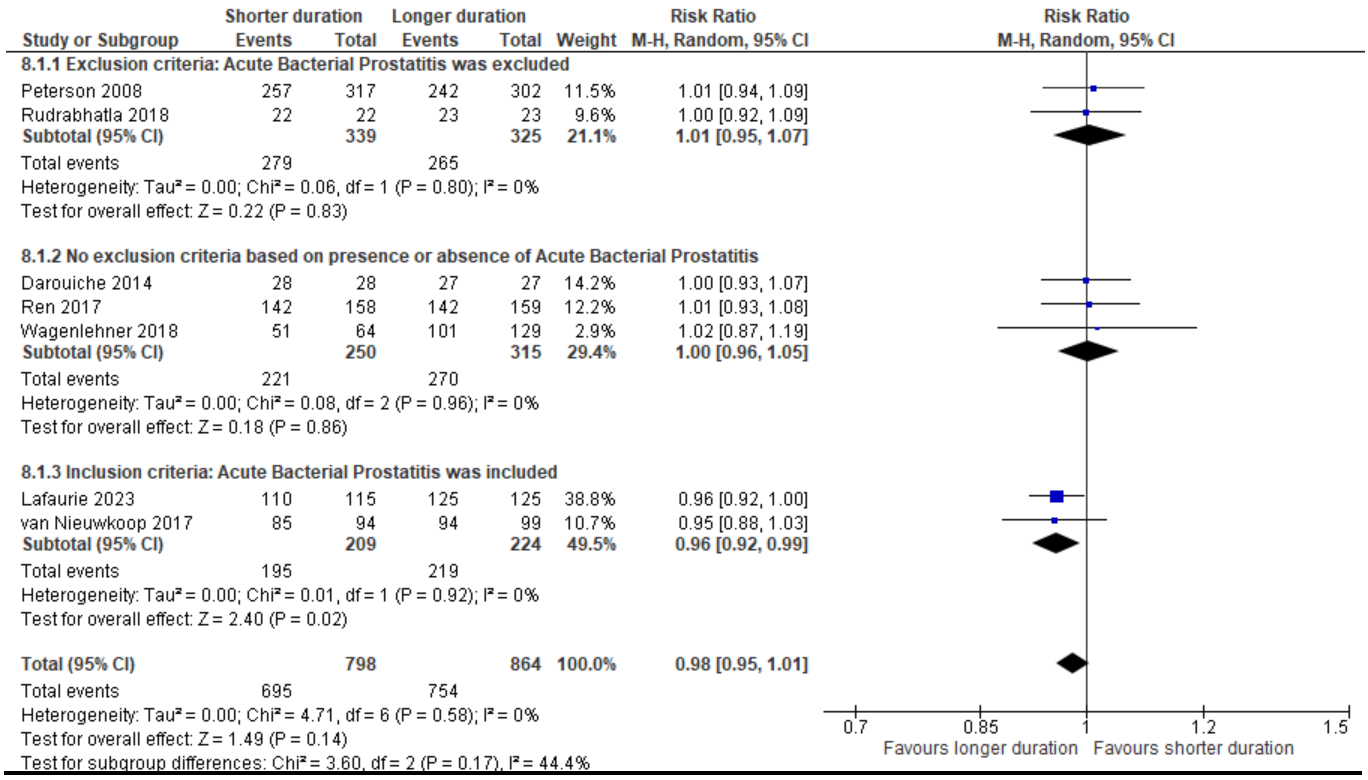
Based on our pre-determined decision threshold (non-inferiority margin) of 10% for clinical and microbiological cure, the evidence suggests that a shorter course of antimicrobial therapy in men with febrile UTI from these two trials [11,13] may decrease clinical cure (up to 6 weeks after starting therapy) (RD: -6.1%; 95% CI: -12.7% to 4.1% / RR: 0.94; 95% CI: 0.87 to 1.01; very low CoE) as well as microbiological cure (at TOC) (RD: -14.5%; 95% CI: -23.1% to -5.9%/ RR: 0.85; 95%CI: 0.76 to 0.94; very low CoE), but this evidence is very uncertain due to the high risk of bias in these studies and the small sample size (N= 326). In terms of risk of bias, in the larger study the men in the shorter course group had more comorbidities, thus randomization may have failed.[11]

The subgroup of men with acute prostatitis in these studies may be driving the conclusion that shorter course therapy is not non-inferior for febrile UTI in men (**Supplementary Table 6:** Studies of duration of treatment for cUTI including men). This point about prostatitis is important, so we have illustrated it with both a table (**Supplementary Table 6**) and a Forest plot (**Figure 2.3**). Lafaurie et al. (2023)[11] provided data on a subgroup of 27 patients suspected of having acute prostatitis on rectal examination (personal communication), and this analysis showed a non-significant reduction in clinical cure with shorter course of antimicrobial therapy

was observed (RD: -20.0%; 95% CI: -51.7% to 11.8% / RR: 0.77; 95% CI: 0.49 to 1.20). Recurrence of infection (up to 12 week of follow-up) was comparable between groups (RD: -2.4%; 95%CI: -7.7% to 2.8% / RR: 0.50; 95%CI: 0.10 to 2.49) but this evidence was based on a very small number of patients from only one study and might have been influenced by a higher rate of lost-to-follow-up in the short course group.[11]

Detection of a painful prostate on rectal examination is probably the most practical way to diagnose acute prostatitis. In current clinical practice, digital rectal examination is rarely done in men with UTI, and the diagnostic accuracy of the digital rectal examination for prostatitis is not well defined.[29] Acute prostatitis may have a spectrum of severity, in which most men with febrile UTI have some early degree of prostate involvement, as marked by elevated prostate specific antigen, but only in a few of them does the entire gland become infected and inflamed.[30]

Figure 2.3: Subgroup analysis of the impact of prostatitis on effectiveness of shorter versus longer treatment duration for cUTI: Forest plots for clinical cure (n=7 trials, 2000-2024). In these Forest plots the trials are groups by whether the eligibility criteria for enrolling males assessed the presence/absence of acute bacterial prostatitis and either did or did not exclude men if prostatitis was suspected. Trials that did not comment on presence or absence of prostatitis in men are grouped under "No exclusion criteria based on the presence or absence of acute bacterial prostatitis."



Patients with indwelling catheters

Our systematic review of the literature only included one trial that compared duration of therapy in UTI associated with indwelling urinary catheters,[8] and this was a study exclusively in hospitalized patients with spinal cord injury. Two other trials included some patients with indwelling catheters.[9,13] Generalizability of our meta-analysis to patients with indwelling catheters is unclear due to the limited data and the unique characteristics of the population.

Other considerations

Stewardship considerations

In general, shorter courses of antibiotics reduce complications, and reduce pressure for resistance as well as reduce adverse outcomes.[31] Risk of superinfection (secondary infection with another organism) and resistance increases substantially with extended duration of therapy,[3] and *C. difficile* superinfection risk is clearly extended with antibiotics.[32] A meta-analysis which included 35 systematic reviews reported that each day of antibiotic therapy was associated with 4% increased odds of experiencing an adverse event and a 3% increased odds of development of antimicrobial resistance.[3]

Reducing the duration of therapy to the shortest effective course is therefore very important to antibiotic stewardship. Both IDSA guidelines on establishing antibiotic stewardship programs and the CDC Core Elements of Hospital Antibiotic Stewardship Programs recommend implementing strategies to reduce antibiotic therapy to the shortest effective duration.[6,7] A narrative review of over 120 randomized clinical trials showed that shorter-course therapy was as effective as longer-course therapy for a variety of infections, although the durations and types of infections treated were heterogeneous.[33] Many studies show infections are often treated for longer than guidelines recommend, and evidence demonstrates that each additional day of antibiotics increases the risk of patient harm.[4,5] These risks to individuals are further compounded by the public health concern of increased selection for and transmission of multidrug resistant organisms. Furthermore, providing patients with longer courses of oral antibiotics has the practical concern of creating more opportunities for patients to store leftover antibiotics at home, if they do not take the entire prescribed course.[34]

Patients' values and preferences

There is little direct evidence addressing antibiotic duration in cUTI and patient well-being. Consultation with patient representatives participating in this guideline revealed a

disconnect between stewardship goals (shorter is better) and their desire to avoid chronic urinary symptoms (longer is better). Several of our patient representatives suffer from chronic UTI symptoms, and their experience may be representative of a subpopulation that is most severely affected by UTI. Our recommendations may be less applicable to this subpopulation. In light of the high value our patient representatives place on avoidance of recurrent infection, we considered recurrence of infection to be a critical outcome for our analyses.

Clinical cure (with relief of symptoms) was the most important outcome from the patient representatives' perspectives, and the duration of antibiotics chosen should be sufficient to achieve clinical cure. They did not feel that a negative urine culture was necessarily a goal, and they lacked confidence in the accuracy of urine cultures at predicting urinary symptoms. They also valued avoiding recurrence of symptomatic infection and readmission to hospital. Patients expressed that if an individual patient has experienced clinical recurrence of cUTI after a shorter duration of therapy, consideration should be given to treating the recurrence with a longer course of antibiotics.

Additional days of IV antibiotics incur all the risks of having an intravenous device in place, including pain and discomfort, reduced mobility, phlebitis, infusion site reactions, cellulitis, and even bacteremia. Antibiotics administered systemically through any route can cause nausea, diarrhea, gastrointestinal upset, metallic taste, or other side effects that limit the patient's ability to function normally. Longer courses of antibiotics are associated with antibiotic-related complications, including hospitalizations and resistant infections.[31]

Costs, Resources, Feasibility and Equity

While few studies have quantified cost/resource reductions associated with a shorter duration of treatment specific to cUTI, the panel judged that a potential reduction in cost and resources favors the shorter course of antibiotics. This can be justified by a reduction in medication costs, length of hospitalization, or outpatient parenteral antibiotic therapy costs.[35-37] The panel could not identify a scenario where a longer duration of therapy would be more feasible, as compared to a shorter course of treatment. Likewise, a longer duration of therapy, if no more effective than a shorter duration of therapy, should not impact equity.

Conclusions and research needs

The guideline panel suggests treating cUTI, including acute pyelonephritis, with a shorter course of antibiotics rather than a longer course. The panel notes that the majority of the patients included in the evidence base were female, outpatients, and treated with a fluoroquinolone. Also, these trials generally enrolled patients whose organisms were susceptible to the antibiotics under study, meaning that effective therapy is likely important to the success of a shorter course of treatment. For patients with uncomplicated Gram-negative bacteremia, the panel suggests a shorter course of antibiotic therapy in patients with cUTI when afebrile, hemodynamically stable and with adequate source control. Source control in the context of urinary-source bacteremia would mean ensuring relief of any urinary obstruction.

Additional clinical trial data would be very helpful to determine the duration of therapy for cUTI using non-fluoroquinolone antibiotics, as the resistance rate to fluoroquinolone antibiotics has increased dramatically since many of the studies supporting this recommendation were conducted. Also, larger studies that address the risk of complications with *E. coli* versus non-*E. coli* organisms would be welcome, as some urinary pathogens may be more likely to persist, invade the bloodstream, or evolve into abscesses. Additional clinical trials are needed to provide increased certainty about the potential for both benefit and harms of shorter treatment in cUTI complicated by bacteremia.

Subgroup analysis of men with cUTI, including those who are febrile and/or bacteremic, suggests that 10-14 days of antibiotics may be associated with higher rates of cure. Additionally, men with cUTI and acute prostatitis may need a longer course of antibiotics than 7 days to avoid clinical failure. Research is needed on when to consider prostatic involvement in men with febrile UTI and how the extent of prostatic involvement impacts the ideal treatment duration. Additional studies are needed to define how to diagnose and manage acute prostatitis.

Additional research into safety of shorter treatment in patients at higher risk of treatment failure or complications of cUTI, such as men (with or without acute prostatitis), patients with indwelling urinary catheters, with sepsis or septic shock on initial presentation, with immunocompromised status, or with severe renal insufficiency is needed to ascertain whether this course of action is acceptable in these scenarios. Infectious diseases practice in general is trending towards more individualized durations of therapy based on patient-specific risk factors for treatment failure, and further research is needed to determine such risk factors in patients presenting with cUTI. Individualized durations of therapy based on symptom resolution should also be studied.

These recommendations are not necessarily applicable to lactating or pregnant women, who were often excluded from the studies evaluated; patients with suppurative or chronic infections, such as renal abscesses, chronic pyelonephritis, and chronic bacterial prostatitis; patients with complete urinary obstruction; patients with nephrostomy tubes; or patients with recent urinary instrumentation, surgery, or lithotripsy (in the prior 7 days). Traditional practice, albeit not informed by comparative data, favors longer durations in these subpopulations widely perceived to be at higher risk for treatment failure with shorter course of antimicrobial therapy.

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